

Topological Data Analysis and Topological Deep Learning Beyond Persistent Homology—A Review

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Abstract

Topological data analysis (TDA) is a rapidly evolving field in applied mathematics and data science that leverages tools from topology to uncover robust, shape-driven insights in complex datasets. The main workhorse is persistent homology, a technique rooted in algebraic topology. Paired with topological deep learning (TDL) or topological machine learning, persistent homology has achieved tremendous success in a wide variety of applications in science, engineering, medicine, and industry. However, persistent homology has many limitations due to its high-level abstraction, insensitivity to non-topological changes, and reliance on

超越持续同调论的拓扑数据分析与拓扑深度学习——综述

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摘要

拓扑数据分析 (TDA) 是应用数学和数据科学中一个快速发展的领域, 它利用拓扑学的工具来揭示复杂数据集中稳健的、由形状驱动的洞见。其主要工具是持续同调, 这是一种植根于代数拓扑学的技术。结合拓扑深度学习 (TDL) 或拓扑机器学习, 持续同调在科学、工程、医学和工业的广泛应用中取得了巨大成功。然而, 由于其高度的抽象性、对非拓扑变化的不敏感性以及对点云数据的依赖, 持续同调存在许多局限性。本文对超越持续同调的 TDA 和 TDL 进行了全面综述。它分析了持续拓扑拉普拉斯算子和狄拉克算子如何提供谱表示, 以捕获拓扑不变量和同伦演化。其他表述则从层理论、迈耶拓扑和交互拓扑的角度进行介绍。对于可微流形上的数据, 综述了植根于微分拓扑的技术, 例如持续德拉姆上同调、持续霍奇拉普拉斯算子和霍奇分解。对于嵌入三维空间的一维 (1D) 曲线, 讨论了来自几何拓扑的方法, 包括多尺度高斯链环积分、持续琼斯多项式和持续霍诺夫同调。本文进一步讨论了针对不同输入数据 (如点云、序列数据、流形上的数据、嵌入三维空间的曲线以及具有额外非几何信息的数据) 如何适当选择拓扑工具。还综述了各种拓扑表示、软件包和机器学习向量化方法。最后, 本综述以结束语收尾。

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point cloud data. This paper presents a comprehensive review of TDA and TDL beyond persistent homology. It analyzes how persistent topological Laplacians and Dirac operators provide spectral representations to capture both topological invariants and homotopic evolution. Other formulations are presented in terms of sheaf theory, Mayer topology, and interaction topology. For data on differentiable manifolds, techniques rooted in differential topology, such as persistent de Rham cohomology, persistent Hodge Laplacian, and Hodge decomposition, are reviewed. For one-dimensional (1D) curves embedded in 3-space, approaches from geometric topology are discussed, including multiscale Gauss-link integrals, persistent Jones polynomials, and persistent Khovanov homology. This paper further discusses the appropriate selection of topological tools for different input data, such as point clouds, sequential data, data on manifolds, curves embedded in 3-space, and data with additional non-geometric information. A review is also given of various topological representations, software packages, and machine learning vectorizations. Finally, this review ends with concluding remarks.

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1 Introduction

Topological data analysis (TDA), rooted in topological techniques, is an emerging approach in applied mathematics and data science. TDA extracts topological information from data that cannot be obtained using statistical, physical, or other mathematical methods. Since topology is one of the most abstract fields in mathematics, TDA inherently reduces the dimensionality of high-dimensional data. Additionally, it simplifies geometric complexity and provides high-order data representations, extending conventional zeroth-order algorithms. Unlike classical topology or homology, TDA employs multiscale analysis, bridging topology and geometry through filtration.

A cornerstone of TDA is persistent homology [112, 333], an algebraic topology tool for analyzing point clouds or discrete data. Persistent homology constructs a family of topological spaces across spatial scales, capturing topological invariants of varying dimensions. For further details, see [44, 116]. Its standard representations, i.e., persistence barcodes [132] and persistence diagrams [112], encode the birth, death, and persistence of features (e.g., connected components, holes, cavities) during filtration. Initially used for qualitative analysis, persistent homology has evolved into a robust tool for quantitative analysis and machine learning [42, 313, 317]. Applications span image processing [17, 72], neuroscience [84], computational chemistry [291], biology [37, 38, 128, 178], nanomaterials [184, 317], crystalline materials [161], and complex networks [156].

Despite its utility, persistent homology has notable constraints [310]. It primarily captures topological invariants, often overlooking geometric shape evolution during filtration when no topological changes occur. Moreover, it struggles with localized topological information in data and cannot directly analyze data on differential manifolds or 1D curves embedded in 3-space (e.g., knots, links).

Efforts have been made to develop persistent topological Laplacians for various data structures, providing representations that can be integrated into machine learning models. For point cloud data and graphs, persistent combinatorial Laplacians were introduced by Wang et al. [302] and have been extensively studied in recent years [143, 195, 217, 303]. The harmonic spectra of persistent combinatorial Laplacians capture changes in topological invariants, which could recover the topological information captured by persistent homology, while the non-harmonic spectra encode geometric shape changes in the filtration process [217]. In addition to persistent combinatorial Laplacians, a variety of persistent topological Laplacians have been proposed for other topological domains. These include persistent sheaf Laplacians [310] for labeled point cloud data, persistent path Laplacians [300] for directed graphs and networks, persistent hypergraph Laplacians [202] and hyperdigraph Laplacians [60] for graphs and networks with complex relationships, persistent directed flag Laplacian for flag complexes [163, 329]. Persistent topological Laplacians have been successfully applied to protein-ligand binding prediction [58, 219, 282], interactomic network modeling [110], gene expression

1 引言

拓扑数据分析 (TDA) 植根于拓扑技术, 是应用数学和数据科学中一种新兴的方法。TDA 从数据中提取无法通过统计、物理或其他数学方法获得的拓扑信息。由于拓扑学是数学中最抽象的领域之一, TDA 本质上降低了高维数据的维度。此外, 它简化了几何复杂性, 并提供高阶数据表示, 从而扩展了传统的零阶算法。与经典拓扑学或同调论不同, TDA 采用多尺度分析, 通过过滤连接拓扑与几何。

TDA 的一个基石是持续同调 [112, 333], 这是一种用于分析点云或离散数据的代数拓扑工具。持续同调在空间尺度上构建一系列拓扑空间, 捕捉不同维度的拓扑不变量。更多细节请参见 [44, 116]。其标准表示, 即持续条形码 [132] 和持续图 [112], 编码了特征 (例如连通分量、孔洞、腔体) 在过滤过程中的诞生、消亡和持续性。持续同调最初用于定性分析, 现已演变为定量分析和机器学习的强大工具 [42, 313, 317]。其应用涵盖图像处理 [17, 72]、神经科学 [84]、计算化学 [291]、生物学 [37, 38, 128, 178]、纳米材料 [184, 317]、晶体材料 [161] 以及复杂网络 [156]。

尽管持续同调具有实用性, 但它也存在显著的局限性 [310]。它主要捕捉拓扑不变量, 常常忽略过滤过程中当没有拓扑变化发生时几何形状的演变。此外, 它在处理数据中的局部拓扑信息方面存在困难, 并且无法直接分析微分流形或嵌入三维空间的一维曲线 (例如纽结、链环) 上的数据。为将各种数据结构开发持久拓扑拉普拉斯算子, 已付出诸多努力, 这些表示可被整合到机器学习模型中。针对点云数据和图, Wang 等人 [302] 引入了持久组合拉普拉斯算子, 并在近年来得到了广泛研究 [143, 195, 217, 303]。持久组合拉普拉斯算子的调谐捕捉了拓扑不变量的变化, 这可以恢复持久同调所捕获的拓扑信息, 而非调谐则编码了过滤过程中的几何形状变化 [217]。除了持久组合拉普拉斯算子外, 针对其他拓扑领域也提出了多种持久拓扑拉普拉斯算子。这些包括用于标记点云数据的持久层拉普拉斯算子 [310]、用于有向图和网络的持久路径拉普拉斯算子 [300]、用于具有复杂关系的图和网络的持久超图拉普拉斯算子 [202] 和持久超有向图拉普拉斯算子 [60], 以及用于旗复形的持久有向旗拉普拉斯算子 [163, 329]。持久拓扑拉普拉斯算子已成功应用于蛋白质-配体结合预测 [58, 219, 282]、相互作用组网络建模 [110]、基因表达分析 [81]、深度突变扫描 [66] 和蛋白质工程 [253]。特别值得一提的是, 持久拓扑拉普拉斯算子最令人印象深刻的成就之一是对新兴主导 SARS-COV-2 变体 BA4 和 BA5 的预测 [65]。

analysis [81], deep mutational scanning [66], and protein engineering [253]. In particular, one of the most impressive achievements of persistent topological Laplacians is the prediction of emerging dominant SARS-COV-2 variants BA4 and BA5 [65].

Another important spectral approach is quantum persistent homology or persistent Dirac, first introduced by Ameyro et al. [9]. In quantum theory, the Dirac operator is for momentum, whereas the Laplacian is for kinetic energy, which reveals their relationship. Persistent Dirac operators have been studied in many topological domains with various applications [287, 308].

For data on (differential) manifolds, a natural approach is persistent de Rham cohomology, rooted in differential topology or differential geometry. However, persistent de Rham cohomology has rarely been implemented independently. In contrast, evolutionary de Rham-Hodge theory, which includes persistent de Rham cohomology, was first proposed by Chen et al. for data on differential manifolds in Lagrangian representation, i.e., tetrahedral meshes [64]. This approach draws on differential geometry or differential topology. Recently, persistent Hodge Laplacians have been developed in the Eulerian representation, i.e., Cartesian grid defined by level-set functions [282]. The similarity and difference between combinatorial Laplacians and Hodge Laplacians have been thoroughly discussed in [260]. However, the term “Hodge Laplacians” has also been used in the combinatorial setting, which can be somewhat misleading. Essentially, although the orthogonal decomposition can be carried out for both combinatorial Laplacians and Hodge Laplacians, only the Hodge decomposition of vector fields on differential manifolds can give rise to curl-free, divergence-free, and harmonic components, i.e., the so-called Helmholtz-Hodge decomposition [181, 324]. Persistent Sheaf theory [309, 319] and the persistent interaction topology [193] can effectively describe the local topology in the data. For manifolds with boundary, a more refined topology-preserving 5-component Hodge decomposition was developed to further split the harmonic component into three subcomponents under certain boundary conditions [248, 249, 281, 283, 324]. Hodge decomposition is an important tool in analyzing vector fields by extracting different dynamic features, and has many applications in computational fluid dynamics [318, 320], geometric modeling [304], and spectral data analysis [171]. Recently, it has been applied to the study of single-cell RNA velocities [281] and has been integrated with deep learning models for medical image analysis [207].

For data on 1D curves embedded in 3-space, natural approaches stem from low-dimensional topology. Multiscale Gauss linking integral method was proposed to deal with knot-like data [270]. Additionally, multiscale Jones polynomial and persistent Jones polynomial have been introduced [277]. Recently, persistent Khovanov homology has been developed [194, 268]. While Khovanov homology bridges geometric topology and algebraic topology, persistent Khovanov homology extends TDA to 1D data embedded in 3-space.

Cang and Wei introduced the first integration of persistent homology and deep neural networks, including topological deep learning (TDL), in

另一种重要的谱方法是量子持久同调或持久狄拉克算子，最初由 Ameyro 等人 [9] 引入。在量子理论中，狄拉克算子对应于动量，而拉普拉斯算子对应于动能，这揭示了它们之间的关系。持久狄拉克算子已在许多拓扑领域中得到研究，并具有多种应用 [287, 308]。对于（微分）流形上的数据，一种自然的方法是持久德拉姆上同调，它植根于微分拓扑或微分几何。然而，持久德拉姆上同调很少被独立实现。相比之下，进化德拉姆-霍奇理论（包含持久德拉姆上同调）由 Chen 等人首次提出，用于处理拉格朗日表示（即四面体网格）中的微分流形数据 [64]。该方法借鉴了微分几何或微分拓扑。最近，持久霍奇拉普拉斯算子在欧拉表示（即由水平集函数定义的笛卡尔网格）中得到了发展 [282]。组合拉普拉斯算子与霍奇拉普拉斯算子之间的相似性和差异性已在 [260] 中进行了深入讨论。然而，“霍奇拉普拉斯算子”这一术语也用于组合设定，这可能有些误导性。本质上，尽管正交分解可以同时组合拉普拉斯算子和霍奇拉普拉斯算子进行，但只有微分流形上向量场的霍奇分解才能产生无旋、无散和调和分量，即所谓的亥姆霍兹-霍奇分解 [181, 324]。持久层理论 [309, 319] 和持久交互拓扑 [193] 可以有效地描述数据中的局部拓扑。对于带边界的流形，发展了一种更精细的保拓扑五分量霍奇分解，以在某些边界条件下将调和分量进一步拆分为三个子分量 [248, 249, 281, 283, 324]。霍奇分解是通过提取不同动态特征来分析向量场的重要工具，在计算流体力学 [318, 320]、几何建模 [304] 和谱数据分析 [171] 中有许多应用。最近，它已被应用于单细胞 RNA 速度的研究 [281]，并与深度学习模型集成用于医学图像分析 [207]。

对于嵌入三维空间的一维曲线数据，自然的方法源于低维拓扑。提出了多尺度高斯环绕积分方法来处理类组结数据 [270]。此外，还引入了多尺度琼斯多项式和持久琼斯多项式 [277]。最近，持久科瓦诺夫同调得到了发展 [194, 268]。虽然科瓦诺夫同调连接了几何拓扑和代数拓扑，但持久科瓦诺夫同调将 TDA 扩展到了嵌入三维空间的一维数据。Cang 和 Wei 于 2017 年首次将持久同调与深度神经网络相结合，包括拓扑深度学习 (TDL) [37]。通过利用拓扑的高层抽象，TDL 显著降低了维度、简化了几何复杂性、捕捉了高阶相互作用，并提供了一个可解释的学习框架。基于持久同调的机器学习模型已成功应用于科学、工程、医学、工业、国防等多个领域 [151, 243]。然而，拓扑本质上简化了数据，这意味着某些信息的不可逆丢失。因此，TDL 的竞争性表现主要涉及本质上复杂的数据，例如在生物科学中发现的数据 [40]。或许 TDL 持续展示其相对于其他竞争方法优势

2017 [37]. By leveraging the high-level abstraction of topology, TDL dramatically reduces dimensionality, simplifies geometric complexity, captures high-order interactions, and offers an interpretable learning framework. Persistent homology-based machine learning models have been successfully applied in various fields in science, engineering, medicine, industry, defense, etc. [151, 243]. However, topology inherently simplifies data, which implies the irreversible loss of certain information. Therefore, competitive performance from TDL mostly involves intrinsically complex data, such as those found in biological sciences [40]. Perhaps some of the most compelling examples of applications in which TDL consistently demonstrates its advantages over other competing methods are the victories of TDL in the D3R Grand Challenges [233, 234], the discovery of SARS-CoV-2 evolution mechanisms [63, 301], and the successful forecasting of SARS-CoV-2 variants BA.2 [62] and BA.4/BA.5 [65] about two months in advance.

Machine learning typically requires all samples to have a uniform input vector. Therefore, featurization or vectorization of TDA information is often needed to convert topological representations, such as persistence barcodes, diagrams, images, landscapes, Betti curves, etc., into fixed-length feature vectors. Many attempts have been made to define kernels [53, 180, 183, 255]. Others focus on developing featurization techniques, such as statistical summaries [6, 14, 15, 37, 40, 42, 167, 330, 331], persistence bins [37], topological images [40], and Wasserstein metrics [40].

Numerous reviews have been conducted on persistent homology and its featurization [8, 21, 70, 77, 116, 250], as well as a dedicated survey on persistent topological Laplacians [310]. However, there is no review of the generalizations of persistent homology and their vectorization. The aim of this paper is to present a high-level overview of various TDA approaches beyond persistent homology and their machine learning featurization methods, without too many technical details or formulas. This work is intended to be accessible to new researchers in the field and to provide a broad perspective on various techniques and their applications.

The rest of the review is organized as follows. Section 2 is devoted to algebraic topology approaches and techniques defined on point-cloud data. These approaches include persistent homology and its generalizations to various topological domains, persistent topological Laplacians on various topological domains, persistent Dirac operators, persistent sheaf theory, persistent Mayer topology, and persistent interactions. Section 3 reviews methods that originate in differential topology or differential geometry. We discuss the de Rham-Hodge theory for data on manifolds with boundary, a commonly used setting in various applications. Persistent de Rham-Hodge theory, including persistent de Rham cohomology and persistent Hodge Laplacian, is discussed. Section 4 is devoted to various geometric topology-rooted TDA methods, such as multiscale Gauss linking integrals, persistent Jones polynomial, persistent Khovanov homology, and Khovanov Laplacian. Vectorization techniques are reviewed in Section 5. We discuss commonly used methods, such as statistical

的一些最引人注目的应用实例包括：TDL 在 D3R Grand Challenges 中的胜利 [233, 234]、SARS-CoV-2 进化机制的发现 [63, 301]，以及成功提前约两个月预测 SARS-CoV-2 变体 BA.2[62] 和 BA.4/BA.5[65]。

机器学习通常要求所有样本具有统一的输入向量。因此，通常需要对 TDA 信息进行特征化或向量化，以将拓扑表示（如持久条形码、图、图像、景观、Betti 曲线等）转换为固定长度的特征向量。已有许多尝试来定义核函数 [53, 180, 183, 255]。其他研究则侧重于开发特征化技术，例如统计摘要 [6, 14, 15, 37, 40, 42, 167, 330, 331]、持久性分箱 [37]、拓扑图像 [40] 以及 Wasserstein 度量 [40]。

已有大量综述对持久同调及其特征化进行了探讨 [8, 21, 70, 77, 116, 250]，以及一篇专门关于持久拓扑拉普拉斯算子的综述 [310]。然而，目前尚无关于持久同调的推广及其向量化的综述。本文旨在对超越持久同调的各种 TDA 方法及其机器学习特征化方法提供一个高层次的概述，而不涉及过多的技术细节或公式。这项工作旨在使该领域的新研究人员能够理解，并提供对各种技术及其应用的广泛视角。本文其余部分的组织结构如下。第 2 节专门介绍定义于点云数据上的代数拓扑方法与技术。这些方法包括持久同调及其在各种拓扑域上的推广、各种拓扑域上的持久拓扑拉普拉斯算子、持久狄拉克算子、持久层理论、持久迈耶拓扑以及持久相互作用。第 ?? 节回顾源于微分拓扑或微分几何的方法。我们讨论了带边界流形上数据的德拉姆-霍奇理论，这是各类应用中常用的设定。文中探讨了持久德拉姆-霍奇理论，包括持久德拉姆上同调与持久霍奇拉普拉斯算子。第 3 节致力于介绍各种基于几何拓扑的 TDA 方法，例如多尺度高斯环绕积分、持久琼斯多项式、持久霍瓦诺夫同调以及霍瓦诺夫拉普拉斯算子。

向量化技术在第 4 节中予以评述。我们讨论了常用方法，例如统计摘要、贝蒂曲线、持久景观图与持久图像。我们还提供了适用于谱方法的相关特征化技术，包括特征值、特征向量与霍奇分解。此外，我们探讨了针对不同类型数据集进行有效分析的 TDA 方法选择问题。最后，在第 5 节中，我们给出总结性评论。

2 代数拓扑方法

近年来，点云数据的拓扑数据分析发展迅速，提供了一系列建立在代数拓扑工具之上的方法。这些方法通过过滤过程提取拓扑和几何特征，从而提供点云数据的多尺度表征。由此产生的特征提供了关于底层数据的丰

summaries, Betti curves, persistence landscapes, and persistence images. We also provide relevant featurization techniques applicable to spectral methods, including eigenvalues, eigenvectors, and Hodge decompositions. In addition, we discuss the choice of TDA methods for effective analysis of different types of datasets. Finally, in Section 6, we provide concluding remarks.

2 Algebraic topology approaches

Topological data analysis for point cloud data has grown rapidly in recent years, providing a range of methods built upon tools in algebraic topology. These methods offer multiscale characterizations of point cloud data by extracting topological and geometrical features through a filtration process. The resulting features provide rich information on the underlying data and have been successfully integrated into many downstream analyses, including classification, clustering, and other machine learning tasks.

The most fundamental method, persistent homology [112, 333], captures the persistence of topological features, such as connected components, holes, and cavities, on different scales in a filtration process. However, it has several limitations as it focuses mostly on the topology without considering the geometric evolution in the filtration or additional structures associated with data points, and it can be computationally expensive for large datasets, see Section 2.1.4.

Recently developed methods, such as persistent combinatorial Laplacians [302] with their variations [60, 163, 202, 300, 329], and the persistent Dirac operator [9, 10], along with their variations [286, 287], generalize persistent homology and address some of its limitations. Persistent combinatorial Laplacians [302] integrate the multiscale analysis with combinatorial Laplacians, allowing for the analysis of both topological and geometric or combinatorial structures across different scales. Their harmonic spectra fully recover the topological information from persistent homology, while the non-harmonic spectra offer additional geometric or combinatorial features of the data. The persistent Dirac [9, 10], whose square yields the persistent combinatorial Laplacians, provides efficient algorithms based on quantum computing with exponential speed-up compared to classical methods in topological data analysis. There are also other formulations based on alternative homology theories, such as the persistent sheaf Laplacians [309], which further extend the persistent combinatorial Laplacians, allowing the analysis of labeled or weighted datasets; the Mayer topology [269, 287]; and the persistent interaction topology [193, 194]. These tools offer new perspectives for capturing topological features that are not accessible through standard homology.

In the following, we first review persistent homology with its variations and limitations, then turn to the persistent combinatorial Laplacians and their various variations for analyzing point cloud data with additional structures. We

富信息, 并已成功集成到许多下游分析中, 包括分类、聚类和其他机器学习任务。

最基本的方法是持续同调 [112, 333], 它捕捉了在过滤过程中不同尺度上拓扑特征 (如连通分量、孔洞和腔体) 的持续性。然而, 它存在一些局限性, 因为它主要关注拓扑, 而没有考虑过滤过程中的几何演化或与数据点相关的附加结构, 并且对于大型数据集计算成本可能很高, 参见第 2.1.4 节。

最近发展的方法, 例如持续组合拉普拉斯算子 [302] 及其变体 [60, 163, 202, 300, 329], 以及持续狄拉克算子 [9, 10] 及其变体 [286, 287], 推广了持续同调并解决了其部分局限性。持续组合拉普拉斯算子 [302] 将多尺度分析与组合拉普拉斯算子相结合, 允许分析不同尺度上的拓扑和几何或组合结构。它们的调谐完全恢复了持续同调的拓扑信息, 而非调谐则提供了数据的额外几何或组合特征。持续狄拉克算子 [9, 10] 的平方产生持续组合拉普拉斯算子, 它提供了基于量子计算的高效算法, 与拓扑数据分析中的经典方法相比具有指数级加速。此外, 还有基于其他同调理论的表述, 例如持续层拉普拉斯算子 [309], 它进一步扩展了持续组合拉普拉斯算子, 允许分析带标签或带权重的数据集; 迈耶拓扑 [269, 287]; 以及持续交互拓扑 [193, 194]。这些工具为捕捉标准同调无法访问的拓扑特征提供了新的视角。

接下来, 我们首先回顾持续同调及其变体和局限性, 然后转向持续组合拉普拉斯算子及其各种变体, 用于分析具有附加结构的点云数据。接着, 我们描述持续狄拉克算子及其变体。最后, 我们讨论其他表述, 包括持续层拓扑、持续迈耶拓扑和持续交互拓扑。

2.1 持久同调

持久同调是单纯同调的一种多尺度扩展, 它是代数拓扑学中的一个工具, 用于根据单纯复形的同调群对其进行分类。同调群度量了底层单纯复形中的拓扑特征, 例如连通分支、空洞和腔体。它们的秩由贝蒂数给出, 用于计数这些拓扑特征的数量。例如, 第 0 个贝蒂数提供了连通分支的数量, 第 1 个贝蒂数给出了圆环或空洞的数量, 而第 2 个贝蒂数则代表腔体的数量。

然而, 单纯同调不能直接应用于点云数据, 因为除非明确构建, 否则此类数据没有定义良好的拓扑结构。此外, 它对尺度敏感, 因为不同的尺度选择可能导致不同的同调群。持久同调通过引入过滤和持续性来解决这些局限性, 从而为底层数据提供更稳健的表征。

then describe the persistent Dirac operator and its variations. Finally, we discuss other formulations, including the persistent sheaf topology, the persistent Mayer topology, and the persistent interaction topology.

2.1 Persistent homology

Persistent homology is a multiscale extension of simplicial homology, which is a tool in algebraic topology that classifies simplicial complexes based on their homology groups. The homology groups measure the topological features in the underlying simplicial complex, such as connected components, holes, and cavities. Their ranks, given by Betti numbers, count the number of these topological features. For example, the 0th Betti number provides the number of connected components, the 1st Betti number gives the number of circles or holes, and the 2nd Betti number represents the number of cavities.

However, simplicial homology does not directly apply to point-cloud data, as such data do not have a well-defined topological structure unless explicitly constructed. In addition, it is sensitive to scales, as different choices of scales can lead to different homology groups. Persistent homology addresses these limitations by introducing filtration and persistence, providing a more robust characterization of the underlying data.

2.1.1 Simplicial homology

The simplicial complexes are made of simplices, where a k -simplex is defined to be the convex hull of $k+1$ affinely independent points $v_0, v_1, \dots, v_k$, and has dimension k . Examples include a point (0-simplex), a line segment (1-simplex), a triangle (2-simplex), and a tetrahedron (3-simplex). These points $v_0, v_1, \dots, v_k$ are called the vertices of the k -simplex, and the simplices formed by a subset of the set of all these vertices are called faces.

A simplicial complex K is then defined to be a finite set of simplices satisfying the gluing conditions, where all faces of a simplex are also in K , and the intersection of any two simplices in K is either empty or a common face of both. The dimension of K is given by the largest dimension of its simplices. A linear combination of all k -simplices of K with coefficients in a field (commonly $\mathbb{Z}_2$) defines a k -chain. The set of all k -chains, together with addition, forms an abelian group $\mathcal{C}_k(K)$, called the k -th chain group. The set of all k -simplices of K serves as a basis of $\mathcal{C}_k(K)$.

To define the boundary operator ∂ , each simplex of K needs to be oriented (not necessary when using $\mathbb{Z}_2$ coefficients as $-1 = +1$). An oriented k -simplex is represented as an ordered list of its vertices, denoted by $[v_0, v_1, \dots, v_k]$. The orientation is crucial for constructing the chain complex and defining homology, as it ensures that $\partial\partial = 0$. The k -th boundary operator $\partial_k : \mathcal{C}_k(K) \rightarrow \mathcal{C}_{k-1}(K)$ is a linear map that acts on each k -simplex by mapping it to an alternating sum of its $(k-1)$ -faces, where each is obtained by removing one vertex from

2.1.1 单纯同调

单纯复形由单纯形构成, 其中 k -单纯形定义为 $k+1$ 个仿射独立点 $v_0, v_1, \dots, v_k$ 的凸包, 其维数为 k 。例子包括点 (0-单纯形)、线段 (1-单纯形)、三角形 (2-单纯形) 和四面体 (3-单纯形)。这些点 $v_0, v_1, \dots, v_k$ 称为 k -单纯形的顶点, 由所有这些顶点集合的子集形成的单纯形称为面。

单纯复形 K 则定义为满足粘合条件的有限单纯形集合, 其中每个单纯形的所有面也属于 K , 且 K 中任意两个单纯形的交集要么为空, 要么是两者的公共面。 K 的维数由其单纯形的最大维数给出。 K 的所有 k -单纯形在某个域 (通常为 $\mathbb{Z}_2$) 上的线性组合定义了一个 k -链。所有 k -链的集合, 连同加法运算, 构成一个阿贝尔群 $\mathcal{C}_k(K)$, 称为第 k 个链群。 K 的所有 k -单纯形的集合构成了 $\mathcal{C}_k(K)$ 的一组基。

为了定义边缘算子 ∂ , K 中的每个单纯形都需要被定向 (当使用 $\mathbb{Z}_2$ 系数时, 由于 $-1 = +1$, 定向不是必需的)。一个有向 k -单纯形表示为其顶点的有序列表, 记作 $[v_0, v_1, \dots, v_k]$ 。定向对于构造链复形和定义同调至关重要, 因为它确保了 $\partial\partial = 0$ 。第 k 个边缘算子 $\partial_k : \mathcal{C}_k(K) \rightarrow \mathcal{C}_{k-1}(K)$ 是一个线性映射, 作用于每个 k -单纯形, 将其映射为其 $(k-1)$ -面的交错和, 其中每个面通过从 k -单纯形中移除一个顶点得到, 即,

$$\partial_k[v_0, v_1, \dots, v_k] = \sum_i (-1)^i [v_0, \dots, \hat{v}_i, \dots, v_k]$$

边缘算子满足 $\partial_k \partial_{k+1} = 0$ 。链复形由通过边缘算子连接的一系列链群给出:

$$\dots \xrightarrow{\partial_{k+2}} \mathcal{C}_{k+1} \xrightarrow{\partial_{k+1}} \mathcal{C}_k \xrightarrow{\partial_k} \mathcal{C}_{k-1} \xrightarrow{\partial_{k-1}} \dots$$

K 的第 k 个单纯同调群定义为边缘算子 ∂_k 的核模去 ∂_{k+1} 的像的商群:

$$H_k(K) = \ker \partial_k / \operatorname{im} \partial_{k+1},$$

其中 $\ker \partial_k$ 和 $\operatorname{im} \partial_{k+1}$ 分别称为 k -闭链群和 k -边缘链群。同调群 $H_k(K)$ 捕捉了那些不是 k -边缘链的 k -闭链, 即单纯复形中的 k 维洞。其秩, 称为第 k 个贝蒂数 β_k , 用于计数这些洞的数量。

2.1.2 过滤与持续性

过滤由一系列嵌套的单纯复形序列给出。对于点云数据, 需要通过构建单纯复形来表示数据点及其在不同维度上的连接, 从而构造这样的过滤。最常用的三种复形是 Vietoris-Rips 复形 [295]、Čech 复形 [114] 和 Alpha

the k -simplex, i.e.,

$$\partial_k[v_0, v_1, \dots, v_k] = \sum_i (-1)^i [v_0, \dots, \hat{v}_i, \dots, v_k]$$

The boundary operator satisfies $\partial_k \partial_{k+1} = 0$. A chain complex is given by a sequence of chain groups connected by the boundary operator:

$$\dots \xrightarrow{\partial_{k+2}} \mathcal{C}_{k+1} \xrightarrow{\partial_{k+1}} \mathcal{C}_k \xrightarrow{\partial_k} \mathcal{C}_{k-1} \xrightarrow{\partial_{k-1}} \dots$$

The k -th simplicial homology group of K is defined as the quotient of the kernel of the boundary operator ∂_k modulo the image of ∂_{k+1} :

$$H_k(K) = \ker \partial_k / \operatorname{im} \partial_{k+1},$$

where $\ker \partial_k$ and $\operatorname{im} \partial_{k+1}$ are called the group of k -cycles and the group of k -boundaries, respectively. The homology group $H_k(K)$ captures the k -cycles that are not k -boundaries, i.e., the k -dimensional holes in the simplicial complex. Its rank, known as the k -th Betti number β_k , counts the number of these holes.

2.1.2 Filtration and persistence

A filtration is given by a nested sequence of simplicial complexes. For point cloud data, one needs to construct such a filtration by building simplicial complexes to represent data points and their connections in various dimensions. The three most commonly used complexes are the Vietoris-Rips complex [295], the Čech complex [114], and the Alpha complex [113]. There are also other constructions such as witness complex [88], Cover complex [51], and Tangential complex [159].

The Vietoris-Rips complex [295] is constructed by considering pairwise intersections of balls centered at each data point within a fixed distance, leading to a simplex whenever its vertices have pairwise overlapping regions. The Čech complex [114] requires a nonempty intersection of all balls in a set, meaning that a simplex is included only if all corresponding balls intersect. The Alpha complex can be constructed in a similar manner to the Čech complex. However, each ball is restricted to the Voronoi cell associated with each point. The Alpha complex [113] is a subset of the Čech complex but has the same homology. By varying the radius of the balls, one obtains a filtration of simplicial complexes. The Čech complex involves checking for a large number of intersections and thus comes with significantly higher computational costs for large datasets, making the Vietoris-Rips complex the most common choice in real applications. An alternative to the Čech complex is the Alpha complex, which is particularly computationally more efficient when the ambient space has dimension 2 or 3 [114].

复形 [113]。此外还有其他构造方法，例如见证复形 [88]、覆盖复形 [51] 和切向复形 [159]。

Vietoris-Rips 复形 [295] 的构造方法是：考虑以每个数据点为中心、在固定距离内的球的两两相交，只要其顶点之间存在两两重叠的区域，就形成一个单形。Čech 复形 [114] 要求一个集合中所有球的交集非空，这意味着只有当所有对应球都相交时，才包含一个单形。Alpha 复形的构造方式与 Čech 复形类似，但每个球被限制在与每个点关联的 Voronoi 胞腔内。Alpha 复形 [113] 是 Čech 复形的一个子集，但具有相同的同调。通过改变球的半径，可以得到一个单纯复形的过滤。Čech 复形需要检查大量的相交情况，因此对于大型数据集计算成本显著更高，这使得 Vietoris-Rips 复形成为实际应用中最常见的选择。Čech 复形的一个替代方案是 Alpha 复形，当环境空间的维度为 2 或 3 时，其计算效率尤其更高 [114]。

设 K 是由点云构造出的最大单纯复形。那么该点云的一个过滤是 K 的子复形的一个嵌套序列：

$$\emptyset \subset K^0 \subset K^1 \dots \subset K^m = K.$$

记 $i^{i,j}$ 为从 K_i 到 K_j 的包含映射，其中 $i < j$ 。它在同调群上诱导的映射 $f_k^{i,j} : H_k(K^i) \rightarrow H_k(K^j)$ 追踪了 k 维空洞在过滤过程中的演化，更准确地说，是追踪 $H_k(K^i)$ 中的 k 维同调类是否仍然存在于 $H_k(K^j)$ 中。记 $\partial_k^i : \mathcal{C}_k(K^i) \rightarrow \mathcal{C}_{k-1}(K^i)$ 为单纯复形 K^i 上的第 k 个边界算子。第 k 个持续同调群定义为 $f_k^{i,j}$ 的像：

$$H_k^{i,j} = \operatorname{im} f_k^{i,j} = \ker \partial_k^i / (\operatorname{im} \partial_{k+1}^j \cap \ker \partial_k^i),$$

其秩导出对应的第 k 个持续贝蒂数 $\beta_k^{i,j} = \operatorname{rank} H_k^{i,j}$ 。拓扑特征的出生和死亡可以通过检查持续同调群中的元素来定义。如果一个特征首次出现在 K^i ，则称其出生于 K^i ；如果它在 K^j 处与一个在 K^j 之前就已存在的特征合并，则称其死亡于 K^j 。特征 γ 的持续性（即拓扑特征的寿命）由 $j-i$ 给出，如果它在过滤中永不死亡，则被视为无限。

这些拓扑特征的出生和死亡信息催生了持续同调的各种表示方法，例如持续性条形码 [132]、持续性图 [112]、持续贝蒂数、贝蒂曲线、持续性景观 [34]、持续性图像 [4]。关于这些特征化技术的更多细节将在第 4 节中讨论。持久同调已成功应用于从图像处理到复杂网络等多个领域 [17, 37, 38, 72, 84, 128, 156, 161, 178, 184, 291]。其最显著的成就之一是与深度学习模

Let K be the largest simplicial complex constructed from a point cloud. Then a filtration of this point cloud is a nested sequence of the subcomplexes of K :

$$\emptyset \subset K^0 \subset K^1 \cdots \subset K^m = K.$$

Denote by $i^{i,j}$ the inclusion maps from K_i to K_j with $i < j$. Its induced map $f_k^{i,j} : H_k(K^i) \rightarrow H_k(K^j)$ on the homology groups tracks how the k -dimensional holes evolve across the filtration, more precisely, if the k -dimensional homology classes in $H_k(K^i)$ still exist in $H_k(K^j)$. Denote by $\partial_k^i : C_k(K^i) \rightarrow C_{k-1}(K^i)$ the k -th boundary operator on simplicial complex K^i . The k -th persistent homology group is defined to be the image of $f_k^{i,j}$:

$$H_k^{i,j} = \text{im } f_k^{i,j} = \ker \partial_k^i / \left(\text{im } \partial_{k+1}^j \cap \ker \partial_k^i \right),$$

and its rank leads to the corresponding k -th persistent Betti number $\beta_k^{i,j} = \text{rank } H_k^{i,j}$. The birth and death of a topological feature can then be defined by examining the elements in the persistent homology groups. A feature is said to be born at K^i if it first appears at K^i , and it is said to die at K^j if it merges with a feature at K^j that already exists before K^j . The persistence of γ , denoting the lifetime of a topological feature, is then given by $j - i$, and considered infinite if it never dies in the filtration.

These birth and death information of topological features gives rise to various representations of persistent homology, such as persistence barcodes [132], persistence diagrams [112], persistent Betti numbers, Betti curves, persistence landscapes [34], persistence images [4]. More details about these featurization techniques will be discussed in Section 5.

Persistent homology has been successfully applied in various fields, ranging from image processing to complex networks [17, 37, 38, 72, 84, 128, 156, 161, 178, 184, 291]. The most significant achievements would be its integration with deep learning models, which has led to exceptional performance in the D3R grand challenges [233, 234] and in predicting the evolutionary mechanism of SARS-CoV-2 [63, 301].

2.1.3 Other variations

In addition to the standard formulation of persistent homology for simplicial complexes, several variations have been proposed for analyzing data with different structures or for addressing specific application needs. We briefly review these methods below.

Multiparameter persistent homology

The multiparameter persistent homology was first explored in [46], while a variation in size theory was introduced in [127]. The framework extends classical homology by considering two or more parameters instead of one, and it is

型的结合, 这在 D3R 大挑战 [233, 234] 以及预测 SARS-CoV-2 的进化机制 [63, 301] 中取得了卓越的性能。

2.1.3 其他变体

除了单纯复形持久同调的标准表述外, 针对具有不同结构的数据或满足特定应用需求, 人们提出了若干变体。下面我们简要回顾这些方法。

多参数持久同调

多参数持久同调最初在 [46] 中被探讨, 而尺寸理论中的一个变体则在 [127] 中被引入。该框架通过考虑两个或更多参数而非单一参数来扩展经典同调理论, 适用于分析含有异常值或密度变化的数据、配备实值函数的数据以及具有较大局部噪声的函数数据。更多介绍性资源, 请参见 [31, 103, 169, 263]。关于计算和软件的讨论可参见 [31]。

Zig-zag 持久同调

Zig-zag 持久同调在 [45] 中被引入, 用于处理更一般的设置, 其中空间序列通过任意方向的映射连接, 这与经典持久同调中包含映射沿同一方向进行形成对比。该框架功能强大, 因为它允许过滤中的单纯复形随着单形的插入和删除而演化, 这适用于分析动态点云或具有变化顶点连接的动态图 [67, 95, 96, 154]。已提出了多种用于计算 zigzag 持久性的软件 [47, 97]。该框架也已扩展到多参数设置 [96, 98]。

参数化同调

参数化同调 [50] 将水平集 zigzag 持久同调扩展到连续参数设置。该框架利用矩形测度理论 [56] 定义了四个不同的连续参数持久图, 每个图对应于水平集 zigzag 条形码中的四种条形类型之一。这些图编码了同调特征, 并捕获了关于每个特征在其定义区间两端点如何消失的详细信息。

局部化同调

局部化同调 [334] 被提出用于识别拓扑属性 (如隧道和空洞) 的位置。给定一个具有覆盖的拓扑空间, 局部化同调工具可用于确定拓扑特征在空间内的位置 (特定的覆盖元素)。具体而言, 它构造了一个更大的爆破复形 [267], 该复形与原始空间同伦等价, 并且在覆盖元素之间具有清晰的

suitable for analyzing data with outliers or variations in density, data equipped with real-valued functions and functional data with large local noise. For more introductory resources, see [31, 103, 169, 263]. Discussions on the computation and software can be found in [31].

Zig-zag persistent homology

The Zig-zag persistent homology was introduced in [45] to handle a more general setting where the sequence of spaces is connected with maps in arbitrary directions, in contrast to the classical persistent homology where the inclusion maps follow in the same direction. The framework is powerful since it allows simplicial complexes in the filtration to evolve with the insertion and deletion of simplices, which is suitable for analyzing dynamic point clouds or dynamic graphs with varying vertex connections [67, 95, 96, 154]. Various software for computing the zigzag persistence have been proposed [47, 97]. The framework has also been extended to the multiparameter setting [96, 98].

Parametrized homology

Parametrized homology [50] extends levelset zigzag persistent homology to a continuous-parameter setting. This framework leverages the theory of rectangle measures [56] to define four distinct continuous-parameter persistence diagrams, each corresponding to one of the four bar types in the levelset zigzag barcode. These diagrams encode homological features and capture detailed information about how each feature disappears at both endpoints of its defining interval.

Localized homology

Localized homology [334] is proposed for identifying the locations of topological attributes such as tunnels and voids. Given a topological space with a cover, the localized homology tool can be used to determine the location (the specific cover element) of topological features within the space. Specifically, it constructs a larger blowup complex [267] that is homotopy equivalent to the original space and has clear boundaries between cover elements. The homology bases are then computed on the blowup complex to localize the features to specific cover elements. An efficient algorithm is also provided for practical applications.

Evolutionary homology

Cang et al. [41] introduced Evolutionary Homology (EH), a method generating time-dependent topological invariants ("evolutionary barcodes") for individual components within physical systems. EH works by coupling differential equations or chaotic oscillators to model system interactions, creating coupled dynamics. On the resulting trajectories of these oscillators, they define simplices, simplicial complexes, algebraic groups, and topological persistence using a time evolution-based filtration. This process reveals the topology-function

边界。然后在爆破复形上计算同调基，从而将特征定位到特定的覆盖元素。还为实际应用提供了一种高效算法。

演化同调

Cang 等人 [41] 引入了演化同调 (EH)，这是一种为物理系统内的单个组件生成时间依赖的拓扑不变量（“演化条形码”）的方法。EH 通过耦合微分方程或混沌振荡器来模拟系统相互作用，从而创建耦合动力学。在这些振荡器产生的轨迹上，他们使用基于时间演化的过滤定义了单形、单纯复形、代数群和拓扑持久性。这一过程揭示了单个组件的拓扑-功能关系，并且对于分析点云数据中的局部拓扑特别有效。EH 已成功应用于生物物理学中的实际问题。

多分辨率持续同调

多分辨率持续同调是为分析过大规模数据集而提出的 [315, 316]。其核心思想是通过采用由分辨率参数调制的刚性-密度函数，来匹配数据中感兴趣的尺度。通过调整这些参数，原始的大规模数据集可以在一个合适的较低分辨率下表示，从而实现高效的拓扑分析。

多维持续同调

多维持续同调最初于 2007 年被引入，用于分析多重滤过的拓扑结构 [46]。后来，另一种版本的多维持续同调 [314] 被提出，用于将动态数据和单分子数据都转化为基于图像表示。对于动态数据，其图像表示是通过在时间步上对每个数据帧重复应用持续同调来构建的。对于单分子数据，持续同调被应用于分子的体积密度表示在多个空间尺度上，并将得到的拓扑特征聚合以形成最终的图像。

加权持续同调

加权持续同调用于研究加权数据，例如加权单纯复形，其中每个单形都被赋予一个权重。主要有两种方法来捕获权重信息：一种方法使用权重来细化滤过过程 [115, 247]，而另一种方法则通过纳入权重来修正链复形的边缘算子 [87, 257]。加权持续同调已被应用于各种生物分子数据分析应用 [220, 251]。

relationship of individual components and is particularly effective for analyzing local topology within point cloud data. EH was successfully applied to realistic problems in biophysics.

Multiresolution persistent homology

Multiresolution persistent homology was proposed for analyzing excessively large datasets [315, 316]. The idea is to match the scale of interest in the data by employing the rigidity-density functions modulated through resolution parameters. By tuning these parameters, the original large-scale datasets can be represented at an appropriate lower resolution, enabling efficient topological analysis.

Multidimensional persistence

The multidimensional persistence was first introduced in 2007 for analyzing the topology of multifiltrations [46]. Later, another version of multidimensional persistence [314] was proposed for transforming both dynamic and single molecular data into image-based representations. For dynamical data, the image representation is constructed by repeatedly applying persistent homology to each data frame across time steps. For single molecular data, persistent homology is applied at multiple spatial scales to the volumetric density representation of the molecule, and the resulting topological features are aggregated to form the final image.

Weighted persistent homology

The weighted persistent homology is used to study the weighted data, such as weighted simplicial complexes, where each simplex is assigned a weight. There are primarily two types of methods for capturing the weight information: one approach uses weights to refine the filtration process [115, 247], while the other revises the boundary operators of the chain complexes by incorporating the weights [87, 257]. The weighted persistent homology has been applied to various biomolecular data analysis applications [220, 251].

Persistent path homology

Persistent path homology has been introduced for analyzing directed networks that incorporate asymmetric directional information [69], which integrates the filtration method in TDA with the path homology theory. Path homology [137], recently renamed GLMY homology [158], has been extensively developed in both theory [104, 138, 140, 186, 187] and applications [59, 124, 135, 312].

Persistent hypergraph homology

Persistent hypergraph homology [33] is a generalization of the standard persistent simplicial homology to hypergraphs. Using embedded homology theory, hypergraph homology is suitable for addressing systems with incomplete information [201], such as the coauthor network, where a three-person collaboration

持续路径同调

持续路径同调已被引入用于分析包含非对称方向信息的有向网络 [69], 它将 TDA 中的滤过方法与路径同调理论相结合。路径同调 [137], 最近更名为 GLMY 同调 [158], 在理论 [104, 138, 140, 186, 187] 和应用 [59, 124, 135, 312] 方面都得到了广泛发展。

持续超图同调

持续超图同调 [33] 是标准持续单纯同调到超图的推广。利用嵌入同调理论, 超图同调适用于处理信息不完整的系统 [201], 例如合著者网络, 其中可能存在三人合作但并非所有两两合作都存在的情况。超图的稳定性结果 [256] 和离散莫尔斯理论 [258] 已经建立, 并且持续超图同调的算法实现 [205] 也已开发出来。

持续超超图同调

持续超超图同调 [136] 提供了一个统一且广义的框架, 将持续单纯同调、持续超图同调和持续 Δ -集合同调都作为其特例。与单纯复形和超图相比, 超超图的一个显著特点是每个超边并非由其顶点唯一确定, 这意味着存在额外的内部结构。这使其成为一个灵活的框架, 能够捕捉各种系统的复杂性 [123, 206]。

基于图复形的持续同调

在拓扑数据分析中, 为数据构建合适的单纯复形, 其重要性堪比机器学习中的数据表示。最常用的构造是 Vietoris-Rips 复形 [332] 和 Alpha 复形 [113], 这主要得益于现有成熟且用户友好的软件包。一些研究探索了使用其他基于图的复形, 例如旗复形 [156, 210]、Dowker 复形 [203]、邻域复形 [200]、Hom-复形 [204] 以及其他复形 [136, 165]。针对特定应用选择最合适的复形, 而非仅仅使用流行的选择, 值得进一步研究。

扩展持续同调

持续图之间的距离常被用来比较不同拓扑空间的拓扑特征。然而, 具有不同贝蒂数的拓扑空间的持续图, 在任何滤过下都可能导致无限距离。扩展持续同调 [74, 292] 通过扩展滤过过程来解决这一限制, 使得所有同调类都被赋予一个出生时刻和一个死亡时刻。

exists but not all pairwise collaborations exist. A stability result [256] and a discrete Morse theory [258] for hypergraphs have been established, and an algorithmic implementation for persistent hypergraph homology [205] has also been developed.

Persistent super-hypergraph homology

Persistent super-hypergraph homology [136] provides a unified and generalized framework that encompasses persistent simplicial homology, persistent hypergraph homology, and persistent Δ -set homology as special cases. Compared with a simplicial complex and a hypergraph, a distinct characteristic of super-hypergraph is that each super-edge is not uniquely determined by its vertices, implying the presence of additional internal structure. This makes it a flexible framework to capture the complexity of various systems [123, 206].

Graph-complex-based persistent homology

In TDA, modeling data with an appropriate simplicial complex is as critical as data representation in machine learning. The most commonly used constructions are the Vietoris-Rips complex [332] and the Alpha complex [113], mainly due to the availability of well-established and user-friendly software packages. Several studies have explored the use of other graph-based complexes, such as the flag complex [156, 210], Dowker complex [203], neighborhood complex [200], Hom-complex [204], and others [136, 165]. Selecting the most suitable complex for a given application, rather than just using the popular choices, is worth further investigation.

Extended persistent homology

Distances between persistence diagrams are often used to compare topological features of different topological spaces. However, persistent diagrams of topological spaces with different Betti numbers can lead to infinite distances under any filtration. Extended persistent homology [74, 292] addresses this limitation by extending the filtration process so that all homology classes are assigned a birth and a death.

Join persistent homology

Join persistent homology [305] is based on the join operation in topology. The idea is that different topological features can be captured from multiple perspectives by employing various types of unit in the join process. This allows the method to reveal diverse structural information within the data. Join persistent homology also provides a solution to the issue of infinite distance between persistence diagrams of spaces with different Betti numbers.

Persistent cohomology

Persistent cohomology is a sister counterpart to persistent homology, offering computational advantages in time and space. In practice, people often

联持续同调

联持续同调 [305] 基于拓扑学中的联运算。其思想是，通过在联过程中使用不同类型的单元，可以从多个视角捕捉不同的拓扑特征。这使得该方法能够揭示数据中多样的结构信息。联持续同调也为具有不同贝蒂数的空间之间持续图距离无限的问题提供了一个解决方案。

持续上同调

持续上同调是持续同调的姊妹理论，在时间和空间上具有计算优势。在实践中，人们通常先计算持续上同调，然后通过对偶性推导出持续同调 [90]，因为前者计算起来可能更高效。GUDHI 软件包 [211] 就是一个例子。许多应用依赖于原始的持续上同调算法 [89]。一些研究探索了将领域特定信息融入上同调类，以增强应用中背景相关信息的表征 [39, 201]。

凯莱持续同调

凯莱持续同调是基于群分次 [27] 构建的。与标准的 $\mathbb{Z}$ -分次持续同调类似，它也具有分解结果。关键区别在于其能够捕捉更灵活的持续模式。它不仅允许持续现象出现在区间和网格上，还允许其出现在由底层群分次引入的更复杂结构上。

莫尔斯理论

莫尔斯理论 [231] 提供了一个分析和刻画形状拓扑的框架。其核心思想是将形状表示为一个配备莫尔斯函数的拓扑空间，该函数通过其临界点捕捉关键的拓扑特征。离散莫尔斯理论已与持久同调相结合，用于无权无向网络的分析 [168]。莫尔斯理论的计算算法已被开发 [145]。莫尔斯分解的持久同调已被应用于组合动力学 [102]。在莫尔斯理论衍生的各种描述子中，里布图可能是应用最广泛的 [29, 130, 254, 323]。里布图的一个关键特性是能够将形状的拓扑结构编码成一维图，且独立于底层流形的内在维度。

康莱指数

动力系统分析的大部分内容涉及不变集的存在性与结构。康莱指数理论由康莱及其追随者发展 [61, 76, 225]，是研究此类不变集的重要拓扑工具。康莱指数理论不分析任意的不变集，而是专注于孤立不变集，这类集合比一般不变集对连续扰动更具鲁棒性。对于一个孤立不变集 S ，可以从其指数对构造一个商空间， S 的同伦康莱指数是该商空间的同伦型，而 S

compute the persistent cohomology and subsequently derive persistent homology via the duality [90], as the former can be more efficient to compute. An example is the GUDHI package [211]. Many applications rely on the original persistent cohomology algorithm [89]. Several studies have explored incorporating domain-specific information into cohomology classes to enhance the characterization of background-related information in applications [39, 201].

Cayley-persistence

Cayley-persistence is formulated based on the group grading [27]. Similar to the standard $\mathbb{Z}$ -graded persistence, it has the decomposition results. The key difference is its ability to capture more flexible persistence patterns. It allows the emergence of persistence not only on the intervals and grids, but also on more complex structures introduced by the underlying group grading.

Morse theory

Morse theory [231] provides a framework for analyzing and characterizing the topology of shapes. The central idea is to represent a shape as a topological space equipped with a Morse function that captures critical topological features through its critical points. Discrete Morse theory is integrated with persistent homology for unweighted and undirected networks [168]. Computational algorithm for Morse theory has been developed [145]. The persistent Homology of Morse decompositions has been applied to combinatorial dynamics [102]. Among the various descriptors derived from Morse theory, the Reeb graph is probably the most widely used [29, 130, 254, 323]. A key property of the Reeb graph is its ability to encode the topological structure of a shape into a one-dimensional graph, independent of the intrinsic dimension of the underlying manifold.

Conley index

Much of dynamical system analysis involves the existence and structure of invariant sets. Conley index theory, developed by Conley and his followers [61, 76, 225], is an important topological tool for studying such invariant sets. Rather than analyzing arbitrary invariant sets, Conley index theory focuses on isolated invariant sets, which are more robust to continuous perturbations than general invariant sets. For an isolated invariant set S , a quotient space can be constructed from its index pair, the homotopy Conley index of S is the homotopy type of this quotient space, and the cohomological Conley index of S is the Alexander-Spanier cohomology of the quotient space with integer coefficients. The Conley index is invariant under certain deformations of the dynamical system and can be used to establish the existence of fixed points and periodic orbits of the dynamical system. Recently, the Conley index has been extended to the persistence setting for data analysis [100, 101].

的上同调康莱指数是该商空间具有整数系数的亚历山大-斯潘尼尔上同调。康莱指数在动力系统的某些形变下保持不变，可用于确定动力系统的不动点和周期轨道的存在性。最近，康莱指数已被扩展至持久化设定以用于数据分析 [100, 101]。

持久等变上同调

对于一个带有群作用的数据集，持久等变上同调不仅捕捉数据集的底层形状，还捕捉群作用的性质，即数据中固有的对称性。在 [5] 中，作者明确描述了圆周在其 Vietoris-Rips 度量加厚上的作用的持久等变上同调。特别地，他们推导了这些 Vietoris-Rips 加厚的持久等变上同调的显式公式。当等变上同调基于自由群作用时，可将其视为扭曲上同调的一个特例，其中的扭曲源于群作用。已有多项研究在不同背景下探索了扭曲（上）同调的理论 [139, 188, 322]。

偏序集上的持续性

偏序集上的持续性 [175] 是一个通用框架，其中持续性模定义在偏序集上，多参数持续性作为其特例，其索引偏序集为 $\mathbf{R}^n$ 。作为多参数持续性中最著名的同调不变量，秩不变量 [46] 通过自然地扩展其定义域，已演变为适用于偏序集上持续性的广义秩不变量 [244]。特别地，对于任何有限维向量空间的区间可分解持续性模，其条形码可以通过容斥原理 [174] 从广义秩不变量中提取。这意味着广义秩不变量是区间可分解持续性模的一个完备不变量，而秩不变量并非区间可分解多参数持续性模的完备不变量 [32]。

2.1.4 其他视角

尽管持久同调具有极大的实用性，但仍有许多方向需要对其进行扩展。这些扩展与主流代数拓扑学（即实际空间和流形的拓扑学，而非点云的拓扑学）中的大量工作相平行。以下是其中几个方向。

- **谱分析：**注意，持久同调未能捕捉到过滤过程中同伦几何形状的变化，这意味着它无法区分具有相同拓扑结构的不同形状 [153]。基于拉普拉斯算子的谱分析，已经发展出多种框架来解决这一局限性 [310]。这些方法不仅从持久同调中恢复拓扑信息，还能捕捉描述过滤过程中同伦几何形状变化的额外信息。
- **分析方法：**从该学科的早期开始，人们就理解到，在流形的情况下，存在使用微分形式计算某种版本的同调（系数为 $\mathbb{R}$ 的同调）的方法。这些

Persistent equivariant cohomology

For a dataset with a group action on it, persistent equivariant cohomology captures not only the underlying shape of the dataset but also properties of the group action, namely, the symmetries inherent in the data. In [5], the authors provide an explicit description of the persistent equivariant cohomology of the circle action on the Vietoris-Rips metric thickening of the circle. In particular, they derive explicit formulas for the persistent equivariant cohomology of these Vietoris-Rips thickening. When equivariant cohomology is based on a free group action, it can be viewed as a special case of twisted cohomology, with the twist arising from the group action. Various studies have explored the theory of twisted (co)homology in different contexts [139, 188, 322].

Persistence over posets

Persistence over posets [175] is a general framework where persistence modules are defined on posets, with multiparameter persistence as a special case where the indexing poset is $\mathbf{R}^n$. As the most known invariant for multiparameter persistence, the rank invariant [46] has evolved into the generalized rank invariant [244] for persistence over posets by naturally extending its domain. In particular, for any interval decomposable persistence module of finite-dimensional vector spaces, its barcode can be extracted from the generalized rank invariant by the principle of inclusion-exclusion [174]. This implies that the generalized rank invariant is a complete invariant for interval decomposable persistence modules, while the rank invariant is not a complete invariant for interval decomposable multiparameter persistence modules [32].

2.1.4 Additional perspectives

Despite the great utility of persistent homology, there are a number of directions in where it is desirable to extend it. These extensions parallel much of the work that has been done in mainstream algebraic topology, i.e., the topology of actual spaces and manifolds rather than point clouds. Here are a few of them.

- **Spectral analysis:** Note that persistent homology fails to capture the homotopic geometric shape changes within a filtration, meaning that it cannot distinguish between different shapes with the same topology [153]. Several frameworks have been developed to address this limitation based on the spectral analysis of Laplacians [310]. These methods not only recover topological information from persistent homology but also capture additional information that describes the homotopic geometric shape changes in the filtration.
- **Analytic methods:** From the early days of the subject, it was understood that in the case of manifolds, there are approaches to computing a version of homology (homology with $\mathbb{R}$ coefficients) using differential forms. These methods rely on *de Rham cohomology* [306]. The advantage of these methods is that they tie the homological qualitative information together with the

方法依赖于德拉姆上同调 [306]。这些方法的优点在于，它们将同调的定性信息与流形上的分析联系起来。另一个优点是，霍奇定理 [306] 提供了在上同调类中选择一个调和代表元，从而明确了该类的意义。这一分析中的一个关键要素是霍奇拉普拉斯算子，它是微分形式上的一个偏微分算子。

- **层:** 在 20 世纪早期同调方法初步发展之后，到 1950 年，人们逐渐认识到，为了更有效地利用同调方法来理解空间，需要更复杂的概念，即由 J. Leray 开创的以层为系数的上同调 [185]。这一概念对于将拓扑技术扩展到代数几何和数论至关重要 [92, 93, 311]。
- **从局部到整体的计算:** 使用线性代数方法直接计算拓扑不变量，除了最小的玩具示例外，通常是难以处理的。因此，代数拓扑学家被迫开发并行计算的方法，并利用问题中固有的结构，例如纤维化结构。这些思想的例子包括使用迈耶-菲托里斯长正合序列和谱序列、纤维化的塞尔谱序列、艾伦伯格-穆尔谱序列 [216]。
- **先验知识:** 拓扑问题通常配备有先验信息，例如到某个参考基空间的映射，或者空间内某些子空间的选择。同调方法已经得到改进，以在更精细的不变量中捕捉这些额外信息。

为了应对持久同调无法捕捉过滤过程中同伦几何变化的局限性，近期研究引入了持久组合拉普拉斯算子 [302]，以及其他一些持久拉普拉斯算子的变体 [60, 163, 202, 300, 309, 329]，这些算子通过谱同时编码了数据的拓扑结构以及几何或组合结构。与持久同调相比，这些持久拉普拉斯算子能够对数据集进行更深入的分析。它们的调和谱（即零特征值）完全恢复了持久同调产生的拓扑输出，而非调和谱（即非零特征值）则捕捉了关于数据形状的额外几何或组合信息。这些额外信息对于描述过滤过程中持久同调无法检测到的同伦几何形状变化特别有用。Wei 和 Wei [310] 对持久拉普拉斯算子进行了全面的综述。

这些持久拓扑拉普拉斯算子可以视为组合拉普拉斯算子 [111] 的多尺度表述，后者起源于谱图论，并将图拉普拉斯算子 [176] 的概念推广到更高维的单纯复形。研究组合拉普拉斯算子的谱对于理解数据的形状至关重要，近年来受到了广泛关注 [134, 144, 155]。

接下来，我们首先概述组合拉普拉斯算子，然后讨论持久组合拉普拉斯算子，最后介绍几种其他变体的持久拓扑拉普拉斯算子。

analysis on the manifold. Another advantage is that the *Hodge theorem* [306] provides a choice, within a cohomology class, of a harmonic representative, which yields clarity about the significance of the class. A key ingredient within this analysis is the Hodge Laplacian, a partial differential operator on differential forms.

- **Sheaves:** After the initial development of homological methods in the early part of the 20th century, by 1950 it became understood that in order to gain more power in the use of the homological methods for the understanding of spaces, one needed the more complex notion of *cohomology with coefficients in a sheaf* pioneered by J. Leray [185]. This notion became critical to the extension of topological techniques to algebraic geometry and number theory [92, 93, 311].
- **Local to global computation:** The direct computation of topological invariants using the linear algebraic methods, is unmanageable except in the smallest toy examples. For this reason, algebraic topologists were forced to develop methods for parallelizing the computations, and for using structures inherent in the problem, such as a fibration structure. Examples of these ideas are the use of Mayer-Vietoris long exact sequences and spectral sequences, the Serre spectral sequence for a fibration, Eilenberg-Moore spectral sequences [216].
- **A priori knowledge:** Often a topological problem is equipped with a priori information, such as a map to a reference base space or some choices of subspaces within the space. Homological methods have been refined to capture this additional information in more subtle invariants.

2.2 Persistent combinatorial Laplacians

To address the limitation that persistent homology fails to capture the homotopic geometric changes during a filtration, recent advances have introduced persistent combinatorial Laplacians [302], and some other variations of persistent Laplacians [60, 163, 202, 300, 309, 329] that encode in spectra both the topological and the geometric or combinatorial structures of data. Compared to persistent homology, these persistent Laplacians enable deeper analysis of datasets. Their harmonic spectra (i.e., zero eigenvalues) fully recover the topological outputs from the persistent homology, while the non-harmonic spectra (i.e., nonzero eigenvalues) capture additional geometric or combinatorial information about the shape of data. This additional information is particularly useful in describing the homotopic geometric shape changes in a filtration, which the persistent homology fails to detect. A comprehensive survey of persistent Laplacians is given by Wei and Wei [310].

These persistent topological Laplacians can be viewed as multiscale formulations of combinatorial Laplacians [111], which originate from spectral graph theory and generalize the notion of graph Laplacians [176] to higher-dimensional simplicial complexes. The study of the spectra of combinatorial Laplacians is crucial in understanding the shape of data and has gained much attention in recent years [134, 144, 155].

2.1.5 组合拉普拉斯算子

组合拉普拉斯算子将经典的图拉普拉斯算子推广到单纯复形 [111]。与仅编码节点间连接的图拉普拉斯算子相比, 组合拉普拉斯算子是使用边界算子构建的, 这些边界算子捕获了包括顶点、边、三角形以及更高维结构在内的各维单形之间的关系。

第 k 个组合拉普拉斯算子 L_k 作用于第 k 个链群 $C_k(K)$ 上, 定义如下:

$$L_k = \partial_{k+1} \partial_{k+1}^* + \partial_k^* \partial_k,$$

其中 ∂_k^* 是边界算子 ∂_k 关于链群 $C_k(K)$ 上定义的内积的伴随算子。注意, 这里的组合拉普拉斯算子 L_k 是定义在链群上的, 相应的构造也可以在余链群上进行, 参见 [155]。组合拉普拉斯算子 L_k 是半正定对称的, 因此其谱仅由非负实特征值组成。当 K 是一个图时, $L_0 = \partial_1 \partial_1^*$ 简化为图拉普拉斯算子。此外, 组合拉普拉斯算子导出了一个组合版本的霍奇分解:

$$C_k = \text{im } \partial_{k+1} \oplus \ker L_k \oplus \text{im } \partial_k^*,$$

更多细节参见 [191]。

当链群 $C_k(K)$ 上的内积由克罗内克 δ 函数 $\delta_{i,j}$ 定义时, 即 $C_k(K)$ 中的所有基元素被视为相互正交, 边界算子 ∂_k 可以表示为一个大小为 $N_{k-1} \times N_k$ 的矩阵, 其伴随算子 ∂_k^* 则为该矩阵的转置。这里 N_k 表示 k -单形的数量。

如 [111] 所示, 单纯复形 K 上第 k 个拉普拉斯算子的核对应于其第 k 个单纯同调群:

$$\ker L_k = H_k(K),$$

从而捕获了单纯复形的拓扑信息。其维数由第 k 个贝蒂数给出。此外, 非调和谱捕获了底层结构的几何或组合特征。特别地, 如图论中所指出的, 第一个非零特征值描述了一个图的连通性。

2.1.6 持久组合拉普拉斯算子

为了将持久性的思想融入组合拉普拉斯算子, Wang 等人 [302] 引入了持久组合拉普拉斯算子, 这是首个在理论和实践上建立持久同调与组合拉普拉斯算子之间联系的工作。持久拉普拉斯算子的概念也出现在一篇研讨会笔记中 [190]。关于持久组合拉普拉斯算子的全面理论研究可参见 [217], 其稳定性结果则见于 [195]。

Below, we begin with an overview of combinatorial Laplacians, then discuss the persistent combinatorial Laplacians, and finally cover several other variations of persistent topological Laplacians.

2.2.1 Combinatorial Laplacians

The combinatorial Laplacians extend the classical graph Laplacians to simplicial complexes [111]. Compared to graph Laplacians, which only encode connections between nodes, the combinatorial Laplacians are constructed using the boundary operators that capture the relationships between simplices even in higher dimensions, including vertices, edges, triangles, and higher-dimensional structures.

The k -th combinatorial Laplacian L_k on the k -th chain group $\mathcal{C}_k(K)$ is given as follows:

$$L_k = \partial_{k+1} \partial_{k+1}^* + \partial_k^* \partial_k,$$

where ∂_k^* is the adjoint operator of the boundary operator ∂_k with respect to an inner product defined on the chain group $\mathcal{C}_k(K)$. Note that here the combinatorial Laplacian L_k is defined on the chain group, a corresponding construction can also be made on the cochain group, see [155]. The combinatorial Laplacian L_k is positive semi-definite symmetric, and thus its spectrum consists of only non-negative real eigenvalues. In the case that K is a graph, $L_0 = \partial_1 \partial_1^*$ reduces to the graph Laplacian. In addition, the combinatorial Laplacians lead to a combinatorial version of Hodge decomposition:

$$\mathcal{C}_k = \text{im } \partial_{k+1} \oplus \ker L_k \oplus \text{im } \partial_k^*,$$

see [191] for further details.

In the case where the inner product on the chain group $\mathcal{C}_k(K)$ is defined by the Kronecker delta $\delta_{i,j}$, i.e., all basis elements in $\mathcal{C}_k(K)$ are considered mutually orthogonal, the boundary operator ∂_k can be represented as a matrix of size $N_{k-1} \times N_k$ and its adjoint ∂_k^* given as the transpose of the matrix. Here N_k denotes the number of k -simplices.

It has been shown in [111] that the kernel of the k -th Laplacian on a simplicial complex K corresponds to its k -th simplicial homology group:

$$\ker L_k = H_k(K),$$

thereby capturing the topological information of the simplicial complex. Its dimension is given by the k -th Betti number. In addition, the non-harmonic spectra capture the geometrical or combinatorial features of the underlying structure. In particular, as noted in graph theory, the first nonzero eigenvalue describes the connectedness of a graph.

给定滤过中的一对单纯复形 $K^i \subset K^j$, 第 k 个链群 $\mathcal{C}_k(K^i)$ 上的第 k 个持久组合拉普拉斯算子定义如下:

$$L_k^{i,j} = \partial_{k+1}^{i,j} (\partial_{k+1}^{i,j})^* + (\partial_k^i)^* \partial_k^i,$$

其中 ∂_k^i 是 K^i 上的第 k 个边界算子, $\partial_{k+1}^{i,j}$ 称为持久边界算子, 由 ∂_{k+1}^j 限制到第 $(k+1)$ 个链群 $\mathcal{C}_{k+1}(K^j)$ 的一个子群 $\mathcal{C}_{k+1}^{i,j}$ 上给出, 使得其元素的像在 $\mathcal{C}_k(K^i)$ 中, 而 $(\partial_k^i)^*$ 和 $(\partial_{k+1}^{i,j})^*$ 分别是它们的伴随算子。当 $i = j$ 时, 第 k 个持久边界算子等于标准边界算子, 即 $\partial_k^{i,i} = \partial_k^i$, 因此第 k 个持久组合拉普拉斯算子退化为 K^i 上的标准第 k 个组合拉普拉斯算子。持久组合拉普拉斯算子 $L_k^{i,j}$ 与组合拉普拉斯算子类似, 是半正定对称的, 其特征值为实数且非负。该算子也导出一个持久的霍奇分解 [190], 如下所示:

$$\mathcal{C}_k(K^i) = \text{im } \partial_{k+1}^{i,j} \oplus \ker L_k^{i,j} \oplus \text{im } (\partial_k^i)^*.$$

持久组合拉普拉斯算子的实现已作为一个名为 HERMES 的开源软件包提供 [303]。该算法基于 alpha 复形滤过的一种实用构造 [113]。通过为最终的完整 alpha 复形 (即 Delaunay 剖分 [91]) 构造一个单一的第 k 个边界算子以及某些投影矩阵, $(\partial_{k+1}^{i,j})^*$ 和 $(\partial_k^i)^*$ 的矩阵表示都简单地由表示 $\partial_{k+1}^{i,j}$ 和 ∂_k^i 的矩阵的转置给出。计算持久组合拉普拉斯算子的另一种算法可参见 [217]。

一个直接的结果是, 第 k 个持久组合拉普拉斯算子的核同构于第 k 个持久同调群 [190, 217]:

$$\ker L_k^{i,j} = H_k^{i,j},$$

因此其维数即为第 k 个持久贝蒂数。因此, 持久组合拉普拉斯算子的调和谱恢复了来自持久同调的拓扑信息, 而非调和谱则提供了数据形状的额外几何或组合演化特征。持久组合拉普拉斯算子已被应用于各种生物学问题, 例如蛋白质热稳定性 [302]、蛋白质-配体结合 [219] 以及蛋白质-蛋白质结合问题 [307] 等。特别是, 它们还成功用于预测 SARS-CoV-2 变体 BA.4 和 BA.5 [65]。相较于持续同调, 持久组合拉普拉斯算子在蛋白质工程中的优势已得到证实 [253]。

2.1.7 其他变体

我们现在概述为分析具有其他附加结构 (例如方向信息或更复杂关系) 的点云数据而提出的各种持久组合拉普拉斯算子变体。

2.2.2 Persistent combinatorial Laplacians

To incorporate the ideas of persistence into the combinatorial Laplacians, Wang et al. [302] introduced the persistent combinatorial Laplacians, which is the first work, both theoretically and practically, to establish a connection between the persistent homology and the combinatorial Laplacians. A notion of persistent Laplacian also appeared in a seminar note [190]. A comprehensive theoretical study of persistent combinatorial Laplacians can be found in [217] and a stability result is found in [195].

Given a pair of simplicial complexes $K^i \subset K^j$ in a filtration, the k -th persistent combinatorial Laplacian on the k -th chain group $\mathcal{C}_k(K^i)$ is defined as follows:

$$L_k^{i,j} = \partial_{k+1}^{i,j} \left(\partial_{k+1}^{i,j} \right)^* + (\partial_k^i)^* \partial_k^i,$$

where ∂_k^i is the k -th boundary operator on K^i , $\partial_{k+1}^{i,j}$ is called the persistent boundary operator, given by the restriction of ∂_{k+1}^j to a subgroup $\mathcal{C}_{k+1}^{i,j}$ of the $(k+1)$ -th chain group $\mathcal{C}_{k+1}(K^j)$ so that the images of its elements are in $\mathcal{C}_k(K^i)$, and $(\partial_k^i)^*$ and $\left(\partial_{k+1}^{i,j} \right)^*$ are their adjoint operators, respectively. In the case that $i = j$, the k -th persistent boundary operator equals the standard boundary operator, i.e., $\partial_k^{i,i} = \partial_k^i$, and thus the k -th persistent combinatorial Laplacian reduces to the standard k -th combinatorial Laplacian on K^i . The persistent combinatorial Laplacian $L_k^{i,j}$, analogous to the combinatorial Laplacian, is positive semi-definite symmetric, with its eigenvalues real and non-negative. This operator also results in a persistent Hodge decomposition [190] as follows:

$$\mathcal{C}_k(K^i) = \text{im } \partial_{k+1}^{i,j} \oplus \ker L_k^{i,j} \oplus \text{im } (\partial_k^i)^*.$$

The implementation of persistent combinatorial Laplacians has been made available as an open-source software package, called HERMES [303]. The algorithm is based on a practical construction of filtration of alpha complexes [113]. By constructing a single k -th boundary operator for the final complete alpha complex (i.e., the Delaunay tessellation [91]) and certain projection matrices, the matrix representations of $\left(\partial_{k+1}^{i,j} \right)^*$ and $(\partial_k^i)^*$ are both given simply by the transpose of the matrices representing $\partial_{k+1}^{i,j}$ and ∂_k^i . An alternative algorithm for computing persistent combinatorial Laplacians can be found in [217].

An immediate result is that the kernel of the k -th persistent combinatorial Laplacians is isomorphic to the k -th persistent homology group [190, 217]:

$$\ker L_k^{i,j} = H_k^{i,j},$$

and thus its dimension is the k -th persistent Betti number. The harmonic spectra of persistent combinatorial Laplacians, therefore, recover the topological

持久路径拉普拉斯算子

持久路径拉普拉斯算子 [300] 由 Wang 和 Wei 引入, 用于分析具有方向信息的点数据, 例如有向图和网络。该框架建立在路径同调 [137] 的基础上, 后来被扩展到有向网络上的持久框架 [69]。针对加权有向图的加权路径同调在 [192] 中被引入。最近, 在 [99] 中提出了一种计算 1 维持久路径同调的有效算法。

持久超（有向）图拉普拉斯算子

持久超图拉普拉斯算子 [202] 和持久超有向图拉普拉斯算子 [60] 分别针对具有和不具有方向信息的复杂系统提出。这些构造基于嵌入同调中的下确界和上确界链复形 [33], 从而能够定义相应的下确界和上确界拉普拉斯算子。特别地, 当超图退化为单纯复形时, 下确界和上确界链复形重合, 恢复了单纯拉普拉斯算子作为特例。

持久跳跃路径拉普拉斯算子

持久跳跃路径拉普拉斯算子 [199] 在两个方面推广了图的 k -跳跃路径拉普拉斯算子 [118]: 首先, 从图扩展到单纯复形和超图; 其次, 将持久性纳入框架。通过选择不同的跳跃参数, 该方法提供了对数据结构内成对、多体、直接和间接相互作用的全面描述。

持久有向旗复形拉普拉斯算子

持久有向旗复形拉普拉斯算子 [163] 基于有向图的旗复形构建, 该复形仅包含满足特定顺序约束的完全子图。通过将有向旗复形与标准单纯拉普拉斯算子相结合, 它为分析呈现非对称方向关系的复杂网络提供了一个高效工具。持久有向旗复形拉普拉斯算子已应用于复杂的生物分子数据 [329]。

用于单纯映射的持久拉普拉斯算子

用于单纯映射的持久拉普拉斯算子推广了定义在具有包含映射的滤链上的标准持久拉普拉斯算子 [143]。它保持了与持久同调的联系, 其中持久拉普拉斯算子的零度等于与单纯映射诱导的空间序列相关联的持久贝蒂数。一个关键区别在于, 其持久性不仅取决于单纯复形序列 (如标准设置中那样), 还取决于单纯映射本身的结构。

information from the persistent homology, while the non-harmonic spectra provide additional geometric or combinatorial evolutionary features of the shape of the data.

Persistent combinatorial Laplacians have been applied to various biological problems, such as protein thermal stability [302], protein-ligand binding [219], and protein-protein binding problems [307], etc. In particular, they have also been successfully used to forecast the SARS-CoV-2 variants BA.4 and BA.5 [65]. The advantage of persistent combinatorial Laplacians over persistent homology has been demonstrated in protein engineering [253].

2.2.3 Other variations

We now outline various variations of persistent combinatorial Laplacians proposed for analyzing point-cloud data with other additional structures, such as directional information or more complex relationships.

Persistent path Laplacians

Persistent path Laplacians [300] was introduced by Wang and Wei for analyzing point data with directional information, such as directed graphs and networks. The framework builds upon the path homology [137], which was later extended to a persistent framework on directed networks [69]. A weighted path homology for weighted digraphs was introduced in [192]. More recently, an efficient algorithm for computing the 1-dimensional persistent path homology was proposed in [99].

Persistent hyper(di)graph Laplacians

Persistent hypergraph Laplacians [202] and hyperdigraph Laplacians [60] are proposed for complex systems with and without directional information, respectively. These constructions are based on the infimum and supremum chain complexes in embedded homology [33], enabling the definition of the corresponding infimum and supremum Laplacians. Particularly, when the hypergraph reduces to a simplicial complex, the infimum and supremum chain complexes coincide, recovering the simplicial Laplacian as a special case.

Persistent hopping path-Laplacian

Persistent hopping path Laplacian [199] generalizes the k -hopping path Laplacian of graphs [118] in two aspects: first, by extending from graphs to simplicial complexes and hypergraphs, and second, by incorporating persistence into the framework. By selecting different hopping parameters, this method provides a comprehensive description of pairwise, many-body, direct, and indirect interactions within the data structures.

Persistent directed flag Laplacian

Persistent directed flag Laplacian [163] is constructed based on the flag complex of directed graphs, which only includes those complete subgraphs that

多尺度霍赫希尔德拉普拉斯算子

多尺度霍赫希尔德拉普拉斯算子 [150] 通过将霍赫希尔德拉普拉斯算子与多尺度分析相结合, 为分析有向图数据提供了一个高效工具。特别地, 霍赫希尔德拉普拉斯算子基于有向图的截断路径代数, 因为有向环的有向图的完整路径代数是无限维的, 这使得它不适合计算和实际应用。与组合拉普拉斯算子类似, 霍赫希尔德拉普拉斯算子的核同构于相应维度的霍赫希尔德上同调。

2.2 持久狄拉克算子

近年来, 量子算法被引入用于高效计算拓扑和几何特征, 以应对持久同调的计算复杂性, 特别是针对大规模点云数据 [9, 16, 28, 179, 208]。这些方法允许在量子计算机上以量子并行方式访问数据, 并将组合单纯复形编码为量子态, 从而实现了相对于经典拓扑数据分析技术的指数级加速。这些方法的核心是单纯复形上的狄拉克算子 [106], 其平方恰好给出所有阶的组合拉普拉斯算子。它捕获了与组合拉普拉斯算子相同的信息, 但作为一个线性算子, 它对局部拓扑、组合或几何特征更为敏感, 从而提供了额外的信息。狄拉克算子已被发现在分析拓扑信号方面特别有用 [36], 并且也已应用于分子表示 [308]。

持久狄拉克算子 [9, 10] 将狄拉克算子扩展到持久性设置, 用于分析点云数据。其平方产生持久组合拉普拉斯算子, 因此可以看作是持久组合拉普拉斯技术的一种推广, 它在捕获相同拓扑信息的同时, 编码了底层数据更多的局部拓扑和几何结构。狄拉克算子也被扩展用于追踪路径和超图的持久性 [286]。基于 N -链复形, 持久迈耶-狄拉克算子 [287] 已被开发出来, 扩展了基于持久狄拉克技术的范围并提高了其适用性。

在下文中, 我们简要回顾狄拉克算子和持久狄拉克算子。我们还将讨论针对具有附加结构数据的其他变体。

2.2.1 狄拉克算子

在组合情形中, 狄拉克算子也像组合拉普拉斯算子那样, 将单纯复形 K 中相邻维度的单形连接起来。它定义为边界算子与其伴随算子的和, 即 $D = \partial + \partial^*$ 。它可以视为定义在全链复形 (即所有链群的直和 $\mathcal{C}(K) = \bigoplus_k \mathcal{C}_k(K)$) 上的组合拉普拉斯算子的平方根。当用埃尔米特表示表达时,

satisfy specific order constraints. By integrating the directed flag complex with the standard simplicial Laplacian, it provides an efficient tool for analyzing complex networks that exhibits asymmetric directional relationships. Persistent directed flag Laplacian has been applied to complex biomolecular data [329].

Persistent Laplacian for simplicial maps

Persistent Laplacian for simplicial maps generalizes the standard persistent Laplacian defined on a filtration with inclusion maps [143]. It maintains a connection to persistent homology, where the nullity of the persistent Laplacian equals the persistent Betti numbers associated with the simplicial map-induced space sequence. A key difference is that its persistence depends not only on the sequence of simplicial complexes, as in the standard setting, but also on the structures of the simplicial maps themselves.

Multiscale Hochschild Laplacian

Multiscale Hochschild Laplacian [150] offers an efficient tool for analyzing digraph data by integrating Hochschild Laplacians with multiscale analysis. Particularly, the Hochschild Laplacian is based on the truncated path algebras of the digraphs, as the full path algebra of a digraph with directed cycles is infinite-dimensional, making it unsuitable for computation and practical applications. Similar to the combinatorial Laplacians, the kernel of the Hochschild Laplacian is isomorphic to the Hochschild cohomology in the corresponding dimension.

2.3 Persistent Dirac operator

Recently, quantum algorithms have been introduced for the efficient computation of topological and geometrical features to address the computational complexity of persistent homology, especially for large point-cloud data [9, 16, 28, 179, 208]. These methods allow the data to be accessed in quantum parallel on a quantum computer and the combinatorial simplicial complexes to be encoded as quantum states, enabling an exponential speed-up over the classical topological data analysis techniques. Central to these methods is the Dirac operator [106] on simplicial complexes, whose square yields exactly the combinatorial Laplacians of all degrees. It captures the same information as combinatorial Laplacians but provides additional information as it is more sensitive to local topological and combinatorial or geometrical features as a linear operator. The Dirac operator has been found particularly useful in analyzing topological signals [36], and has also been applied to molecular representations [308].

The persistent Dirac operator [9, 10] extends the Dirac operator to the persistent setting for analyzing point-cloud data. Its square yields persistent combinatorial Laplacians and thus can be seen as a generalization of the persistent combinatorial Laplacian technique, capturing the same topological information while encoding more local topological and geometric structures of

其平方得到一个块矩阵, 其对角线由不同阶数的组合拉普拉斯算子给出:

$$D^2 = \begin{pmatrix} 0 & \partial_1 & 0 & \cdots \\ \partial_1^* & 0 & \partial_2 & \cdots \\ 0 & \partial_2^* & 0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}^2 = \begin{pmatrix} L_1 & 0 & 0 & \cdots \\ 0 & L_2 & 0 & \cdots \\ 0 & 0 & L_3 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

与组合拉普拉斯算子 (它保持阶数并将 k -链映射到 k -链) 相比, 狄拉克算子将 k -链映射到 $(k-1)$ -链和 $(k+1)$ -链。此外, 狄拉克算子是一阶微分算子, 这意味着与作为二阶算子的拉普拉斯算子相比, 它对局部几何和拓扑结构更为敏感。狄拉克算子的核编码了单纯复形在所有阶数上的贝蒂数, 而非零特征值是组合拉普拉斯算子特征值的平方根。

在拓扑数据分析中使用狄拉克算子的关键优势在于其适用于高效的量子计算。Lloyd 等人 [208] 提出了首个利用狄拉克算子估计贝蒂数的量子算法, 相比拓扑数据分析中的经典算法实现了指数级加速。该方法构建一个量子态来表示每个尺度下的单纯复形, 然后使用量子相位算法 [235] 来寻找埃尔特表示下狄拉克算子 D 的核的维数。该方法提供了贝蒂数的多尺度版本, 而无需显式地追踪滤过中的持续性。

2.2.2 持久狄拉克算子

Ameneyro 等人 [9] 首次将狄拉克算子扩展到持久化设置中, 用于在量子计算中通过狄拉克算子跨不同尺度计算点云的持久性。该框架利用单个 k 维边界算子 ∂_k , 该算子使用定义在滤集中所有可能 k -单形集合上的单量子比特泡利 X 门进行编码, 同时还利用了一个投影矩阵 P_k^i , 该矩阵从此集合映射到单纯复形 K^i 中的 k -单形集合。对于滤集中存储为量子态的一对单纯复形 $K^i \subset K^j$, 其 k 维持久边界算子定义为 $\partial_k^{i,j} = P_{k-1}^i \partial_k P_k^j$ 。[9] 中定义了作为厄米算子的 k 维持久狄拉克算子为

$$D_k^{i,j} = \begin{pmatrix} 0 & \partial_k^{i,i} & 0 \\ (\partial_k^{i,i})^* & 0 & \partial_{k+1}^{i,j} \\ 0 & (\partial_{k+1}^{i,j})^* & 0 \end{pmatrix} - \xi \begin{pmatrix} P_{k-1}^i & 0 & 0 \\ 0 & -P_k^i & 0 \\ 0 & 0 & P_{k+1}^j \end{pmatrix}$$

the underlying data. The Dirac operator has also been extended for tracking the persistence of paths and hypergraphs [286]. The Persistent Mayer Dirac [287] has been developed based on N -chain complexes, expanding the scope and improving the applicability of persistent Dirac-based techniques.

In the following, we briefly review the Dirac operator and the persistent Dirac operator. We also discuss the other variations for data with additional structures.

2.3.1 Dirac operator

In the combinatorial case, the Dirac operator also connects simplices of consecutive dimensions in a simplicial complex K as the combinatorial Laplacians. It is defined as the sum of the boundary operator and its adjoint, given by $D = \partial + \partial^*$. It can be viewed as the square root of the combinatorial Laplacian defined in the full chain complex, i.e., the direct sum of all chain groups $\mathcal{C}(K) = \bigoplus_k \mathcal{C}_k(K)$. When expressed in the Hermitian representation, its square leads to a block matrix with diagonals given by the combinatorial Laplacians in different degrees:

$$D^2 = \begin{pmatrix} 0 & \partial_1 & 0 & \cdots \\ \partial_1^* & 0 & \partial_2 & \cdots \\ 0 & \partial_2^* & 0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}^2 = \begin{pmatrix} L_1 & 0 & 0 & \cdots \\ 0 & L_2 & 0 & \cdots \\ 0 & 0 & L_3 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Compared to the combinatorial Laplacian, which preserves the degree and maps a k -chain to a k -chain, the Dirac operator maps a k -chain to both $(k-1)$ - and $(k+1)$ -chains. In addition, the Dirac operator is a first-order differential operator, which means that it is more sensitive to local geometric and topological structures compared to the Laplacians, which are second-orders. The kernel of the Dirac operator encodes the Betti numbers of a simplicial complex across all degrees, and the nonzero eigenvalues are the square roots of those of the combinatorial Laplacians.

The key advantage of using the Dirac operator in topological data analysis lies in its suitability for efficient quantum computations. Lloyd et al. [208] introduced the first quantum algorithm for estimating Betti numbers by using Dirac operators, with an exponential speed-up over the classical algorithm in topological data analysis. The method constructs a quantum state that represents the simplicial complex at each scale, and then use a quantum phase algorithm [235] to find the dimension of the kernel of the Dirac operator D in Hermitian representation. The method provides a multiscale version of Betti numbers without explicitly tracking the persistence in filtration.

2.3.2 Persistent Dirac operator

The Dirac operator was extended by Amenyro et al. [9] to the persistent setting for the first time in a quantum implementation for computing the

其中 $\xi \in \mathbb{R}$, 其平方由下式给出

$$(D_k^{i,j})^2 = \begin{pmatrix} \partial_k^{i,i} (\partial_k^{i,i})^* & 0 & \partial_k^{i,i} \partial_{k+1}^{i,j} \\ 0 & L_k^{i,j} & 0 \\ (\partial_{k+1}^{i,j})^* (\partial_k^{i,i})^* & 0 & (\partial_{k+1}^{i,j})^* \partial_{k+1}^{i,j} \end{pmatrix} + \xi^2 \begin{pmatrix} P_{k-1}^i & 0 & 0 \\ 0 & P_k^i & 0 \\ 0 & 0 & P_{k+1}^j \end{pmatrix}$$

根据其定义, 持久狄拉克算子的平方不会像标准情况那样得到一个对角块矩阵, 而是仅其中一个块提供了经过 ξ^2 平移的 k 维持久组合拉普拉斯算子。由此可见, k 维持久组合拉普拉斯算子的特征值对应于 k 维狄拉克算子平方特征值的 ξ^2 平移。在计算持久贝蒂数时, 平移量 ξ 被选为狄拉克算子的正特征值。如 [9] 所述, 设置 $\xi = 0$ 会导致对拉普拉斯算子核中态的重计数, 从而得到错误的贝蒂数估计。

持久狄拉克算子已被应用于分析时间序列数据 [10]。Hayakawa [149] 提出了另一种计算持久贝蒂数的量子实现方法, 该方法基于持久组合拉普拉斯算子的块编码。虽然这些方法为持久拓扑特征提供了高效计算, 但它们的实现需要容错量子计算机, 而这在近期内不太可能用于现实的大数据应用。

2.2.3 其他变体

这里我们介绍几种最近针对不同结构数据开发的持久狄拉克算子的变体。

持久路径狄拉克与超图狄拉克算子

路径狄拉克与超图狄拉克算子及其持久化扩展在文献 [286] 中被提出, 用于从路径复形和超图中提取有价值的信息。它们在分子结构分析中的应用也已得到探索。

持久迈耶狄拉克算子

N 链复形上的迈耶狄拉克算子与持久迈耶狄拉克算子在文献 [287] 中被提出并应用于分子结构分析。请注意, 迈耶同调 [215] 不同于标准同调。在此语境下, 边界算子满足 $\partial^N = 0$ (其中 $N \geq 2$), 而非通常的条件 $\partial^2 = 0$ 。关于基于迈耶拓扑的工具描述, 请参见第 2.3.2 节。

persistence of point clouds using the Dirac operator across different scales. The framework utilizes a single k -th boundary operator ∂_k encoded using 1-qubit Pauli X gates defined on the set of all possible k -simplices in the filtration, and also a projection matrix P_k^i from this set to the set of k -simplices in the simplicial complex K^i . The k -th persistent boundary operator is then given as $\partial_k^{i,j} = P_{k-1}^i \partial_k P_k^j$ for a pair of simplicial complexes $K^i \subset K^j$ stored as quantum states in the filtration. The k -th persistent Dirac operator as Hermitian is defined in [9] as

$$D_k^{i,j} = \begin{pmatrix} 0 & \partial_k^{i,i} & 0 \\ (\partial_k^{i,i})^* & 0 & \partial_{k+1}^{i,j} \\ 0 & (\partial_{k+1}^{i,j})^* & 0 \end{pmatrix} - \xi \begin{pmatrix} P_{k-1}^i & 0 & 0 \\ 0 & -P_k^i & 0 \\ 0 & 0 & P_{k+1}^j \end{pmatrix}$$

with $\xi \in \mathbb{R}$, and its square is given by

$$(D_k^{i,j})^2 = \begin{pmatrix} \partial_k^{i,i} (\partial_k^{i,i})^* & 0 & \partial_k^{i,i} \partial_{k+1}^{i,j} \\ 0 & L_k^{i,j} & 0 \\ (\partial_{k+1}^{i,j})^* (\partial_k^{i,i})^* & 0 & (\partial_{k+1}^{i,j})^* \partial_{k+1}^{i,j} \end{pmatrix} + \xi^2 \begin{pmatrix} P_{k-1}^i & 0 & 0 \\ 0 & P_k^i & 0 \\ 0 & 0 & P_{k+1}^j \end{pmatrix}$$

Based on its definition, the square of the persistent Dirac does not lead to a diagonal block matrix as in the standard case, but just one of the blocks provides the k -th persistent combinatorial Laplacian shifted by ξ^2 . It follows that the eigenvalues of the k -th persistent combinatorial Laplacian correspond to a ξ^2 shift of the squared eigenvalues of the k -th Dirac operator. In the computation of persistent Betti numbers, the shift ξ is chosen to be the positive eigenvalue of the Dirac operator. As noted in [9], setting $\xi = 0$ results in overcounting of the states in the kernel of the Laplacian and thus an incorrect estimate of Betti numbers.

The persistent Dirac operator has been applied for analyzing time series data [10]. Another quantum implementation of computing the persistent Betti numbers was proposed by Hayakawa [149], which is based on the block-encoding of persistent combinatorial Laplacians. While these methods provide efficient computations of the persistent topological features, their realization requires fault tolerant quantum computers, which are unlikely to be available in the near future for realistic big data applications.

2.3.3 Other variations

Here we provide a few variations of the persistent Dirac operator recently developed for data with different structures.

霍瓦诺夫狄拉克算子

霍瓦诺夫狄拉克算子 [162] 被提出用于分析纽结与链环的全局拓扑特征。它基于霍瓦诺夫同调，这是几何拓扑中研究纽结与链环的一个强大拓扑不变量。此主题将在后面的第 3.2 节进一步讨论。

2.3 其他表述

在本节中，我们概述了拓扑数据分析中最近发展的一些表述，这些表述利用替代的同调理论，扩展了用于分析点云数据的标准同调。

2.3.1 持久层理论

前述方法，如持久同调和持久组合拉普拉斯算子，已被证明在分析数据的拓扑结构方面非常强大 [253]。然而，它们在描述局部特征及其之间的关系方面仍然存在局限，而这些关系在实际应用中通常起着至关重要的作用。

层理论为理解拓扑空间提供了一个更强大、更通用的框架。特别是，它使得研究定义在空间上的结构的局部到全局性质成为可能。近年来，它在拓扑数据分析中找到了多种应用 [78, 79, 133, 147, 148, 321]，特别是通过胞腔层的概念 [271]，这对研究人员来说更易于理解，也更容易进行实际计算实现。对于一个单纯复形，一个胞腔层可以被视为每个单形分配一个向量空间，连同称为限制映射的线性变换，这些映射通过面关系定义在向量空间之间。这些限制映射必须满足某些一致性条件，从而允许从局部特征确定全局信息。这使得胞腔层对于分析具有局部化性质的数据非常有价值，并可以从中推断出全局特征。

由于限制映射的方向性，层自然地与上同调相关联。一致性条件导致层上链复形上存在一个良定义的上边缘算子 δ ，满足 $\delta\delta = 0$ ，从而定义了层上同调，其中标准上同调是其特例。关于持久同调的层理论视角首次在 [83] 中进行了研究，后来在 [261, 319] 中被扩展到持久化设置。[146] 中也提出了持久层的概念。

通过在上链群上引入内积，可以定义层拉普拉斯算子 [148]。它们的核对应于层上同调群，其维数由贝蒂数给出。通过结合持久性，Wei 和 Wei [309] 提出了持久层拉普拉斯算子，以解决持久组合拉普拉斯算子在处理带有标签或权重的数据点时的局限性。该框架推广了持久组合拉普拉斯算子，因为当所有数据点被分配相同的量时，后者可以视为前者的特例。给定一

Persistent path Dirac and hypergraph Dirac operators

The path Dirac and hypergraph Dirac operators, along with their persistence extensions, were introduced in [286] for extracting valuable information from path complexes and hypergraphs. Their application to molecular structure analysis has also been explored.

Persistent Mayer Dirac operators

The Mayer Dirac and persistent Mayer Dirac operators on N -chain complexes were proposed in [287] and applied to molecular structure analysis. Note that Mayer homology [215] differs from the standard homology. In this context, the boundary operator satisfies $\partial^N = 0$ with $N \geq 2$ instead of the usual condition $\partial^2 = 0$. For a description of tools based on Mayer topology, see Section 2.4.2.

Khovanov Dirac

The Khovanov Dirac [162] was proposed for analyzing the global topological features of knots and links. It is based on Khovanov homology, a powerful topological invariant in geometric topology for studying Knots and links. This topic will be discussed further later in Section 4.3.

2.4 Other formulations

In this section, we outline formulations recently developed in topological data analysis that utilize alternative homology theories, extending the standard homology for analyzing point-cloud data.

2.4.1 Persistent sheaf theory

The aforementioned methods, such as persistent homology and persistent combinatorial Laplacians, have proven to be powerful for analyzing the topological structure of data [253]. However, they remain limited in their ability to describe the local features along with the relationships among them, which often play a crucial role in practical applications.

Sheaf theory provides a more powerful and general framework for understanding topological spaces. In particular, it enables the study of local-to-global properties of structures defined on spaces. In recent years, it has found various applications in topological data analysis [78, 79, 133, 147, 148, 321], particularly through the notion of cellular sheaves [271], which is more accessible to researchers and easier to implement for practical computations. For a simplicial complex, a cellular sheaf can be thought of as an assignment of a vector space to each simplex, together with linear transformations, called restriction maps, defined between the vector spaces by face relations. These restriction maps must satisfy certain consistency conditions, allowing global information to be determined from local features. This makes cellular sheaves valuable for analyzing data with localized properties, from which global features can be inferred.

个滤过中的一对层，持久边缘算子的定义方式与持久组合拉普拉斯算子的情况类似。主要区别在于数据的底层结构以及使用上同调而非同调，同时还有一个由受一致性规则约束的标量乘法定义的上边缘算子。持久层拉普拉斯算子遵循相同的公式，其核同构于持久上同调群，维数由持久贝蒂数给出。此外，持久层拉普拉斯算子的矩阵表示可以按照与持久组合拉普拉斯算子相同的方式计算 [309]。

2.3.2 持久迈耶拓扑

需要注意的是，拓扑数据分析中所有前述方法都依赖于标准链复形的构造，其中边界算子满足 $\partial^2 = 0$ 。这些方法在揭示数据底层结构更复杂信息方面可能存在局限，因为边界算子仅捕获了相邻维度单形之间的关系。

为了解决这个问题，Shen 等人 [269] 引入了持久迈耶同调和持久迈耶拉普拉斯算子，将迈耶同调 [215] 推广到持久化设置，从而能够研究跨多个维度（而不仅仅是相邻维度）的单形之间的连接。该方法建立在一个 N -链复形之上，其边界算子满足更一般的条件 $\partial^N = 0$ ，其中 $N \geq 2$ 。这扩展了当前基于标准同调（其中 $\partial^2 = 0$ ）的现有框架。关于迈耶同调与标准同调之间关系的讨论，请参见 [278]。

与标准情况类似，持久迈耶拉普拉斯算子的核对应于持久迈耶同调群，其秩由持久迈耶贝蒂数给出。然而，与仅由过滤参数控制的标准持久同调不同，每个度 k 的持久迈耶同调和持久迈耶拉普拉斯算子还受一个额外参数的控制。因此，该框架能够编码关于底层结构更丰富的拓扑、几何或组合信息，特别是跨不同维度单形之间的关系。持久迈耶拉普拉斯算子的进一步推广，称为持久迈耶狄拉克算子，已在 [287] 中提出，它对局部特征更为敏感。

持久迈耶拉普拉斯算子和迈耶狄拉克算子已应用于分子结构分析的研究 [269, 287]。持久迈耶同调已被用作预测蛋白质-配体结合亲和力的多通道特征 [122]。

2.3.3 持久交互拓扑

在代数几何中，相交理论研究给定簇内子簇的相交。在拓扑学中，相交（上）同调研究奇异空间。在离散背景下，交互涉及理解复杂系统内各组成部分如何相互影响。这一概念在复杂网络、分子相互作用以及生物系统中生物体相互作用等领域尤为重要。拓扑数据分析中的传统工具通常缺

Due to the direction of the restriction maps, sheaves are naturally associated with cohomology. The consistency conditions lead to a well-defined coboundary operator δ on sheaf cochain complexes satisfying $\delta\delta = 0$, which defines the sheaf cohomology, with the standard cohomology being a special case. A sheaf-theoretical perspective on persistent homology was first investigated in [83] and later extended to the persistent setting in [261, 319]. A notion of persistent sheaves was also proposed in [146].

By introducing an inner product on the cochain groups, the sheaf Laplacians [148] can be defined. Their kernels correspond to the sheaf cohomology groups with dimensions given by the Betti numbers. By incorporating persistence, Wei and Wei [309] proposed persistent sheaf Laplacians to address the limitation of persistent combinatorial Laplacians in handling data points with labels or weights. The framework generalizes persistent combinatorial Laplacians, as the latter can be seen as a special case of the former when all data points are assigned the same quantity. Given a pair of sheaves in a filtration, the persistent boundary operator is defined analogously to the case of persistent combinatorial Laplacians. The main difference lies in the underlying structure of the data and the use of cohomology instead of homology, along with a coboundary operator defined by scalar multiplication subject to consistency rules. The persistent sheaf Laplacians follow with the same formulas, with kernels isomorphic to the persistent cohomology groups and dimensions given by the persistent Betti numbers. In addition, the matrix representations of persistent sheaf Laplacians can be calculated in the same way as for the persistent combinatorial Laplacians [309].

2.4.2 Persistent Mayer topology

Note that all aforementioned methods in topological data analysis rely on the construction of the standard chain complexes, where the boundary operator satisfies $\partial^2 = 0$. These methods might be limited to revealing more intricate information about the underlying structure of data as the boundary operator captures only the relationship between simplices of consecutive dimensions.

To address this, Shen et al. [269] introduced the persistent Mayer homology and persistent Mayer Laplacians, which generalize the Mayer homology [215] to the persistent setting, allowing the study of connections between simplices across multiple dimensions rather than just adjacent ones. The approach is built on an N -chain complex with the boundary operator satisfying a more general condition $\partial^N = 0$ with $N \geq 2$. This extends the current existing frameworks based on standard homology where $\partial^2 = 0$. For a discussion about the relationship between Mayer homology and the standard homology, see [278].

Analogous to the standard case, the kernels of the persistent Mayer Laplacians correspond to the persistent Mayer homology groups, with ranks given by the persistent Mayer Betti numbers. However, in contrast to the standard persistent homology governed solely by the filtration parameter, the persistent

乏捕捉此类交互的能力，因为它们通常关注全局拓扑特征，而非聚焦于系统内元素之间的局部交互。

最近，受交互上同调概念 [177] 的启发，Liu 等人 [193] 发展了交互拓扑，以深入理解拓扑空间之间的交互，并利用拓扑不变量来刻画这些交互。该理论建立在交互空间范畴之上。在论文中，他们引入了交互空间的概念，研究了其同伦和奇异同调，并证明了交互奇异同调是同伦不变量。此外，他们提出了交互单纯同调和上同调，以促进在点云数据分析中的应用。

基于交互同调，Liu 等人 [194] 提出了持久交互同调和持久交互拉普拉斯算子，并证明了其稳定性。该论文还提供了计算有限点集交互同调和交互拉普拉斯算子的算法。该框架的优势通过对分子内不同类型原子间相互作用的分析得到了展示。

在本节中，我们将注意力转向微分流形上数据的微分拓扑分析。该方法利用微分拓扑的工具，如向量场、微分形式、拉普拉斯-贝尔特拉米算子以及霍奇拉普拉斯算子，并结合其多尺度分析的扩展，来理解微分流形的底层结构。这些工具通过德拉姆-霍奇理论——20 世纪数学的一个里程碑理论——与底层流形的拓扑学紧密相连，因此非常适合分析自然表示为微分流形的数据。例子包括二维曲面、三维实体以及图像数据，例如来自磁共振成像（MRI）或计算机断层扫描（CT）的二维切片或三维体积。这些数据常见于计算机图形学、生物建模和医学成像等领域。

尽管在许多情况下，针对点云数据的代数拓扑分析技术可以通过点云表示应用于数据，但将数据建模为连续或微分流形更为合适。这些数据在观测上本质是连续的，尽管在实际中它们被表示为离散点。仅依赖采样点可能会因其不同的组合表示而导致不一致性，从而在使用代数和组合技术时产生不同的结果。一个更自然的解决方案是在理论上将数据建模为连续或微分流形，仅在实现时进行离散化。这种方式确保了无论采用何种离散化，分析都保持一致性和稳定性，并且结果是收敛的。

此外，微分流形的研究自然地包含了边界条件，例如法向（狄利克雷）和切向（诺伊曼）边界条件，这在边界具有物理或几何意义的应用中（如电磁学和流体动力学）极为重要。这些边界条件提供了约束，支配系统在边界附近的行为，并在求解边值问题时确保解的存在性和唯一性。它还保证了微分算子的适定性以及霍奇拉普拉斯算子在带边界流形上进行微分运算的能力。然而，对于组合拉普拉斯算子而言，并没有内在的域边界概念。

Mayer homology and persistent Mayer Laplacians for each degree k are controlled by an additional parameter. The framework thus enables the encoding of richer topological and geometric or combinatorial information about the underlying structure, in particular, the relationship between simplices across different dimensions. A further generalization of persistent Mayer Laplacians, called persistent Mayer Dirac, has been proposed in [287], which is more sensitive to local features.

Persistent Mayer Laplacians and the Mayer Dirac have been applied to the study of molecular structure analysis [269, 287]. Persistent Mayer homology has been used as multichannel features for predicting protein-ligand binding affinities [122].

2.4.3 Persistent interaction topology

In algebraic geometry, intersection theory studies the intersection of subvarieties within a given variety. In topology, intersection (co)homology studies singular spaces. In the discrete setting, interaction involves understanding how components within a complex system influence each other. This concept is particularly important in fields such as complex networks, molecular interactions, and organism interactions in biological systems. Traditional tools in topological data analysis often lack the ability to capture such intersections, as they typically focus on global topological features instead of focusing on localized interactions between elements in the system.

Recently, motivated by the notion of interaction cohomology [177], Liu et al. [193] developed the interaction topology to gain insight into the interactions between topological spaces and characterize such interactions using topological invariants. The theory is built on the category of interaction spaces. In the paper, they introduced the concept of interaction spaces, studied their homotopy and singular homology, and proved that interaction singular homology is a homotopy invariant. In addition, they proposed the interaction simplicial homology and cohomology to facilitate applications in point-cloud data analysis.

Based on interaction homology, Liu et al. [194] proposed the persistent interaction homology and persistent interaction Laplacians, and demonstrate their stability. The paper also provides an algorithm for computing the interaction homology and interaction Laplacians for finite point sets. The strength of the framework is demonstrated through the analysis of interactions among different types of atoms within molecules.

3 Differential topology approaches

In this section, we shift our attention to differential topology analysis of data on differential manifolds. The approach uses tools from differential topology, such as vector fields, differential forms, Laplace-Beltrami operators, and Hodge Laplacians, along with their extensions to multiscale analysis, to understand

由于复杂的边界条件, 德拉姆-霍奇理论的实现多年来一直具有挑战性。早期的方法, 如标准有限元方法 [71], 不足以捕捉流形拓扑, 并且可能导致在法向或切向边界条件下向量拉普拉斯算子的特征值完全错误 [11]。随着最近的进展, 例如有限元外微积分 (FEEC) [12] 和离散外微积分 (DEC) [94], 该理论的实际实现已成为可能。这些技术确保离散化与连续理论保持一致, 从而能够准确计算底层流形的上同调。基于这些技术, 已经开发了用于计算霍奇分解、霍奇拉普拉斯算子及其多尺度扩展的计算框架。这些进展使得德拉姆-霍奇理论能够应用于拓扑数据分析, 为分析流形上数据的拓扑和几何特征开辟了新的方向。下面, 我们首先回顾带边界流形上的德拉姆-霍奇理论, 包括德拉姆上同调、霍奇拉普拉斯算子以及霍奇分解, 并介绍其近期实现。接着, 我们简要讨论持久德拉姆上同调 [64]。我们还将重点介绍近期发展的持久霍奇拉普拉斯算子 [64, 282] 在捕捉二维和三维紧致域拓扑与几何特征方面的应用。我们同时涵盖了德拉姆-霍奇理论的拉格朗日表述 [64] 和欧拉表述 [282], 这分别取决于底层流形是使用单纯网格式离散化, 还是通过水平集函数在笛卡尔网格中表示为区域。

2.4 带边流形的 de Rham-Hodge 理论

de Rham-Hodge 理论通过微分形式为理解流形的拓扑和几何结构提供了一个基础框架。对于带边流形的情况, 其表述尤为微妙, 此时必须施加适当的边界条件, 例如法向和切向边界条件 [272]。这些边界条件对于该理论的核心结果, 即调和形式与上同调之间的对应关系, 至关重要。

微分形式的概念起初可能显得抽象。然而, 它们与向量微积分中熟悉的概念密切相关。例如, 在 $\mathbb{R}^3$ 中, 一个 0-形式或 3-形式就是一个标量场, 而一个 1-形式或 2-形式则对应一个向量场。一般而言, 微分形式可以看作是微分流形上标量场和向量场的推广。微分形式与标量场或向量场之间的这些对应关系, 为理解 1-形式的边界条件提供了直观认识: 一个 1-形式是法向的, 如果其对应的向量场与边界垂直; 是切向的, 如果该场与边界相切。法向和切向边界条件的正式定义, 以及微分形式与 $\mathbb{R}^3$ 中向量微积分对应物之间的完整对应关系, 可以在 [324] 中找到。这两种边界条件在 Hodge 星算子下彼此对偶。

微分算子 d^k (即外微分) 和余微分算子 δ^k 是构建 de Rham-Hodge 理论的两个基本算子。我们在此简要回顾它们及其与向量微积分中经典算子的对应关系, 以便直观理解。微分算子 d^k 将 k -形式映射到 $(k+1)$ -形式, 并

the underlying structure of differential manifolds. These tools are deeply connected to the topology of the underlying manifolds through de Rham-Hodge theory, a landmark theory of the 20th century mathematics, and thus are well-suited for analyzing data naturally represented as differential manifolds. Examples include 2D surfaces, 3D solids, and image data such as 2D slices or 3D volumes from magnetic resonance imaging (MRI) or computed tomography (CT) scans. These data are commonly found in fields such as computer graphics, biological modeling, and medical imaging.

Although in many scenarios, techniques in algebraic topology analysis for point cloud data can be applied to data using point cloud representation, it is more appropriate to model the data as continuous or differential manifolds. These data are naturally continuous in observation, even though they are practically represented as discrete points. Relying solely on sampled points can lead to inconsistencies due to their varying combinatorial representations, resulting in different outcomes using algebraic and combinatorial techniques. A more natural solution is to model the data theoretically as continuous or differential manifolds, and perform discretization only in the implementation. This way ensures that the analysis remains consistent and stable regardless of different discretizations, and the results are convergent.

In addition, the study of differential manifolds naturally incorporates the boundary conditions, such as the normal (Dirichlet) and tangential (Neumann) boundary conditions, which are significantly important in applications where the boundary carries physical or geometric meaning, such as electromagnetism and fluid dynamics. These boundary conditions provide constraints that govern the behavior of a system near the boundary, and ensure the existence and uniqueness of solutions when solving boundary value problems. It also ensures the well-posedness of differential operators and Hodge Laplacians' capability to perform differential calculus on manifolds with boundary. However, there is no intrinsic notion of domain boundary for combinatorial Laplacians.

The implementation of de Rham Hodge theory has been challenging for years due to the complicated boundary conditions. Earlier methods, such as the standard finite element method [71], are insufficient to capture the manifold topology and can lead to entirely wrong eigenvalues for the vector Laplacians under normal or tangential boundary conditions [11]. With recent advances, such as the finite element exterior calculus (FEEC) [12] and the discrete exterior calculus (DEC) [94], the practical realization of the theory has become possible. These techniques ensure that the discretizations are consistent with the continuous theory and thus allow for accurate computation of the cohomologies of the underlying manifolds. Based on these techniques, computational frameworks have been developed for computing Hodge decompositions, Hodge Laplacians and their multiscale extension. These developments have enabled the application of de Rham Hodge theory in topological data analysis, opening new directions for analyzing the topological and geometric features of data on manifolds.

满足 $d^k d^{k-1} = 0$ 。它概括并统一了向量分析中的梯度 ∇ 、旋度 $\nabla \times$ 和散度 $\nabla \cdot$ 的概念，它们分别对应于在 $\mathbb{R}^3$ 中将 d 应用于 0-形式、1-形式和 2-形式。余微分 δ^k 可以定义为 d^{k-1} 关于由微分形式空间上的 Hodge 星算子诱导的 Hodge L^2 内积的伴随算子。它将 k -形式映射到 $(k-1)$ -形式，并满足 $\delta^{k-1} \delta^k = 0$ 。在 $\mathbb{R}^3$ 中，当分别应用于 2-形式、1-形式和 0-形式时，它对应于 $-\nabla$ 、 $\nabla \times$ 和 $-\nabla \cdot$ 。

2.4.1 de Rham 上同调

设 Ω_n^k 和 Ω_t^k 分别为定义在 m 维光滑流形 M 上的法向 k -形式空间和切向 k -形式空间。关于它们的正式定义，请参见 [324]。这两个空间在 Hodge 星算子下互为对偶，即 $\Omega_n^k \cong \Omega_t^{m-k}$ 。微分形式的 de Rham 上同调定义在由微分 d^k 和余微分 δ^k 连接的两个序列上：

$$\begin{aligned} \dots &\xrightarrow{d^{k-2}} \Omega_n^{k-1} \xrightarrow{d^{k-1}} \Omega_n^k \xrightarrow{d^k} \Omega_n^{k+1} \xrightarrow{d^{k+1}} \dots, \\ \dots &\xleftarrow{\delta^{k-1}} \Omega_t^{k-1} \xleftarrow{\delta^k} \Omega_t^k \xleftarrow{\delta^{k+1}} \Omega_t^{k+1} \xleftarrow{\delta^{k+2}} \dots. \end{aligned}$$

这两个复形是良定义的，因为 d 保持法向边界条件，而 δ 保持切向边界条件。由此得到的上同调分别对应于第 k 阶相对 de Rham 上同调 $H_{dR}^k(M, \partial M)$ 和第 k 阶绝对 de Rham 上同调 $H_{dR}^k(M)$ ，如下所示：

$$\begin{aligned} H_{dR}^k(M, \partial M) &= \ker d^k / \operatorname{im} d^{k-1}, \\ H_{dR}^k(M) &= \ker \delta^k / \operatorname{im} \delta^{k+1}, \end{aligned}$$

其中 $\ker d^k$ (相应地, $\ker \delta^k$) 称为闭 (相应地, 余闭) k -形式空间, 而 $\operatorname{im} d^{k-1}$ (相应地, $\operatorname{im} \delta^{k+1}$) 是恰当 (相应地, 余恰当) k -形式空间。 $H_{dR}^k(M, \partial M)$ 和 $H_{dR}^k(M)$ 的秩分别由第 $(m-k)$ 个 Betti 数 β_{m-k} 和第 k 个 Betti 数 β_k 给出。这里，这些算子的定义域被限制在相应的边界条件下。关于绝对和相对 de Rham 上同调及其对应同构的详细讨论，请参阅 [266]。

2.4.2 Hodge 拉普拉斯算子

第 k 个 Hodge 拉普拉斯算子 Δ^k 定义在 k -形式空间上，具体如下：

$$\Delta^k = d^{k-1} \delta^k + \delta^{k+1} d^k.$$

当 $k = 0$ 时，Hodge 拉普拉斯算子退化为作用在流形上函数空间的 Laplace-Beltrami 算子 $\delta^1 d^0$ 。在存在边界的情况下，Hodge 拉普拉斯算子

Below, we start with a review of de Rham-Hodge theory for manifolds with boundary, including the de Rham cohomology, Hodge Laplacians, and Hodge decomposition, along with their recent implementations. We then provide a short discussion on persistent de Rham cohomology [64]. We also focus on the recently developed persistent Hodge Laplacians [64, 282] in capturing the topological and geometric features of 2D and 3D compact domains. We include both the Lagrangian formulation [64] and the Eulerian formulation [282] of the de Rham-Hodge theory, depending on whether the underlying manifolds are discretized using simplicial meshes or represented as regions in a Cartesian grid defined by level-set functions, respectively.

3.1 De Rham-Hodge theory for manifolds with boundary

The de Rham-Hodge theory offers a foundational framework for understanding the topological and geometrical structures of manifolds through differential forms. Its formulation is particularly subtle in the case of manifolds with boundary, where appropriate boundary conditions, such as normal and tangential boundary conditions [272], must be imposed. These boundary conditions are essential for the core results of the theory, i.e., the correspondence between harmonic forms and the cohomology.

The notion of differential forms might be abstract at first. However, they are closely related to familiar concepts in vector calculus. For example, in $\mathbb{R}^3$, a 0- or 3-form is just a scalar field, while a 1- or 2-form corresponds to a vector field. In general, differential forms can be viewed as a generalization of scalar and vector fields on differential manifolds. These correspondences between differential forms and scalar or vector fields provide an intuitive understanding of the boundary conditions in the case of 1-forms: a 1-form is normal if its corresponding vector field is normal to the boundary, or tangential if the field is tangential to the boundary. The formal definitions of normal and tangential boundary conditions, and also a complete correspondence between differential forms and their counterparts in vector calculus in $\mathbb{R}^3$ can be found in [324]. The two types of boundary conditions are dual to each other under the Hodge star operator.

The differential d^k (i.e., the exterior derivative) and the codifferential operator δ^k , are two fundamental operators in building the de Rham-Hodge theory. We briefly recall them here with their correspondences with classical operators in vector calculus for an intuitive understanding. The differential d^k maps k -forms to $(k+1)$ -forms and satisfies $d^k d^{k-1} = 0$. It generalizes and unifies the notions of gradient ∇ , curl $\nabla \times$, and divergence $\nabla \cdot$ in vector analysis, which correspond to d applied to 0-, 1-, and 2-forms in $\mathbb{R}^3$, respectively. The codifferential δ^k can be defined as an adjoint operator of d^{k-1} with respect to the Hodge L^2 inner product induced from the Hodge star operator on the space of differential forms. It maps k -forms to $(k-1)$ -forms and satisfies $\delta^{k-1} \delta^k = 0$. In $\mathbb{R}^3$, it corresponds to $-\nabla$, $\nabla \times$, and $-\nabla \cdot$ when applied to 2-, 1-, and 0-forms, respectively.

Δ^k 与流形拓扑没有直接对应关系，因为其核是无限维的。然而，我们可以分别通过将 Hodge 拉普拉斯算子 Δ^k 限制在法向形式空间和切向形式空间上，来定义 Hodge 拉普拉斯算子 Δ_n^k 和 Δ_t^k 。这两种类型的 Hodge 拉普拉斯算子具有有限维的核。空间 $\ker \Delta_n^k$ 对应于相对 de Rham 上同调，而 $\ker \Delta_t^k$ 对应于绝对 de Rham 上同调 [126]。此外，这两个核通过 Poincaré 对偶性相关联，从而得到以下恒等式：

$$\ker \Delta_n^k \cong H_{dR}^k(M, \partial M) \cong H_{dR}^{m-k}(M) \cong \ker \Delta_t^{m-k}$$

$\ker \Delta_n^k$ 中的元素称为法向调和 k -场，而 $\ker \Delta_t^k$ 中的元素称为切向调和 k -场。这些恒等式使得我们可以使用任意一种边界条件来研究底层流形及其边界的拓扑结构。

需要注意的是，与组合情形类似，Hodge 拉普拉斯算子的调和谱捕获了底层流形的拓扑信息，其零特征值的数量由 Betti 数给出。然而，Hodge 拉普拉斯算子的非调和谱编码了丰富的几何信息，例如底层流形的度量和曲率。这与组合情形有根本性的不同，在组合情形中，非调和谱仅反映组合或离散结构信息，而非光滑的几何特征。关于组合拉普拉斯算子与 Hodge 拉普拉斯算子之间相似性与差异的深入讨论，请参见 [260]。

Hodge 拉普拉斯算子的实现已通过 FEEC 在 [12] 中以及通过 DEC [260, 324] 实现，两种方法都提供了准确的特征值结果并保持了底层域的上同调。此外，文献 [260] 引入了一种边界诱导图（BIG）拉普拉斯算子的概念，它定义在具有适当边界条件的笛卡尔域上。它是离散化 Hodge 拉普拉斯算子的一个特例，计算更简单，同时能给出数据的正确拓扑维数。零阶 Hodge 拉普拉斯算子，即 Laplace-Beltrami 算子，已被广泛应用于研究流形的形状。例如，ShapeDNA [259] 利用该算子的谱为形状提供了一个简单的表示。全局点签名（GPS）[262] 使用该算子的缩放特征函数向量对形状进行编码。ShapeDNA 采用标准的有限元方法，而 GPS 则使用 DEC 中的工具。

2.4.3 Hodge 分解

Hodge 分解是 Hodge 理论中的一个核心结果，它将微分 k -形式空间分解为相互正交子空间的直和。微分形式的一个三分量 Hodge 分解，称为 Hodge Morrey 分解 [230]，给出如下：

$$\Omega^k = d\Omega_n^{k-1} \oplus \delta\Omega_t^{k+1} \oplus \mathcal{H}^k$$

3.1.1 De Rham cohomology

Let Ω_n^k and Ω_t^k be the space of normal k -forms and the space of tangential k -forms over a smooth manifold M of dimension m , respectively. For their formal definitions, see [324]. These two spaces are dual to each other under the Hodge star operator, i.e., $\Omega_n^k \cong \Omega_t^{m-k}$. The de Rham cohomologies of differential forms are defined on two sequences connected by the differential d^k and the codifferential δ^k :

$$\begin{aligned} \dots &\xrightarrow{d^{k-2}} \Omega_n^{k-1} \xrightarrow{d^{k-1}} \Omega_n^k \xrightarrow{d^k} \Omega_n^{k+1} \xrightarrow{d^{k+1}} \dots, \\ \dots &\xleftarrow{\delta^{k-1}} \Omega_t^{k-1} \xleftarrow{\delta^k} \Omega_t^k \xleftarrow{\delta^{k+1}} \Omega_t^{k+1} \xleftarrow{\delta^{k+2}} \dots. \end{aligned}$$

These two complexes are well-defined as d preserves the normal boundary conditions and δ preserves the tangential boundary conditions. The resulting cohomologies correspond to the k -th relative de Rham cohomology $H_{dR}^k(M, \partial M)$ and the k -th absolute de Rham cohomology $H_{dR}^k(M)$ as follows:

$$\begin{aligned} H_{dR}^k(M, \partial M) &= \ker d^k / \operatorname{im} d^{k-1}, \\ H_{dR}^k(M) &= \ker \delta^k / \operatorname{im} \delta^{k+1}, \end{aligned}$$

where $\ker d^k$ (resp. $\ker \delta^k$) is called the space of closed (resp. coclosed) k -forms and $\operatorname{im} d^{k-1}$ (resp. $\operatorname{im} \delta^{k+1}$) is the space of exact (resp. coexact) k -forms. The ranks of $H_{dR}^k(M, \partial M)$ and $H_{dR}^k(M)$ are given by the $(m-k)$ -th Betti number β_{m-k} and the k -th Betti number β_k , respectively. Here the domains of these operators are restricted under the corresponding boundary conditions. For a detailed discussion of the absolute and relative de Rham cohomology and the corresponding isomorphisms, refer to [266].

3.1.2 Hodge Laplacians

The k -th Hodge Laplacian Δ^k on the space of k -forms is defined as follows:

$$\Delta^k = d^{k-1}\delta^k + \delta^{k+1}d^k.$$

When $k = 0$, the Hodge Laplacian reduces to the Laplace-Beltrami operator $\delta^1 d^0$ acting on the space of functions defined on the manifold. In the presence of the boundary, the Hodge Laplacian Δ^k has no direct correspondence with the manifold topology, as its kernel is infinite-dimensional. However, one can define the Hodge Laplacians Δ_n^k and Δ_t^k by restricting the Hodge Laplacian Δ^k to the space of normal forms and to the space of tangential forms, respectively. These two types of Hodge Laplacians lead to finite-dimensional kernels. The space $\ker \Delta_n^k$ corresponds to the relative de Rham cohomology, and $\ker \Delta_t^k$ to the absolute de Rham cohomology [126]. In addition, these two kernels are related by the Poincaré duality, leading to the following identifications:

$$\ker \Delta_n^k \cong H_{dR}^k(M, \partial M) \cong H_{dR}^{m-k}(M) \cong \ker \Delta_t^{m-k}$$

这里 $\mathcal{H}^k = \ker d^k \cap \ker \delta^k \subset \ker \Delta^k$, 称为调和 k -场空间。当应用于区域上的向量场时, 此分解表明任何向量场都可以正交分解为一个无旋分量、一个无散分量和一个调和分量, 从而导出经典的 Helmholtz-Hodge 分解 [181]。空间 $\mathcal{H}^k$ 等于 Hodge 拉普拉斯算子的核 $\ker \Delta^k$, 并且同构于第 k 个 de Rham 上同调群。然而, 在存在边界的情况下, 空间 $\mathcal{H}^k$ 是无限维的 [266], 与流形拓扑没有联系。研究表明, 通过施加边界条件, 该空间可以进一步分裂, 从而揭示相对上同调和绝对上同调 [126]。特别地, 对于欧几里得空间中的紧致区域, 存在一个五分量保拓扑 Hodge 分解 [272], 给出如下:

$$\Omega^k = d\Omega_n^{k-1} \oplus \delta\Omega_t^{k+1} \oplus \mathcal{H}_n^k \oplus \mathcal{H}_t^k \oplus (d\Omega_n^{k-1} \cap \delta\Omega_t^{k+1}),$$

其中 $\mathcal{H}_n^k = \ker \Delta_n^k$ 和 $\mathcal{H}_t^k = \ker \Delta_t^k$ 分别是 $\mathcal{H}^k$ 在法向形式空间和切向形式空间上的限制, 如前所述, 它们编码了底层流形的相对上同调和绝对上同调。

对于 Hodge 分解的计算框架, 我们在此仅关注完整的五分量保拓扑 Hodge 分解。关于经典的三分量 Helmholtz-Hodge 分解的综述可参见 [26]。向量场的五分量 Hodge 分解的离散化已通过其微分形式表示被提出, 适用于四面体网格 [324] 以及适用于由笛卡尔网格中水平集函数定义的二维和三维紧致区域 [281, 283]。这两种方法都基于 DEC。其他方法, 假设分段常数向量场, 已针对曲面三角形网格 [249] 以及四面体网格 [248] 被提出。关于 $\mathbb{R}^3$ 中紧致区域的五分量 Hodge 分解在向量场和标量场方面的讨论可参见 [43]。

2.5 持久 de Rham 上同调

持久 de Rham 上同调最初由 Chen 等人在 [64] 中引入。该工作详细讨论了在法向和切向边界条件下调和形式的持久性。持久 de Rham 上同调在理论和实际应用中具有与持久同调对于点云数据同等的重要性。它作为一种基础工具展现出潜力, 并为分析可表示为微分流形的数据的拓扑结构开辟了新的方向。

为了定义持久 de Rham 上同调, 我们考虑一个由水平集函数 f 的子水平集定义的流形 M , 即对于 $c \in \mathbb{R}$, $M = f^{-1}([-\infty, c])$ 。通过改变 f 的等值, 可以得到一个带有包含映射的微分流形滤链如下:

$$M_0 \hookrightarrow M_1 \cdots \hookrightarrow M_{s-1} \hookrightarrow M_s,$$

The elements in $\ker \Delta_n^k$ are called the normal harmonic k -fields, and those in $\ker \Delta_t^k$ are called the tangential harmonic k -fields. These identifications enable the use of either type of boundary conditions for the study of the topology of the underlying manifold and its boundary.

Note that similar to the combinatorial case, the harmonic spectra of Hodge Laplacians capture the topological information about the underlying manifolds with the number of zero-eigenvalues given by the Betti numbers. However, the non-harmonic spectra of the Hodge Laplacians encode rich geometric information such as the metric and the curvature of the underlying manifolds. This is fundamentally different from the combinatorial case, where the non-harmonic spectra reflect only the combinatorial or the discrete structural information rather than the smooth geometric features. For a thorough discussion about the similarity and difference between the combinatorial Laplacians and Hodge Laplacians, see [260].

The implementation of Hodge Laplacians has been realized through FEEC in [12] and through DEC [260, 324], with both approaches providing accurate eigenvalue results and preserving the cohomology of the underlying domain. In addition, a notion of boundary-induced graph (BIG) Laplacians has been introduced [260], defined on Cartesian domains with appropriate boundary conditions. It is a special case of the discretized Hodge Laplacians and is simpler to compute while delivering correct topological dimensions of the data. The Hodge Laplacian of degree 0, i.e., the Laplace-Beltrami operator, has been widely applied to the study of the shapes of manifolds. For example, ShapeDNA [259] provides a simple representation of a shape using the spectra of the operator. Global point signature (GPS) [262] encodes a shape using a vector of scaled eigenfunctions of the operator. ShapeDNA utilizes standard finite element approaches, while GPS employs tools in DEC.

3.1.3 Hodge decomposition

The Hodge decomposition is a central result in Hodge theory, which decomposes the space of differential k -forms into a direct sum of mutually orthogonal subspaces. A 3-component Hodge decomposition of differential forms, called the Hodge Morrey decomposition [230], is given as follows:

$$\Omega^k = d\Omega_n^{k-1} \oplus \delta\Omega_t^{k+1} \oplus \mathcal{H}^k$$

Here $\mathcal{H}^k = \ker d^k \cap \ker \delta^k \subset \ker \Delta^k$, known as the space of harmonic k -fields. When applied to vector fields on a domain, this decomposition states that any vector field can be orthogonally decomposed into a curl-free, a divergence-free, and a harmonic component, leading to the classical Helmholtz-Hodge decomposition [181]. The space $\mathcal{H}^k$ is equal to the kernel of the Hodge Laplacian $\ker \Delta^k$ and isomorphic to the k -th de Rham cohomology group. However, in the presence of a boundary, the space $\mathcal{H}^k$ is infinite-dimensional [266], providing no connections to the manifold topology. It has been shown that this space can be further split by enforcing boundary conditions, revealing the relative

其中每个 M_i 是对应于不同等值的 f 的子水平集。序列中任意一对流形 $M_i \hookrightarrow M_j$ 的包含映射在法向微分形式 $\Omega_n^k(M_i) \rightarrow \Omega_n^k(M_j)$ 上诱导了一个满足特定条件的唯一延拓。该映射进一步在上同调群上诱导了一个映射 $\psi_{i,j}^k : H_{dR,i}^k(M, \partial M) \rightarrow H_{dR,j}^k(M, \partial M)$, 用于追踪拓扑特征的演化, 即法向调和场, 因为 $H_{dR,i}^k(M, \partial M) \cong \mathcal{H}_{n,i}^k$ 。这里 $\mathcal{H}_{n,i}^k$ 表示 M_i 上的法向调和 k -场空间。然后, 可以将持久相对 de Rham 上同调定义为该映射的像:

$$H_{dR,i,j}^k(M, \partial M) = \text{im } \psi_{i,j}^k = \ker d_i^k / (\text{im } d_j^{k-1} \cap \ker d_i^k).$$

它的秩给出了第 $(m-k)$ 个持久 Betti 数 $\beta_{m-k}^{i,j}$ 。随后, 可以通过检查持久上同调群中的元素, 类似于组合情形, 相应地定义法向调和场的诞生、死亡以及持久性。请注意, 这里仅考虑了法向微分形式的空间。在切向边界条件下的对偶表述可以类似地构建 [64]。

2.6 持久霍奇拉普拉斯算子

在同一篇论文 [64] 中, Chen 等人首次从理论上提出了持久霍奇拉普拉斯算子。该论文还提出了一种使用离散外微积分在拉格朗日表述下对多尺度霍奇拉普拉斯算子的实现, 即使用单纯形网格离散化的流形。随后, Su 等人在 [282] 中提出了持久霍奇拉普拉斯算子的计算框架, 该框架在欧拉表述下, 将离散外微积分从单纯形网格适配到规则网格。此背景下的流形被建模为笛卡尔网格中由水平集函数的等值线或等值面所界定的区域。这些发展基于将 de Rham 上同调与微分流形在滤过中的演化相结合, 可视为早期关于立方体背景下持久同调工作的扩展 [298, 299]。

给定上一小节提到的滤过中的任意一对流形 $M_i \hookrightarrow M_j$, 在法向 k -形式空间 $\Omega^k(M_i)$ 上的第 k 个持久霍奇拉普拉斯算子定义如下:

$$\Delta_{i,j}^k = d_{i,j}^{k-1} \delta_{i,j}^k + \delta_i^{k+1} d_i^k,$$

其中 d_i^k 和 δ^k 分别是 M_i 上法向形式的第 k 个外微分算子与余微分算子, 而 $d_{i,j}^{k-1}$ 是从 M_j 上的法向 $(k-1)$ -形式到 M_i 上的法向 k -形式的持久微分算子, $\delta_{i,j}^k$ 是其伴随算子。持久霍奇拉普拉斯算子可以类似地定义在切向形式空间上。然而, 由于法向与切向边界条件的对偶性, 它们是等价的。当 $i = j$ 时, 持久霍奇拉普拉斯算子退化为限制在法向形式上的通常的霍奇拉普拉斯算子。 $\Delta_{i,j}^k$ 的核定义了持久调和场。它同构于持久相对 de Rham 上同调, 其秩导出持久贝蒂数。

and absolute cohomology [126]. In particular, there is a 5-component topology-preserving Hodge decomposition for compact domains in the Euclidean spaces [272], given as follows:

$$\Omega^k = d\Omega_n^{k-1} \oplus \delta\Omega_t^{k+1} \oplus \mathcal{H}_n^k \oplus \mathcal{H}_t^k \oplus (d\Omega^{k-1} \cap \delta\Omega^{k+1}),$$

where $\mathcal{H}_n^k = \ker \Delta_n^k$ and $\mathcal{H}_t^k = \ker \Delta_t^k$ are the restrictions of $\mathcal{H}^k$ to the space of normal forms and the space of tangential forms, respectively, encoding the relative and absolute cohomology of the underlying manifold as discussed previously.

For the computational frameworks of Hodge decomposition, we focus here only on the complete topology-preserving 5-component Hodge decomposition. A review of the classical 3-component Helmholtz-Hodge decomposition is available in [26]. The discretization of the 5-component Hodge decomposition of vector fields has been proposed using their differential form representations for tetrahedral meshes [324] and for 2D and 3D compact domains defined by level set functions in Cartesian grids [281, 283]. Both methods are based on DEC. Other approaches, assuming piecewise-constant vector fields, have been developed for surface triangle meshes [249] and also tetrahedral meshes [248]. A discussion of this 5-component Hodge decomposition for compact domains in $\mathbb{R}^3$ in terms of vector and scalar fields can be found in [43].

3.2 Persistent de Rham cohomology

Persistent de Rham cohomology was first introduced by Chen et al. in [64]. The work provides a detailed discussion about the persistence of harmonic forms under normal and tangential boundary conditions. The persistent de Rham cohomology holds the same importance both in theory and practical applications as persistent homology does for point-cloud data. It shows promise as a foundation tool and opens new directions for analyzing the topological structures of data that can be represented as differential manifolds.

To define the persistence de Rham cohomology, we consider a manifold M , defined by a sublevel set of a level set function f , i.e., $M = f^{-1}([-\infty, c])$ for $c \in \mathbb{R}$. By varying the isovalues of f , a filtration of differential manifolds with inclusion maps can be obtained as follows:

$$M_0 \hookrightarrow M_1 \cdots \hookrightarrow M_{s-1} \hookrightarrow M_s,$$

where each M_i is the sublevel set of f corresponding to a different isovalue. The inclusion map for any pair of manifolds $M_i \hookrightarrow M_j$ in the sequence induces a unique extension on the normal differential forms $\Omega_n^k(M_i) \rightarrow \Omega_n^k(M_j)$ satisfying certain conditions. This map further induces a map on the cohomology groups $\psi_{i,j}^k : H_{dR,i}^k(M, \partial M) \rightarrow H_{dR,j}^k(M, \partial M)$ that tracks the evolution of topological features, namely, normal harmonic fields as $H_{dR,i}^k(M, \partial M) \cong \mathcal{H}_{n,i}^k$. Here $\mathcal{H}_{n,i}^k$ denotes the space of normal harmonic k -fields on M_i . One can then

我们现在转向离散化框架，并根据底层流形的离散化方式，简要讨论使用离散外微积分的拉格朗日与欧拉表述。

2.6.1 拉格朗日表述

在拉格朗日表述中，底层流形被离散化为单纯形网格，具体而言，在二维中是三角形网格，在三维中是四面体网格。一个离散的 k -形式就是一个向量，其每个条目被分配给一个 k -单纯形。例如，一个离散的 0-形式是一个将值分配给顶点的向量，而 1-形式则具有与边相对应的值。微分是一个稀疏矩阵，由连续维度单纯形之间的带符号关联矩阵的转置给出。余微分是关联矩阵与离散霍奇星算子的乘积，其形式为对角矩阵。关于这些离散算子的更多细节，请参见 [64, 324]。

拉格朗日表述需要对域进行显式剖分，这依赖于高质量的网格生成工具，例如 CGAL [119]、Gmsh [131]、Pymesh [327] 等。这使得在此背景下实现持久霍奇拉普拉斯算子具有挑战性，因为在过滤过程中生成一致的网格可能存在问题，并且得到的离散算子可能不是稀疏矩阵。拉格朗日表述中霍奇拉普拉斯算子的多尺度版本 [64]，除了依赖于这些网格生成工具外，还有几个局限性。由于紧致域的四面体网格表示存在差异，该方法具有敏感性和数值不一致性。该方法在计算上也很昂贵，因为它需要反复对域进行剖分以创建流形在过滤中的演化。这些局限性使得该框架不适合机器学习任务，因为机器学习任务需要计算相同尺度上的特征表示以保证可比性，同时也需要高效性，这对于计算大型数据集的特征至关重要。

2.6.2 欧拉表述

持久霍奇拉普拉斯算子的欧拉表述 [282] 已被引入，并采用了适配的离散外微积分以解决拉格朗日情况下的局限性。该论文提出了一种用于持久霍奇拉普拉斯算子的计算算法。在此设置中，底层流形被建模为标准笛卡尔网格上水平集函数的子水平集，即由水平集函数的等值线或等值面所界的域。从单纯形网格到规则网格的适配离散外微积分遵循与拉格朗日情况类似的结构。请注意，在欧拉表述中，调整水平集函数的等值会自然地导致过滤中嵌套的网格复形序列，从而使得离散持久霍奇拉普拉斯算子的构造成为可能。

欧拉表示具有若干优势：首先，它消除了显式剖分的需求，从而摆脱了对高质量网格生成工具的依赖。其次，所有对象的拉普拉斯算子计算都

define the persistent relative de Rham cohomology as the image of the map:

$$H_{dR,i,j}^k(M, \partial M) = \text{im } \psi_{i,j}^k = \ker d_i^k / (\text{im } d_j^{k-1} \cap \ker d_i^k).$$

Its rank provides the $(m-k)$ -th persistent Betti number $\beta_{m-k}^{i,j}$. The birth and death, as well as the persistence of normal harmonic fields, can then be defined correspondingly by examining the elements in the persistent cohomology groups, analogous to the combinatorial case. Note that only the spaces of normal differential forms are considered here. The dual formulation under the tangential boundary conditions can be constructed in a similar manner [64].

3.3 Persistent Hodge Laplacians

In the same paper [64], Chen et al. first theoretically proposed the persistent Hodge Laplacians. The paper also presents an implementation of multiscaled Hodge Laplacians using DEC for manifolds in the Lagrangian formulation, i.e., manifolds discretized using simplicial meshes. Subsequently, a computational framework of persistent Hodge Laplacians was proposed by Su et al. in [282] in the Eulerian setting with the adapted DEC from simplicial meshes to regular grids. The manifolds in this context are modeled as domains in a Cartesian grid bounded by isocurves or isosurfaces of level-set functions. These developments are based on combining de Rham cohomology with the evolution of differential manifolds in a filtration, and can be viewed as extensions of earlier work on persistent homology in the cubic setting [298, 299].

Given any pair of manifolds $M_i \hookrightarrow M_j$ in the filtration aforementioned in the previous subsection, the k -th persistent Hodge Laplacian on the space of normal k -forms $\Omega^k(M_i)$ is given as follows:

$$\Delta_{i,j}^k = d_{i,j}^{k-1} \delta_{i,j}^k + \delta_i^{k+1} d_i^k,$$

where d_i^k and δ_i^k is the k -th differential and codifferential on normal forms on M_i , respectively, and $d_{i,j}^{k-1}$ is the persistent differential operator from normal $(k-1)$ -forms on M_j to normal k -forms on M_i , with $\delta_{i,j}^k$ its adjoint operator. The persistent Hodge Laplacians can similarly be defined on the space of tangential forms. However, they are equivalent due to the duality of the normal and tangential boundary conditions. When $i = j$, the persistent Hodge Laplacian reduces to the usual Hodge Laplacian restricted to normal forms. The kernel of $\Delta_{i,j}^k$ defines the persistent harmonic fields. It is isomorphic to the persistent relative de Rham cohomology, and its rank leads to the persistent Betti number.

We now turn to the discretization frameworks and briefly discuss the Lagrangian and Eulerian formulations using DEC, depending on how the underlying manifolds are discretized.

可以在预先设计的笛卡尔网格上进行, 因此从不同对象提取的特征对于机器学习任务是-致的且可比较的。此外, 由于顶点、边、面和单元在笛卡尔网格中是固定的, 数据结构和离散算子得到了显著简化。特别是, 通过调整水平集函数的等值, 可以轻松获得流形的演化。这些使得在欧拉表述中计算拉普拉斯算子更加高效。该框架在同一篇论文中通过将其集成到机器学习模型中以预测蛋白质-配体结合亲和力得到了演示。

3 几何拓扑方法

本节重点介绍基于几何拓扑学, 特别是-组理论的方法, 用于分析嵌入三维空间中的曲线, 例如-组、链环、缠结和类链环。此类数据结构常见于化合物 [241]、DNA [13] 和蛋白质结构 [284] 中。组理论研究这些对象在环境同痕下的等价性, 非正式地说, 指的是不涉及切割和重新连接的连续形变。组理论中的关键工具是组不变量, 这些量在这些形变下保持不变, 在区分-组和链环方面起着至关重要的作用。例子包括交叉数 [1]、桥数 [265]、组群 [82]、琼斯多项式 [164]、亚历山大多项式 [7]、Khovanov 同调 [172] 和-组 Floer 同调 [238]。

-组理论已应用于多个领域, 如物理学 [237]、化学 [189] 和生物学 [223, 285?], 但其在实际应用中的成功在一定程度上受到限制。挑战之一在于, -组理论中的经典工具本质上是全局和拓扑的, 对局部几何和拓扑特征提供的洞察有限。这使得这些工具不足以分析生物结构 (如 DNA 和蛋白质), 这些结构通常具有复杂的局部结构和/或片段间的局部缠结。此外, 实际应用中遇到的许多数据集由开放曲线而非闭合环组成, 这使得经典-组不变量不适用。这些挑战凸显了需要开发更有效的工具来分析实际场景中的曲线。

受拓扑数据分析在各个领域成功的启发, 文献 [270] 引入了-组数据分析 (KDA) 的概念。该论文提出了一个称为多尺度-组环绕积分 (mGLI) 的框架, 这是首次将多尺度分析与-组理论工具相结合。继此工作之后, 又提出了其他几种几何拓扑框架, 以解决上述经典方法在分析三维空间中曲线时的局限性。这些包括局部化琼斯多项式 [277]、持久 Khovanov 同调 [268]、Khovanov 拉普拉斯算子以及 Khovanov 狄拉克算子 [162]。

接下来, 我们将探讨这些最近发展的几何拓扑方法。多尺度-组环绕积分 (mGLI) [270] 和局部化琼斯多项式 [277] 分别建立在-组环绕积分 [129] 和开放端-组图的琼斯多项式 [239] 的基础上, 它们是目前已知的仅有的两

3.3.1 Lagrangian formulation

In the Lagrangian formulation, the underlying manifolds are discretized as simplicial meshes, specifically triangular meshes in 2D or tetrahedral meshes in 3D. A discrete k -form is just a vector with each entry assigned to a k -simplex. For example, a discrete 0-form is a vector with values assigned to the vertices, while a 1-form has values corresponding to edges. The differential is a sparse matrix given as the transpose of the signed incidence matrix between simplices of consecutive dimensions. The codifferential is the product of the incident matrix and the discrete Hodge star operators, which takes the form of a diagonal matrix. For further details of these discrete operators, see [64, 324].

The Lagrangian formulation requires explicit tessellation of the domains, which relies on high-quality meshing tools such as CGAL [119], Gmsh [131], Pymesh [327], etc. This makes the implementation of persistent Hodge Laplacians challenging in this context, as generating consistent meshes throughout the filtration can be problematic, and the resulting discrete operators may not be sparse matrices. The multiscale version of Hodge Laplacians [64] in the Lagrangian formulation, in addition to its dependence on these meshing tools, has several limitations. It is sensitive and numerically inconsistent because of variations in tetrahedral mesh representations of a compact domain. The method is also computationally expensive, as it requires repeated tessellation of the domains to create the evolution of manifolds in a filtration. These limitations make the framework unsuitable for machine learning tasks, which require feature representation computed on the same scale for comparability, and also efficiency, which is essential for computing features for large datasets.

3.3.2 Eulerian formulation

The Eulerian formulation of persistent Hodge Laplacians [282] has been introduced with the adapted DEC to address the limitations in the Lagrangian case. A computational algorithm for persistent Hodge Laplacians is presented in the paper. The underlying manifold in this setting is modeled as a sub-level set of a level-set function on a standard Cartesian grid, i.e., a domain bounded by an isocurve or an isosurface of a level set function. The adapted DEC from simplicial meshes to regular grids follows a similar structure as in the Lagrangian case. Note that in the Eulerian formulation, adjusting the isovalues of a level-set function naturally leads to a nested sequence of grid complexes in a filtration, making the construction of discrete persistent Hodge Laplacians feasible.

The Eulerian representation offers several advantages: First, it eliminates the need for explicit tessellation, thereby removing the reliance on high-quality meshing tools. Second, the computation of Laplacians for all objects can be performed on a pre-designed Cartesian grid, the extracted features from different objects are thus consistent and comparable for machine learning tasks. In addition, as the vertices, edges, faces, and cells are fixed in a Cartesian grid, the data structure and the discrete operators are significantly simplified. In

种用于三维空间中开放曲线的单条或成对拓扑复杂度的连续度量。这些方法使得能够分析跨越多个空间尺度的局部和全局缠结特性。持久 Khovanov 同调 [268] 通过引入基于平滑变换的链环图滤过, 扩展了 Khovanov 同调, 揭示了传统不变量常常忽略的更精细的拓扑和几何特征。Khovanov 拉普拉斯算子和狄拉克算子 [162] 通过结合谱信息扩展了 Khovanov 同调, 提供了新的描述符, 这些描述符从同调中恢复已知的不变量, 同时编码了额外的非调和信息。

Shen 等人 [270] 提出了多尺度高斯环绕积分 (mGLI), 为分析开放和闭合曲线中片段的局部结构、连通性和缠结提供了一种多尺度方法。这些局部特征对于理解数据 (如化学化合物和生物分子) 的物理性质和功能至关重要。尽管是局部方法, 但当应用足够大的尺度时, 该方法也能恢复全局拓扑特征。该框架建立在高斯环绕积分的基础上, 高斯环绕积分是一个双线积分, 可提供两条曲线的环绕数, 并已成功应用于理解聚合物 [241] 和蛋白质 [18, 19, 240]。

该方法需要对曲线进行分割, 将给定的一对曲线划分为小片段。这些曲线片段之间的成对高斯环绕积分随后产生一个分割矩阵, 其条目编码了成对环绕信息。对于多尺度分析, 随后会纳入这些条目的权重, 这些权重由距离每个片段特定距离范围内的指示函数定义。通过改变距离范围, 此过程生成一系列分割矩阵, 这些矩阵捕捉了不同空间尺度下的局部缠结。跨尺度的累积积分也为局部结构提供了额外的见解, 并在足够大的尺度上揭示了全局关系。对于一条曲线中的每个片段, 在每一距离范围内计算的其与另一曲线所有片段的局部环绕信息, 可通过统计方法用于生成该片段的多尺度特征。该构造也可应用于单条曲线, 通过考虑同一条曲线片段之间的局部环绕信息。

曲线的分割和距离范围对于确定高斯环绕积分在特征提取中的有效性至关重要。因此, 应根据具体应用仔细选择分割策略和距离范围。该框架已与机器学习模型集成, 以解决各种复杂的生物学问题, 包括蛋白质 B 因子预测、蛋白质-配体结合亲和力预测等。该模型始终优于传统的 TDA 模型, 并在某些任务中表现出卓越的性能 [270]。

通过将经典的高斯环绕积分与多尺度分析相结合, mGLI 为分析曲线型数据的局部和全局结构提供了一个强大的框架, 开辟了数据分析和纽结学习的新领域。该框架通过检查曲线片段的局部缠结, 解决了传统 TDA 技术的局限性, 并且可应用于开放曲线。此外, 它的计算效率高, 在个人

particular, the evolution of manifolds can be easily obtained by adjusting the isovalues of the level-set functions. These make the computation of Laplacians more efficient in the Eulerian formulation. The framework was demonstrated in the same paper by integrating it into machine learning models for predicting the protein-ligand binding affinities.

4 Geometric topology approaches

This section focuses on methods grounded in geometric topology, in particular knot theory, for analyzing curves embedded in 3-space, such as knots, links, tangles, and linkoids. Such data structures are often found in chemical compounds [241], DNA [13], and protein structures [284]. Knot theory studies these objects up to ambient isotopy, informally referring to a continuous deformation that does not involve cutting and rejoining. The key tools in Knot theory are knot invariants, which are quantities that remain unchanged under these deformations and play an essential role in distinguishing knots and links. Examples include the crossing number [1], the bridge number [265], the knot group [82], the Jones polynomial [164], the Alexander polynomial [7], the Khovanov homology [172], and the knot Floer homology [238].

Knot theory has been applied in various fields, such as physics [237], chemistry [189], and biology [223, 264, 285], but its success in practical applications has been somewhat limited. One of the challenges lies in the fact that classical tools in knot theory are inherently global and topological, offering little insight into local geometric and topological features. This makes these tools insufficient for analyzing biological structures, such as DNA and proteins, which often carry intricate local structures and/or local entanglement between segments. In addition, many datasets encountered in real applications consist of open curves rather than closed loops, which makes classical knot invariants inapplicable. These challenges highlight the need to develop more effective tools for analyzing curves in practical settings.

Motivated by the success of topological data analysis for various domains, the concept of knot data analysis (KDA) was introduced in [270]. The paper proposes a framework called multiscale Gauss linking integral (mGLI), which is the first to combine multiscale analysis with tools in knot theory. Following the work, several other geometric topological frameworks have been proposed to address the aforementioned limitations of classical methods in analyzing curves in 3-space. These include a localized Jones polynomial [277], persistent Khovanov homology [268], and Khovanov Laplacians, and Khovanov Dirac operator [162].

In the following, we explore these recently developed geometric topology approaches. The multiscale Gauss linking integral (mGLI) [270] and the localized Jones polynomials [277] build upon the Gauss linking integral [129] and the Jones polynomial of open-ended knot diagrams [239], respectively, which are currently the only two known continuous measures of single or pairwise topological complexity for open curves in 3-space. These approaches enable

计算机上仅需几分钟即可生成特征,这使得分析大型数据集变得切实可行。该方法显示出良好的前景,在各个领域都具有强大的实际应用潜力。

3.1 局部化琼斯多项式

局部化琼斯多项式由 Song 等人 [277] 引入,用于分析不相交开曲线或闭曲线集合的局部和全局纠缠性质。该论文提出了两种模型,称为多尺度琼斯多项式和持久琼斯多项式,它们通过结合空间多尺度分析,扩展了 [20] 中提出的琼斯多项式。[20] 中的方法推广了用于闭曲线的传统琼斯多项式 [164] 以及 [239] 中用于开曲线的方法,为三维空间中开曲线集合的链环复杂性提供了一种连续度量。

同一篇论文中分析了两种提出的局部化琼斯多项式的稳定性,确保了它们在实际应用中的鲁棒性。这些模型也已应用于蛋白质结构分析,提供了超越原子位置本身的纠缠洞察。这些框架展示了在多种实际应用中发挥作用的潜力。

多尺度琼斯多项式

多尺度琼斯多项式 [277] 遵循了与多尺度高斯链环积分 [270] 类似的策略。给定一个具有分割的不相交开曲线或闭曲线集合,该方法为每个曲线段计算一个琼斯多项式,该多项式对应的集合由该曲线段以及所有位于指定距离范围内的其他曲线段组成。通过改变距离区间,该过程为每个段生成一个琼斯多项式向量,从而捕捉多个空间尺度上的拓扑纠缠。通过考虑所有曲线段,该方法最终得到一个琼斯多项式矩阵,编码了曲线集合的局部和全局纠缠性质。在实际应用中,这些琼斯多项式会在一个固定参数处求值,从而得到一个适用于下游分析的实特征矩阵。与多尺度高斯链环积分 [270] 类似,这种特征化依赖于曲线集合分割方式的选择以及距离范围的选择,这两者都可以针对特定应用进行适当调整。

持久琼斯多项式

持久琼斯多项式是通过考虑一个具有分割的不相交开曲线或闭曲线集合的曲线段之间距离矩阵而提出的,该矩阵进而诱导出一个 Vietoris-Rips 复形(也可以使用其他类型的单纯复形)的滤过,其顶点对应于曲线段 [277]。滤过中的每个单形被赋予一个权重,该权重由构成该单形的曲线段集合的琼斯多项式给出,从而产生一个分割的多项式加权滤过。该滤过产生了加权条形码,编码了单形的产生和消亡及其相关的多项式权重。在实

the analysis of local and global entanglement properties across multiple spatial scales. The persistent Khovanov homology [268] extends the Khovanov homology by introducing filtrations of link diagrams based on smoothing transformations, revealing finer topological and geometric features that are often overlooked by traditional invariants. The Khovanov Laplacians and Dirac operator [162] extend the Khovanov homology by incorporating spectral information, offering new descriptors that recover known invariants from homology while encoding additional non-harmonic information.

4.1 Multiscale Gauss linking integral

Shen et al. [270] proposed the multiscale Gauss linking integral (mGLI), offering a multiscale approach for analyzing the local structure, connectivity, and entanglement of segments in both open and closed curves. These local features are crucial for understanding the physical properties and functions of data, such as chemical compounds and biological molecules. Although local, the method can also recover the global topological features when sufficiently large scales are applied. The framework builds on the Gauss linking integral, which is a double line integral that provides the linking number of two curves and has been successfully applied in understanding polymers [241] and proteins [18, 19, 240].

The method requires curve segmentation to partition curves into small segments for a given pair of curves. The pairwise Gauss linking integrals between these curve segments then result in a segmentation matrix whose entries encode the pairwise linking information. For multiscale analysis, weights on these entries, defined by indicator functions in a specific distance range from each segment, are then incorporated. By varying the distance range, this process generates a collection of segmentation matrices that capture the local entanglement across various spatial scales. The cumulative integrals across scales also provide additional insights into local structures, and reveal the global relationships at sufficiently large scales. For each segment in one curve, the local linking information with all segments of the other curve, calculated within each distance range, can be used to generate the multiscale feature of that segment through statistical methods. The construction can also be applied to a single curve by considering the local linking information between segments of the same curve.

The segmentation of curves and the distance range are crucial in determining the effectiveness of Gauss linking integrals for feature extraction. Therefore, the segmentation strategy and the distance range should be chosen carefully based on specific applications. The framework has been integrated with machine learning models to address a variety of complex biological problems, including protein *B*-factor prediction, protein-ligand binding affinity prediction, and others. The model consistently outperforms the traditional TDA models and demonstrates superior performance in certain tasks [270].

By combining the classical Gauss linking integral with multiscale analysis, mGLI offers a powerful framework for analyzing the local and global structures

实际应用中, 琼斯多项式在一个固定参数值处求值。通过引入滤过, 持久琼斯多项式编码了分割内曲线段之间的局部和全局相互作用。与专注于点云数据中孔洞行为的持久同调相比, 持久琼斯多项式捕捉的是 3 维空间中曲线的纠缠。

3.2 持久 Khovanov 同调

Khovanov 同调 [172] 是纽结理论中最重大的进展之一, 它提供了一个强大的链环不变量, 对 Jones 多项式 [164] 进行了范畴化。它揭示了更丰富的信息, 例如挠元。它还能检测平凡结, 这在一般情况下是一个非常困难的问题, 而 Jones 多项式是否能做到这一点仍是一个悬而未决的问题。为了将持久性分析融入曲线型数据中, Shen 等人 [268] 通过定义从链环图解的交叉点平滑变换中获得的链环滤链, 提出了纽结与链环的持久 Khovanov 同调 (或称演化 Khovanov 同调)。这项工作解决了经典纽结理论主要关注全局拓扑性质而可能忽略曲线型数据重要局部特征的局限性。它能够在滤链的各个尺度上对纽结与链环的拓扑结构和几何形状进行更精细的表征。

从概念上讲, 纽结与链环的 Khovanov 同调与单纯复形的经典同调有相似之处, 尽管它们是在不同的背景下发展起来的。两者都是从链复形构造而来, 链复形由一系列通过微分或边界映射连接起来的阿贝尔群序列组成, 这些映射应用两次后为零。一个关键区别在于分次: Khovanov 同调是双分次的, 既有同调分次又有量子分次, 而单纯同调只有单一的同调分次。Khovanov 同调中的分次 Betti 数反映了分次维数, 相比之下, 单纯同调中的经典 Betti 数则计算每个维数下的“洞”的数量。

该论文提出了两种构建链环图解滤链的方法: 基于距离的滤链和解链滤链。其思想是生成一个交叉点的顺序, 这些交叉点通过特定的平滑操作逐步消解, 最终得到一个由不相交圆组成的简单链环图解。这个过程提供了一个滤链。基于距离的滤链考虑以所有交叉点为中心、由单一半径参数确定的圆盘, 如果一个交叉点的圆盘不与任何其他圆盘相交, 则该交叉点被视为孤立的。从一个较大的初始半径开始, 随着半径逐渐减小, 标记孤立的交叉点从而获得顺序。对于解链滤链, 顺序的选择是从一个初始交叉点开始, 沿着曲线的特定方向移动, 标记链环的交叉点。

即使对于平凡结和非打结的链环, 持久 Khovanov 同调也能捕捉到非平凡的演化信息。这些方法的有效性已在同一篇论文中对 SARS-CoV-2 移码假结的分析中得到验证。后来, Liu 等人 [197] 从理论上提出了缠结的持

of curve-type data, opening a new area in data analysis and knot learning. The framework addresses the limitations of traditional TDA techniques by examining the local entanglement of curve segments, and it can be applied to open curves. In addition, it is computationally efficient, requiring only a few minutes on a personal computer to generate the features, which makes it practical for analyzing large datasets. The method shows promise and holds strong potential for practical applications across various fields.

4.2 Localized Jones polynomials

The localized Jones polynomials were introduced by Song et al. [277] for analyzing the local and global entanglement properties of collections of disjoint open or closed curves. The paper proposes two models, termed the multiscale Jones polynomial and the persistent Jones polynomial, which extend the Jones polynomial proposed in [20] by incorporating spatial multiscale analysis. The approach in [20] generalizes the traditional Jones polynomial [164] for closed curves and the one in [239] for an open curve, offering a continuous measure of linking complexity for collections of open curves in 3-space.

The stability of both proposed localized Jones polynomials has been analyzed in the same paper, ensuring their robustness for practical use. These models have also been applied to protein structure analysis, providing insight into the entanglement that goes beyond atomic positions alone. The frameworks demonstrate potential to benefit a range of practical applications.

Multiscale Jones polynomial

The multiscale Jones polynomial [277] follows a similar strategy used for the multiscale Gauss linking integral [270]. Given a collection of disjoint open or closed curves with a segmentation, the method computes, for each curve segment, the Jones polynomial of the set consisting of that curve segment together with all curve segments that lie within a specified distance range from that curve segment. By varying the distance interval, the process generates a vector of Jones polynomials for each segment, which captures the topological entanglement at multiple spacial scales. By considering all curve segments, the method finally leads to a matrix of Jones polynomials, encoding both the local and global entanglement properties for the collection of curves. These Jones polynomials are then evaluated at a fixed parameter for real applications, resulting in a real characteristic matrix suitable for downstream analysis. Analogous to the multiscale Gauss linking integral [270], the featureization depends on the choice of segmentation of the collections of curves, and the selection of distance ranges, both of which can be tuned appropriately for specific applications.

Persistent Jones polynomial

The persistent Jones polynomial was proposed by considering the pairwise distance matrix of curve segments for a collection of disjoint open or closed curves with a segmentation, which then induces a filtration of Vietoris-Rips complexes

久 Khovanov 同调, 侧重于表征曲线型数据中的局部拓扑特征。通过实现对多尺度拓扑不变量的持久性分析, 这些框架提供了强大的工具, 能够捕捉曲线型数据更精细的拓扑细节或局部结构, 为数据科学提供了有前景的几何拓扑方法。

3.3 Khovanov 拉普拉斯算子与 Khovanov 狄拉克算子

受组合拉普拉斯算子、霍奇拉普拉斯算子、狄拉克算子在数据科学中的成功及其与 (上) 同调的联系所启发, Jones 和 Wei [162] 为纽结和链环引入了 Khovanov 拉普拉斯算子和 Khovanov 狄拉克算子, 通过将谱分析纳入纽结与链环的研究, 扩展了 Khovanov 同调。这些算子的调和谱可以恢复 Khovanov 同调中的拓扑不变量, 而非调和谱则揭示了仅通过同调无法获得的额外信息。该框架也可被视为与链环图相关联的有向图上的拉普拉斯算子 [273] 的扩展, 从而在图拉普拉斯算子与纽结的关键性质之间建立了联系。这些图拉普拉斯算子自然地引出了链环的 Seifert 矩阵、Alexander 矩阵和 Goeritz 矩阵, 以及从它们导出的不变量, 包括 Alexander 多项式和 ω -符号差。

Khovanov 拉普拉斯算子与 Khovanov 狄拉克算子的构造, 与单纯复形上的组合拉普拉斯算子和狄拉克算子的构造相类似。由于 Khovanov 同调中的微分仅将同调度增加 1 而保持量子度不变, 我们可以只限制在具有固定量子度的链复形上。该论文通过假设链群的基元素是正交归一的, 从而在其上定义了一个内积, 这进而诱导了微分的唯一伴随算子。随后, Khovanov 拉普拉斯算子使用与组合拉普拉斯算子相同的公式定义, 而狄拉克算子则遵循相同的矩阵结构。与组合情形类似, Khovanov 拉普拉斯算子的核对应于 Khovanov 同调群。

如论文 [162] 所述, Khovanov 拉普拉斯算子和狄拉克算子的构造也可以推广到其他纽结同调理论, 例如纽结 Floer 同调 [238], 该理论将 Alexander 多项式范畴化。这引发了从 Floer 纽结同调导出的拉普拉斯算子与上述有向图上的拉普拉斯算子 [273] 之间可能存在的联系。此外, 类比于组合情形和微分流形情形, 人们可以定义持久 Khovanov 拉普拉斯算子和持久狄拉克算子, 从而进一步推广持久 Khovanov 同调 [268] 的框架。

(other types of simplicial complexes may also be used), with the vertices corresponding to the curve segments [277]. Each facet in the filtration is assigned a weight given by the Jones polynomial of the set of curve segments forming that facet, leading to a weighted filtration of polynomials of the segmentation. This filtration gives rise to weighted barcodes, which encode the birth and death of facets along with their associated polynomial weights. For practical applications, the Jones polynomials are evaluated at a fixed parameter value. By introducing a filtration, the persistent Jones polynomial encodes the local and global interactions among curve segments within a segmentation. In contrast to the persistent homology for point cloud data, which focuses on the behavior of holes, the persistent Jones polynomial captures the entanglement of curves in 3-space.

4.3 Persistent Khovanov homology

Khovanov homology [172] is one of the most significant advances in knot theory, providing a powerful link invariant that categorifies the Jones polynomial [164]. It reveals richer information such as torsion. It can also detect the unknot, which is a very hard problem in general, while it remains an open question whether the Jones polynomial can do so. To incorporate the persistence into the analysis of curve-type data, Shen et al. [268] proposed the persistent Khovanov homology (or evolutionary Khovanov homology) of knots and links by defining filtrations of links obtained through the smoothing transformations of crossings in a link diagram. This work addresses the limitations that classical knot theory primarily focuses on global topological properties and may overlook the important local features of curve-type data. It enables a more refined characterization of the topological structures and geometric shapes of knots and links across various scales in a filtration.

Conceptually, Khovanov homology for knots and links shares similarities with the classical homology for simplicial complexes, although they are developed in different contexts. Both are constructed from chain complexes, which consist of sequences of abelian groups connected by differentials or boundary maps that vanish when applied twice. A key difference lies in the grading, where Khovanov homology is bigraded, with both a homological grading and a quantum grading, whereas the simplicial homology has only a single homological grading. The graded Betti numbers in Khovanov homology reflect the graded dimensions, compared to the classical Betti numbers in simplicial homology, which count the number of holes in each dimension.

The paper presents two approaches for constructing a filtration of link diagrams: distance-based filtration and unzipping filtration. The idea is to generate an order of crossings, which are gradually resolved by a specific smoothing, finally leading to a simple link diagram consisting of disjoint circles. This process provides a filtration. The distance-based filtration considers disks centered at all crossings determined by one single radius parameter, and a crossing is isolated if its disk does not intersect with any other disks. Starting with a large initial radius, the order is then obtained by labeling isolated

4 机器学习特征化

单纯复形是拓扑数据分析中用于建模点云数据最广泛使用的工具，这归功于其概念简单性和计算效率。然而，它们并非对所有数据类型都是普遍最优的。例如，在网络数据中，合著网络更适合建模为超图而非单纯复形。这是因为三位作者之间存在联合发表，并不必然意味着每对作者都曾独立合作过，而这一条件是单纯复形构造所固有的。类似地，流形上的数据，如图像，通常源自光滑结构，因此具有内在的微分性质和边界信息。这些特性无法被单纯复形的纯组合结构完全捕捉。相反，将数据建模为微分流形提供了一个更合适的框架，以纳入光滑性和边界信息。因此，在 TDA 中选择用于建模数据的数学结构，应基于数据的内在本质和特定分析任务的目标。此外，不同的建模选择自然会导致不同类别的拓扑工具和表示。

在本节中，我们首先对 TDA 中常见的各种数据类型进行分类，包括点云数据、距离矩阵数据、网络数据、流形上的数据、序列数据、纽结与链环数据以及带有额外非几何信息的数据。对于每个类别，我们讨论合适的数学模型及其含义。然后，我们介绍从这些数据结构中提取拓扑表示的方法。最后，我们描述将这些拓扑表示转化为适用于机器学习应用的特征的方法。我们主要关注向量化技术，这些技术将拓扑表示转换为固定大小的特征向量。其他策略，如基于核的方法，也会简要讨论。

4.1 输入数据类型与拓扑建模的选择

4.1.1 点云数据

点云数据是众多领域中最常遇到的数据格式之一。例子包括生物分子（如蛋白质、DNA 和 RNA）中的原子，以及各种网络（如社交网络、交通网络和分子相互作用网络）中的节点。使用有向图、单纯复形和超图等数学对象对点云数据进行建模，使得能够应用先进的拓扑工具，包括持久同调和持久组合拉普拉斯算子，进行全面的数据分析。为点云数据选择合适的数学模型取决于数据的性质以及可用的额外信息。

当数据点没有任何额外信息或需要被平等对待时，Vietoris-Rips 复形和 Alpha 复形是合适的；而对于带有额外标签信息（如类型、质量或颜色）的数据点，加权单纯复形或层则更为适合。有向图或超有向图适用于表示具有不对称性或定向关系的数据，例如基因表达数据集，其中方向性信息

crossings as the radius gradually decreases. For the unzipping filtration, the order is chosen by starting with an initial crossing and labeling the crossings of the link by moving in a specific direction along the curve.

The persistent Khovanov can capture nontrivial evolutionary information even for trivial knots and unknotted links. The effectiveness of the methods has been demonstrated in the analysis of the SARS-CoV-2 frameshifting pseudo-knot in the same paper. Later, Liu et al. [197] proposed theoretically the persistent Khovanov homology of tangles, focusing on characterizing local topological characteristics in curve-type data. By enabling a persistence analysis of topological invariants across multiple scales, these frameworks provide powerful tools capable of capturing finer topological details or local structures of curve-type data, offering promising geometric topology approaches in data science.

4.4 Khovanov Laplacians and Khovanov Dirac

Motivated by the success of combinatorial and Hodge Laplacians and Dirac in data science, and their connections to the (co)homology, Jones and Wei [162] introduced the Khovanov Laplacians and Khovanov Dirac for knots and links, extending the Khovanov homology by incorporating the spectral analysis in the study of knots and links. The harmonic spectra of these operators can recover the topological invariants from Khovanov homology, while the non-harmonic spectra reveal additional information not accessible through homology alone. The framework can also be seen as an extension of the Laplacians on directed graphs associated with link diagrams [273], which establishes a connection between graph Laplacians and key properties of knots. These graph Laplacians naturally lead to the Seifert matrix, the Alexander matrix, and the Goeritz matrix of a link, along with the invariants derived from them, including the Alexander polynomial and ω -signature.

The construction of the Khovanov Laplacians and Khovanov Dirac parallels that of the combinatorial Laplacians and the Dirac operator on simplicial complexes. As the differential in Khovanov homology increases only the homological grading by 1 while preserving the quantum grading, one can just restrict to chain complexes with fixed quantum grading. The paper defines an inner product on a chain group by assuming that its basis elements are orthonormal, which then induces a unique adjoint of the differential. The Khovanov Laplacians are subsequently defined using the same formula as for the combinatorial Laplacians, and the Dirac operator follows the same matrix structure. Analogous to the combinatorial case, the kernels of Khovanov Laplacians correspond to the Khovanov homology groups.

As noted in the paper [162], the construction of Khovanov Laplacians and the Dirac operator may also be extended to other knot homology theories, such as the knot Floer homology [238], which categorifies the Alexander polynomial. This raises possible connections between the Laplacians derived from the Floer knot homology and the Laplacians mentioned above on directed graphs [273]. In addition, in analogy with the combinatorial and differential

编码了基因调控。超图和超超图可以有效地处理具有复杂关系或不完整结构的数据点。

4.1.2 距离矩阵数据

对于距离矩阵数据, 必须强调距离度量构建的灵活性与多样性。虽然欧氏距离应用最为广泛, 但它并非在所有场景下都是最优选择。在许多情况下, 修改距离度量或采用其他定义方式, 能够更准确地提取与底层问题相关的拓扑特征。

交互距离 [38] 旨在捕捉两类不同对象之间的相互作用。以蛋白质-配体相互作用为例, 若直接对完整的蛋白质-配体结构应用欧氏距离构建单纯复形, 由于蛋白质内部或配体内部相互作用的贡献占主导地位, 可能会掩盖具有生物学意义的相互作用。为解决此问题, 交互距离对源自不同结构(即蛋白质与配体)的点对赋予原始欧氏距离, 而对同一结构内部的点对则赋予无穷大距离。这一策略确保所构建的单纯复形能专门聚焦于跨结构相互作用, 从而提供更具针对性的蛋白质-配体相互作用表征。

多层次距离 [40] 的提出是为了突出非共价原子相互作用, 例如氢键和范德华力。在此方法中, 键合原子对的距离被设为无穷大, 因为在大多数情况下键合原子比非键合原子更接近, 这使得我们可以专注于非键合、非共价的相互作用。尽管该度量可能违反三角不等式, 但它仍适用于构建 Vietoris-Rips 复形, 因为后者仅需成对距离。此策略不仅能生成编码非共价相互作用的零维拓扑特征, 还能产生更高维的特征, 以捕捉同一分子不同构象之间的微小结构波动。

基于核函数的距离 [40] 被引入用于模拟原子相互作用强度, 因为这些强度通常与欧氏距离并非线性对应。该框架使用径向基函数的负值来定义基于相关函数的过滤矩阵。更强的相互作用对应更小的距离, 并在过滤过程中更早出现。类似地, 在基于柔性刚性指数的距离中, 采用了具有可调尺度参数的洛伦兹型核函数。通过调整该尺度参数, 可以在不同空间分辨率下捕捉结构模式。

基于 KNN 的距离策略 [182] 根据点对之间的相互最近邻关系来分配距离。具体而言, 对于点对 (a, b) , 若 a 是 b 的第 s 个最近邻, 且 b 是 a 的第 t 个最近邻, 则 a 与 b 之间的距离定义为两者中的较小值, 即 $\min(s, t)$ 。随后可从点云得到的距离矩阵构建标准的 Vietoris-Rips 或 Alpha 过滤。该策略已被用于分析单细胞 RNA 测序数据 [80]。

manifold settings, one may define the persistent Khovanov Laplacians and the persistent Dirac, further generalizing the framework of persistent Khovanov homology [268].

5 Machine learning featurization

Simplicial complexes are the most widely used tools for modeling point cloud data in topological data analysis, owing to their conceptual simplicity and computational efficiency. However, they are not universally optimal for all data types. For example, in network data, a coauthorship network is more appropriately modeled as a hypergraph rather than a simplicial complex. This is because the presence of a joint publication among three authors does not necessarily imply that each pair of authors has collaborated independently, a condition inherent in the simplicial complex construction. Similarly, data on manifolds such as images often arise from smooth structures and therefore possess intrinsic differential properties and boundary information. These characteristics cannot be fully captured by the purely combinatorial structure of simplicial complexes. Instead, modeling the data as differential manifolds provides a more suitable framework for incorporating smoothness and boundary information. Thus, the selection of a mathematical structure for modeling data in TDA should be guided by the intrinsic nature of the data and the objectives of the specific analysis task. Moreover, different modeling choices naturally lead to different classes of topological tools and representations.

In this section, we begin by categorizing various data types commonly encountered in TDA, including point-cloud data, distance matrix data, network data, data on manifolds, sequential data, knots and links data, and data with additional non-geometric information. For each category, we discuss appropriate mathematical models and their implications. We then present methods for extracting topological representations from these data structures. Finally, we describe approaches for transforming these topological representations into features suitable for machine learning applications. We mainly focus on vectorization techniques, which convert topological representations into fixed-size feature vectors. Other strategies, such as kernel-based methods, are also briefly discussed.

5.1 Input data types and the choice of topological modeling

5.1.1 Point-cloud data

Point-cloud data is one of the most commonly encountered data formats in a wide range of fields. Examples include atoms in biomolecules such as proteins, DNA, and RNA, as well as nodes in various networks such as social networks, transportation networks, and molecular interaction networks. Modeling point cloud data using mathematical objects such as digraphs, simplicial complexes, and hypergraphs enables the application of advanced topological

4.1.3 网络数据

网络数据由节点和边组成，用于表示实体之间的交互或关系。此类数据广泛出现在众多领域中。例如，在分子网络中，节点对应原子，边对应化学键；在社交网络中，节点代表个体，边表示社交连接；在贸易网络中，节点是国家，边则指示贸易关系。从数学角度，这些网络可以建模为图，从而能够应用图论工具进行成对交互分析。拓扑工具的选择取决于网络的性质以及具体的应用场景。

（有向）无向网络可以自然地建模为一维（有序）单纯复形，其中节点表示 0-单形，边表示 1-单形。为了分析超越成对连接的高阶交互，可以将图扩展为更高维的单纯复形。现有多构造方法：旗复形（或团复形）[163]，它为每个完全子图包含一个单形；邻域复形 [200]，它编码共享的邻域结构；以及 Hom 复形 [204]，可以将其类比解释为卷积神经网络中的卷积核。此外，超图和其他广义数学对象也可直接用于建模高阶交互 [136, 206]。

4.1.4 流形上的数据

流形上数据的常见示例包括图像，例如来自 MRI 和 CT 扫描的 3D 体数据或 2D 切片，以及等值面，例如流体动力学中表示均匀热分布的等温面和分子可视化中模拟分子轨道的电子密度等值面。尽管它们在原始形式上是离散的，但很自然地将其视为流形上的数据，因为它们通常是从连续或光滑结构中采样得到的。在以往的研究中，对此类数据（尤其是图像数据）的拓扑分析通常使用基于点云的技术进行。例如，医学成像中的一种常见方法是从图像中感兴趣区域选取点，然后从这些选取的点构建单纯复形以进行拓扑分析 [275]。此外，立方复形结构也被用于建模图像以进行拓扑分析 [299]。从数学上讲，单纯复形和立方复形都是组合对象，因此它们能够捕捉流形上数据产生的拓扑结构。然而，由于其固有的组合性质，这些表示忽略了底层流形的微分结构和边界条件，从而限制了它们保留此类数据内在微分性质的能力。

因此，将这些数据建模为微分流形提供了一个更合适的框架，以捕捉其内在的微分性质。在此框架下，可以运用微分拓扑或微分几何中的工具，例如微分形式、向量场和微分算子，以促进数据的定量分析。此外，这些数据（例如图像）本身具有将物体与背景分离的边界，使得边界条件至关重要。这些边界条件不仅施加了约束以控制系统在边界附近的行为，还确保了边值问题解的存在性和唯一性。此外，边界条件修改了微分算子，例

tools, including persistent homology and persistent combinatorial Laplacians, for comprehensive data analysis. The selection of an appropriate mathematical model for point cloud data depends on the nature and available additional information of the data.

The Vietoris-Rips complex and Alpha complex are suitable when data points are given without any additional information or when they need to be treated equally, while the weighted simplicial complex or sheaf is better suited for data points that carry additional label information, such as type, mass, or color. The digraphs or hyperdigraphs are appropriate for representing data with asymmetry or directed relations, such as gene expression datasets, where directional information encodes gene regulation. Hypergraphs and super-hypergraphs can be effective for handling data points with complex relationships or incomplete structures.

5.1.2 Distance matrix data

For distance matrix data, it is important to emphasize the flexibility and diversity in the construction of distance metrics. While the Euclidean distance is the most widely used, it is not universally optimal for all applications. In many cases, modifying the distance metric or adopting alternative definitions can lead to a more accurate extraction of topological features relevant to the underlying problem.

The interactive distance [38] is designed to capture the interactions between two distinct types of objects. Taking protein-ligand interactions as an example, directly applying Euclidean distance to construct simplicial complexes from the entire protein-ligand structure can obscure the biologically meaningful interactions due to overwhelming contributions from intra-protein or intra-ligand interactions. To address this, the interactive distance assigns the original Euclidean distance to point pairs that originate from different structures (i.e., protein and ligand), while assigning an infinite distance to point pairs within the same structure. This strategy ensures that the resulting simplicial complex focuses specifically on cross-structure interactions, thereby offering a more targeted characterization of the protein-ligand interactions.

The multi-level distance [40] is proposed to highlight noncovalent atomic interactions, such as hydrogen bonds and van der Waals forces. In this approach, bonded atom pairs are assigned infinite distances since bonded atoms stay closer than non-bonded ones in most cases, allowing us to focus on non-bonded, noncovalent interactions. Although this metric may violate the triangle inequality, it remains valid for constructing Vietoris-Rips complexes, where only pairwise distances are required. This strategy leads to not only zero-dimensional topological features encoding noncovalent interactions but also higher-dimensional features that capture small structural fluctuations among different conformations of the same molecule.

The kernel function-based distance [40] was introduced to model atomic interaction strengths, as they often do not align linearly with the Euclidean distance. The framework uses the negative of a radial basis function to define

如拉普拉斯算子和狄拉克算子,使其不同于在闭流形(无边界)上定义的对算子。值得注意的是,微分流形上的霍奇拉普拉斯算子与定义在单纯复形上的组合拉普拉斯算子在根本上有所不同 [260]。基于微分拓扑方法的有效性已在多种应用中得到证明,例如单细胞 RNA 速度分析中的霍奇分解 [281]、药物设计与发现中的 PHL [282],以及医学图像分析中的 MTDL [207]。

基于这一视角,近期工作已将拓扑和几何先验整合到深度学习架构中。Love 等人引入的拓扑卷积神经网络(TCNN) [209] 用反映图像和视频数据结构的流形参数化滤波器替代了标准卷积层。这种方法保留了拓扑关系,并利用了跨层的几何定位,从而实现了更快的学习、更好的泛化能力和更高的可解释性。TCNN 也能自然地扩展到 3D 数据,凸显了拓扑在深度学习中的更广泛潜力。

4.1.5 序列数据

序列数据广泛出现在多个领域,例如生物学中的蛋白质和 DNA 序列、自然语言处理中的词序列、时间序列分析中的股票价格时序波动以及视频处理中的帧序列。针对拓扑序列分析(TSA),已发展出多种方法。TSA 的一种策略是直接计算成对距离矩阵 [54],然后通过改变距离参数来构建 Vietoris-Rips 滤过。对于时间序列数据,一种方法涉及将曲线分割成短片段,并为每个片段构建一个 Vietoris-Rips 滤过 [326]。还有一些 TSA 方法将序列转换为点云,从而能够直接使用针对点云数据的成熟技术。这些策略包括 Takens 延迟嵌入、滑动窗口方法以及 k -mer 拓扑方法。

在 Takens 延迟嵌入 [288] 中,序列 $x_1, x_2, \dots$ 使用两个参数转换为点:嵌入维度 d 和延迟参数 τ 。例如,当 $d = 2$ 时,结果点云中的每个点形式为 $(x_i, x_{i+\tau})$ 。滑动窗口方法类似地使用两个参数:点维度参数 M 和窗口参数 τ 。例如,如果 $M = 3$,则嵌入空间中的一个点构造为 $(x_i, x_{i+\tau}, x_{i+2\tau})$ 。滑动窗口方法还有其他变体,例如 SW1PerS [246] 和环形滑动窗口嵌入 [245]。

在 k -mer 拓扑方法 [157] 中,参数 k 指定了沿序列滑动的窗口大小,生成一组 k -mer (长度为 k 的子序列),这些 k -mer 被视为一个点云。然后应用元素特定策略来强调特定类型的点。计算这些点之间的成对距离以构建距离矩阵,该矩阵作为后续使用持久同调或持久拉普拉斯等方法进行拓扑分析的基础。

a correlation function-based filtration matrix. Stronger interactions correspond to smaller distances and appear earlier in the filtration process. Similarly, in the flexibility rigidity index-based distance, a Lorentz-type kernel with a tunable scale parameter is employed. By adjusting this scale parameter, one can capture structural patterns at different spatial resolutions.

The KNN-based distance strategy [182] assigns distances between pairs of points based on their mutual nearest-neighbor relationships. Specifically, for a pair of points (a, b) , if a is the s -th nearest neighbor of b , and b is the t -th nearest neighbor of a , then the distance between a and b is defined as the smaller of the two values, i.e., $\min(s, t)$. The standard Vietoris-Rips or Alpha filtration can then be constructed from the resulting distance matrix of the point cloud. This strategy has been used to analyze single-cell RNA sequencing data [80].

5.1.3 Network data

Network data consists of nodes and edges that represent interactions or relationships among entities. Such data arises in a wide range of domains. For example, in molecular networks, nodes correspond to atoms and edges to chemical bonds, in social networks, nodes represent individuals and edges denote social connections, and in trade networks, nodes are countries while edges indicate trading relationships. Mathematically, these networks can be modeled as graphs, enabling the application of tools from graph theory for pairwise interaction analysis. The choice of topological tools depends on the nature of the network and the specific application context.

The (directed) undirected network can be naturally modeled as a one-dimensional (ordered) simplicial complex, where nodes represent 0-simplices and edges represent 1-simplices. To analyze high-order interactions beyond pairwise connections, one can extend the graph to a higher-dimensional simplicial complex. Several constructions are available: the flag complex (or clique complex) [163], which includes a simplex for each complete subgraph, the neighborhood complex [200], which encodes shared neighborhood structures, and the Hom complex [204], which can be interpreted analogously to convolutional kernels in convolutional neural networks. In addition, hypergraphs and other generalized mathematical objects can be employed to model high-order interactions directly [136, 206].

5.1.4 Data on manifolds

Common examples for data on manifolds include images, such as 3D volumes or 2D slices from MRI and CT scans, as well as isosurfaces, such as isothermal surfaces representing a uniform heat distribution in fluid dynamics and electron density isosurfaces modeling molecular orbitals in molecular visualization. Although they are discrete in their raw form, it is natural to treat them as data on manifolds, as they are often sampled from continuous or smooth structures. In previous studies, topological analysis of such data, particularly image data, has often been conducted using point cloud-based

已有研究提出将 Delta 复形方法用于 TSA [198]。其核心思想是将序列的片段视为单纯形，这些单纯形作为 Δ -复形的构建模块。由于 Δ -复形允许重复元素，它们与生物序列的特性相符。与 k -mer 拓扑不同，基于 Δ -复形的 TSA 方法并不系统分析序列中的 k -mer。这些方法最终利用持久同调或持久拉普拉斯，并且对于大型基因组序列，比早期的 k -mer 拓扑方法更快。

TDL 的最新进展引入了整合序列数据中时序和拓扑结构的新方法，特别是在神经尖峰解码的背景下 [226]。这些方法实现了端到端学习，其中拓扑结构是可学习的模型组件 [170, 243]。在许多情况下，数据由时序有序的事件组成，例如神经尖峰，其中每个观测都参与构建一个捕获实体间多路关系（例如，在短时间窗口内一起发放的神经元）的单纯复形。这些复形编码了超越简单成对连接的高阶相互作用，提供了更丰富的结构表示。相较于关注单个时间点或固定大小的窗口，一些拓扑深度学习方法追踪这些单纯复形的时间演化，从而产生一系列拓扑状态。在诸如尖峰解码之类的事件驱动场景中，这种演化通常通过离散时间或状态空间过程来建模。其他方法，[117, 228]，则利用连续时间动力学和偏微分方程来建模拓扑特征在复形上的演化。在这些情况下，每个单纯复形都可以通过为拓扑空间设计的神经架构（例如单纯消息传递网络 [30] 或胞腔复形编码器）嵌入为一个可学习的向量。

其结果是一个统一的、数据驱动的框架，能够同时捕捉时间动态和拓扑结构。在神经解码等应用中，这使得模型能够学习协同激活模式如何随时间变化，并与行为、刺激或潜在神经状态相关联。通过将序列表示为演化的拓扑对象，拓扑深度学习提供了一种灵活且富有表现力的方法来分析结构化时间序列数据，从而能够发现嵌入在时间和拓扑中的复杂模式。

4.1.6 纽结、链环、缠结与辫子

纽结、链环、缠结与辫子是基本的拓扑对象，可以自然地建模为嵌入三维空间中的曲线 [268, 270, 277]。在实际应用中，此类数据结构广泛存在于众多领域。例子包括绳索、鞋带、聚合物、DNA、RNA、核小体和染色体。

将这些对象表示为三维空间中的曲线具有优势，因为它允许使用纽结理论中的工具来分析复杂的结构关系。该方法已在多项研究中被证明是有效的。例如，聚合物链的缠结可以通过其打结和链接特性进行研究，高斯链接积分与聚合物材料的物理性质相关 [241]。此外，研究表明高斯链接积分

techniques. For example, a common approach in medical imaging is to select points from the region of interest in the image and then construct a simplicial complex from these selected points to perform topological analysis [275]. Additionally, cubical complex structures have also been employed to model images for topological analysis [299]. Mathematically, both simplicial complexes and cubical complexes are combinatorial objects, so they are capable of capturing the topological structures arising from the data on manifolds. However, due to their inherently combinatorial nature, these representations neglect the differential structure and boundary conditions of the underlying manifolds, thereby limiting their ability to preserve the differential properties intrinsic to such data.

Therefore, modeling these data as differential manifolds offers a more suitable framework for capturing their intrinsic differential properties. In this setting, tools from differential topology or differential geometry, such as differential forms, vector fields, and differential operators, can be employed to facilitate quantitative analysis of the data. In addition, these data, such as images, inherently have boundaries that separate objects from the background, making the boundary conditions essential. These boundary conditions not only impose constraints that govern the behavior of the system near the boundary but also ensure the existence and uniqueness of solutions in boundary value problems. Moreover, the boundary conditions modify the differential operators, such as the Laplacian operators and the Dirac operator, making them different from their counterparts defined on closed manifolds (without boundary). It is worth noting that the Hodge Laplacians on differential manifolds differ fundamentally from the combinatorial Laplacians defined on simplicial complexes [260]. The effectiveness of differential topology-based methods has been demonstrated in various applications, such as Hodge decomposition in single-cell RNA velocity analysis [281], PHL in drug design and discovery [282], and MTDL [207] in medical image analysis.

Building on this perspective, recent work has integrated topological and geometric priors into deep learning architectures. The Topological Convolutional Neural Network (TCNN) introduced by Love et al. [209] replaces standard convolutional layers with manifold-parameterized filters that reflect the structure of image and video data. This approach preserves topological relationships and leverages geometric localization across layers, leading to faster learning, better generalization, and improved interpretability. TCNNs also extend naturally to 3D data, highlighting the broader potential of topology in deep learning.

5.1.5 Sequential data

Sequential data arise in a wide range of fields, such as protein and DNA sequences in biology, word sequences in natural language processing, temporal stock price fluctuations in time series analysis, and frame sequences in video processing. Many approaches have been developed for topological sequence

与蛋白质缠结有关，特别是在理解蛋白质折叠动力学方面 [240]。它也被应用于 SARS-CoV-2 刺突蛋白的研究 [19]。基于纽结的拓扑框架的有效性也在蛋白质柔性分析、蛋白质-配体相互作用 [270]、蛋白质结构与功能 [277] 以及 SARS-CoV-2 移码假结 [268] 的研究中得到了证实。

目前已开发出诸如 KnotProt [160] 和 LinkProt [85] 等数据库来支持蛋白质纽结与链环的研究。Python 包 Topoly [86] 提供了计算拓扑多项式不变量的工具，用于区分纽结、滑结、链环和空间图。

4.1.7 具有额外非几何信息的数据

对于具有额外非几何信息的数据，我们指的是所有前述数据类型，这些数据在几何结构之外进一步丰富了与背景相关的属性。我们将此设作为一个独立的数据类型，因为此类数据在现实应用中很常见，并且补充信息通常在解决特定领域问题中起着关键作用。例如，蛋白质结构不仅包含原子的三维坐标（几何信息），还包括残基类型、部分电荷和溶剂可及性（非几何生物信息），这些信息对于蛋白质稳定性和蛋白质-配体相互作用至关重要。类似地，在网络数据中，除了节点和边的拓扑结构外，还可能具有节点关联的属性，例如社交网络中的用户兴趣，这对于聚类 and 链接预测任务至关重要。

元素特异性方法 [37, 38, 40] 是一个强大的框架，用于将不同的原子和分子相互作用系统地建模到一个统一的拓扑系统中。例如，给定一个分子原子的点云，可以根据原子的元素类型构建元素特异性单纯复形。这使我们能够专注于特定的原子相互作用，例如共价键、氢键、范德华力和其他非共价相互作用。元素特异性策略与标准持续同调的结合已在各种生物数据分析任务中展现出巨大潜力，例如蛋白质-配体结合亲和力预测、蛋白质-蛋白质相互作用分析和蛋白质稳定性分析。

另一种整合非几何信息的方法是将其编码为权重，并将这些权重整合到基础数学框架中。例如，在富集条形码方法 [39] 中，持续上同调是从蛋白质原子的基础单纯复形建模中计算出来的，额外的生化信息作为权重嵌入到上同调代表元中。这产生了一个富集的或彩色的持续条形码，它同时捕获了拓扑和化学性质。在持续层拉普拉斯方法 [310] 中，原子属性被编码到层构造中，特别是边界算子的构造中。因此，导出的持续层上同调和层拉普拉斯算子将自动携带这些额外信息。

analysis (TSA). One strategy for TSA is to directly compute a pairwise distance matrix [54], which can then be used to construct Vietoris-Rips filtration by varying the distance parameter. For time series data, an approach involves dividing curves into short segments and building a Vietoris-Rips filtration for each segment [326]. There are also TSA approaches to transform the sequence into a point cloud, thereby enabling the direct use of established techniques for point cloud data. These strategies include Takens' delay embedding, the sliding window method, and the k -mer topology approach.

In Takens' delay embedding [288], the sequence $x_1, x_2, \dots$ is transformed into points using two parameters: the embedding dimension d and the delay parameter τ . For instance, with $d = 2$, each point in the resulting point cloud takes the form $(x_i, x_{i+\tau})$. The sliding window method similarly employs two parameters: the point dimension parameter M and the window parameter τ . For example, if $M = 3$, a point in the embedded space is constructed as $(x_i, x_{i+\tau}, x_{i+2\tau})$. There are also other variations for the sliding window method, such as SW1PerS [246] and toroidal sliding window embeddings [245].

In the k -mer topology method [157], the parameter k specifies the window size used to slide along the sequence, generating a collection of k -mers (subsequences of length k), which are treated as a point cloud. The element-specific strategy is then applied to emphasize particular types of points. The pairwise distances between these points are computed to construct a distance matrix, which serves as the foundation for subsequent topological analysis using methods such as persistent homology or persistent Laplacians.

Delta complex approaches have been proposed for TSA [198]. The essential idea is to consider the segments of a sequence as simplices, which serve as the building blocks of a Δ -complex. Since Δ -complexes allow repeated elements, they align with the nature of biological sequences. Unlike k -mer topology, Δ -complex-based TSA methods do not systematically analyze k -mers in a sequence. These approaches eventually utilize persistent homology or persistent Laplacians and are faster than the early k -mer topology method for large genome sequences.

Recent developments in TDL introduce novel methods for integrating temporal and topological structures in sequential data, particularly in the context of neural spike decoding [226]. These approaches enable end-to-end learning where topology is a learnable model component [170, 243]. In many cases, the data consists of temporally ordered events, such as neural spikes, where each observation contributes to the construction of a simplicial complex that captures multi-way relationships among entities (e.g., neurons spiking together within a short time window). These complexes encode higher-order interactions beyond simple pairwise connections, providing a richer structural representation.

Rather than focusing on individual time points or fixed-size windows, some TDL methods track the temporal evolution of these simplicial complexes, producing a sequence of topological states. In event-based settings like spike decoding, this evolution is typically modeled through discrete-time

4.2 拓扑表示方法

4.2.1 持久同调/上同调的表示形式

在域系数下计算时,持久同调与持久上同调会产生相同的条形码 [90],这也是实际应用中最常见的选择。因此,为简化起见,本文仅关注持久同调的情况。一旦为给定数据集建立了合适的拓扑模型,就可以计算相应的持久同调以提取有意义的拓扑表示,例如单纯复形的持久单纯同调、超图的持久超图同调以及层结构的持久层同调。需要注意的是,持久同调可以通过多种形式进行可视化和编码,每种形式都提供了观察拓扑特征的不同视角。接下来,我们总结几种常见的持久同调表示格式。

持久条形码/图

持久同调的两种标准表示形式——持久条形码 [48] 和持久图 [112],都提供了直观的方式来呈现过滤过程中拓扑特征的“诞生”与“消亡”。持久条形码总结了不同维度下拓扑特征(如连通分量、孔洞、腔体)的生命周期。每个条形码由若干水平线段组成,其中每条线段的左端点代表拓扑特征的诞生时间,右端点则表示其消亡时间。持久图与条形码等价,但将相同的拓扑信息编码为二维散点图,其中 x 坐标表示诞生时间, y 坐标表示消亡时间。由于特征的消亡时间总是晚于其诞生时间,持久图中的所有点都位于对角线 $y = x$ 的上方。条形码中较长的线段通常对应显著的拓扑特征,而较短的线段则常被视为噪声。类似地,在持久图中,远离对角线的特征通常被认为是重要的,而靠近对角线的特征则常被当作噪声处理。然而,在考虑物理、化学和生物数据时,这一假设可能并不成立,因为短线段或靠近对角线的特征可能携带重要的物理信息,例如键长、苯环大小及其他性质 [313]。此外,对于多参数持久同调,还存在一种格拉斯曼流形持久图的构造方法 [141, 142]。持久条形码/图能有效捕捉数据的拓扑特征,并已应用于图像块 [2, 49]、手写字母 [75] 以及病毒演化中的异常事件 [54] 等研究中。

持久贝蒂数/贝蒂曲线

持久贝蒂数 [112] 和贝蒂曲线是贝蒂数的多尺度类比,用于统计在过滤过程中持续存在的拓扑特征数量。持久贝蒂数定义为持久同调群的秩,通常可视化为贝蒂曲线,后者提供了基于阶梯函数的持久同调表示。这种表示将持久条形码/图转化为函数表示。函数的向量空间结构使得加法、标量

or state-space processes. Other approaches, [117, 228], model the evolution of topological features over complexes using continuous-time dynamics and partial differential equations. In these cases, each simplicial complex can be embedded as a learnable vector using neural architectures designed for topological spaces, such as simplicial message passing networks [30] or cell complex encoders.

The result is a unified, data-driven framework that captures both temporal dynamics and topological structure. In applications like neural decoding, this enables the model to learn how patterns of co-activation change over time and relate to behavior, stimulus, or latent neural states. By representing sequences as evolving topological objects, TDL provides a flexible and expressive way to analyze structured time-series data, enabling the discovery of complex patterns embedded in both time and topology.

5.1.6 Knots, links, tangles, and braids

Knots, links, tangles, and braids are fundamental topological objects that can be naturally modeled as curves embedded in 3-space [268, 270, 277]. In practical applications, such data structures are commonly found in a wide range of fields. Examples include ropes, shoelaces, polymers, DNA, RNA, nucleosomes, and chromosomes.

The representation of these objects as curves in 3-space has advantages, as it allows the use of tools from knot theory for analyzing complex structure relationships. The approach has proven effective in various studies. For example, the entanglement of polymer chains can be studied through their knotting and linking properties, with the Gauss linking integral correlating with the physical properties of polymeric material [241]. In addition, studies have shown that the Gauss linking integral is related to protein entanglements, particularly in understanding protein folding kinetics [240]. It has also been applied to the study of SARS-CoV-2 spike protein [19]. The effectiveness of the knot-based topological frameworks has also been demonstrated in protein flexibility analysis and protein-ligand interactions [270], protein structure and functions [277], and SARS-CoV-2 frame-shifting pseudo-knot [268].

Databases such as KnotProt [160] and LinkProt [85] have been developed to support the study of knots and links of proteins. The Python package Topoly [86] provides tools for calculating topological polynomial invariants to distinguish knots, slipknots, links, and spatial graphs.

5.1.7 Data with additional non-geometric information

For data with additional non-geometric information, we refer to all the aforementioned data types that are further enriched with background-related attributes beyond geometric structures. We set this as a distinct data type because such data is common in real-world applications, and the supplementary information often plays a crucial role in addressing domain-specific problems. For example, the protein structures not only contain the 3D coordinates of atoms (geometric information) but also include the residue types,

乘法等基本运算成为可能,从而使贝蒂曲线适用于计算均值、方差等统计任务。这些运算在持久条形码/图上并无自然的定义。Betti 曲线简单且易于计算。此外,由于微小扰动通常会引入短暂的生命周期,这些生命周期对整体 Betti 数分布的影响极小,因此它们对噪声也具有鲁棒性。然而,Betti 曲线的一个缺点是其表达能力有限,因为它们仅包含拓扑特征的计数。此外,从图表到曲线的映射不是单射的 [68],这意味着不同的图表可能产生相同的曲线。

持久性景观

一种名为持久性景观的稳定持久同伦表示 [34, 35],提供了持久性图表更结构化、更具表达力的表示,同时保持了与统计和机器学习框架的兼容性。它们被引入以将持久性图表转换为分段线性函数的序列,这可以看作是持久性图表的水平版本。与仅追踪拓扑特征计数的 Betti 曲线相比,持久性景观编码了生灭对其持久性,并与持久性图表保持双射对应关系,这使得后者在需要详细拓扑分析时更受青睐。最近,它已被扩展到多参数情况下的多参数持久性景观 (MPPL) [296]。

持久性景观的一个关键特征是其层次结构,其中不同级别的函数捕获具有不同显著性的拓扑特征。具有最低生命周期下标的函数捕获最持久的拓扑特征,而具有较高生命周期下标的函数则捕获持久性较弱的特征。持久性景观已被用于研究蛋白质结合中的构象变化 [178]、微观结构分析 [109]、从时间序列构建的时间依赖功能网络 [279],以及图像分类任务和轨道分类 [173]。

持久性图像

持久性图像 [4] 作为一种替代的稳定表示被引入,它将持久性图表转换为固定大小的、基于网格的函数表示,这使得它们特别适用于卷积神经网络 (CNN)。持久性图像是通过计算持久性曲面在网格上的积分来构建的,该网格是通过离散化 $\mathbb{R}^2$ 的一个子域获得的。这里,持久性曲面被定义为以持久性图表中每个点为中心的高斯函数的加权和,其中加权函数控制每个点的显著性。在同一论文中,持久性图像已被证明相对于持久性图表之间的 1-Wasserstein 距离是稳定的。

持久性图像的一个关键优势是易于计算,并在三个选择上提供了灵活性:分辨率、分布(及其相关参数)以及加权函数。然而,这些选择是非规范的,这意味着没有指导原则来设置这些参数以在机器学习任务中获得

partial charges, and solvent accessibility (non-geometric biological information), which are critical for protein stability and protein-ligand interactions. Similarly, in network data, beyond the topological structure of nodes and edges, one may have node-associated attributes such as the user interests in social networks, which are critical for the clustering and link prediction tasks.

The element-specific method [37, 38, 40] is a powerful framework for systematically modeling different atomic and molecular interactions into a unified topological system. For example, given a point cloud of molecular atoms, one can construct element-specific simplicial complexes by grouping atoms based on their element types. This enables us to focus on particular atomic interactions, such as covalent bonds, hydrogen bonds, van der Waals forces, and other noncovalent interactions. The integration of element-specific strategy with standard persistent homology has demonstrated great promise in various biological data analysis tasks, such as protein-ligand binding affinity prediction, protein-protein interaction analysis, and protein stability analysis.

Another approach for incorporating non-geometric information is to encode it as weights and integrate these weights into the underlying mathematical frameworks. For example, in the enriched barcode method [39], the persistent cohomology is computed from the underlying simplicial complex modeling of the protein atoms, and the additional biochemical information is embedded as weights into the cohomology representatives. This results in an enriched, or colored, persistence barcode that captures both topological and chemical properties. In the persistent sheaf Laplacian method [310], the atomic properties are encoded into the sheaf construction, particularly, the construction of the boundary operators. Thus the derived persistent sheaf cohomology and sheaf Laplacians will automatically carry this additional information.

5.2 Approaches for topological representations

5.2.1 Representations of persistent homology/cohomology

Persistent homology and persistent cohomology produce identical barcodes when computed with coefficients in a field [90], which is the most common choice in practical applications. Therefore, here we focus only on the case of persistent homology for simplicity. Once an appropriate topological model is established for a given data set, the corresponding persistent homology can be computed to extract meaningful topological representations, such as persistent simplicial homology for simplicial complexes, persistent hypergraph homology for hypergraphs, and persistent sheaf homology for sheaves. Note that persistent homology can be visualized and encoded in various forms, each offering a distinct perspective on the topological features. Next, we summarize several common representation formats of persistent homology.

Persistence barcode/diagram

The two standard representations of persistent homology, persistence barcodes [48] and persistence diagrams [112], both offer intuitive ways to represent the

最佳性能。PI-Net [276], 一个简单的一步可微架构, 是第一个提出使用深度学习直接从数据计算持久性图像的方法。持久性图像也已扩展到多参数设置, 从而产生了多参数持久性图像 (MPPI) [52]。

持久贝蒂数图像

多维持久性的概念在文献 [314] 中被提出, 用于将动态生物分子数据转化为持久贝蒂数图像, 通过编码分子系统随时间的拓扑演化。具体而言, 给定一个表示动态过程的数据帧序列, 对每一帧应用持久同调, 并将得到的持久贝蒂数编码为一维向量。通过在所有时间步上堆叠这些向量, 构建出一个反映系统时间拓扑变化的二维图像。同一论文中提出的多尺度多维持久性进一步在单个图像内编码了不同空间尺度下的分子结构。特别是, 使用具有可调尺度参数的柔性-刚性函数来生成分子的体积密度表示。对于特定选择的尺度参数, 计算持久贝蒂数并将其记录为一维向量, 然后通过在所有尺度上组装这些向量, 形成一个二维、三维或高维图像。

彩色持久条形码

已有多种方法被提出, 用以丰富持久条形码的附加信息, 从而产生了彩色持久条形码。例如, 在持久富集条形码方法 [39] 中, 首先计算持久上同调, 并通过特定函数将与单形相关的补充权重信息纳入上同调代表元, 从而为相应的条形分配颜色。在持久琼斯多项式方法 [277] 中, 为每个单形分配一个琼斯多项式, 并在选定值处计算与条形对应的正单形相关联的多项式, 以确定其颜色。这些增强的条形码有效地编码了背景相关信息, 从而提供了信息更丰富的拓扑表征。

分级持久条形码

最常用的持久条形码源自单分级持久同调。然而, 也存在提供多分级持久同调表示的方法。例如, 在持久迈耶同调方法 [269] 中, N 链复形中的附加分级决定了一种特定类型的边界算子。通过改变分级索引, 可以构造一系列链复形, 从而产生一个持久同调序列, 即一个双分级持久同调。类似地, 在持久科瓦诺夫同调方法 [197] 中, 一个由交叉点的平滑类型决定的附加量子分级被引入到上链复形中, 从而产生一个能捕获更丰富结构信息的双分级持久同调。这些分级持久同调自然地引出了分级持久条形码表示。

birth and death of topological features during the filtration process. Persistence barcodes summarize the lifespans of topological features, such as connected components, holes, and cavities, across different dimensions. Each barcode consists of horizontal line segments, where the left endpoint of each line segment represents the birth time and the right endpoint indicates the death time of a topological feature. A persistence diagram is equivalent to a barcode but encodes the same topological information as a 2D scatter plot, where the x -coordinate represents the birth time and the y -coordinate represents the death time. As the death time of a feature always occurs after its birth time, all points in a persistence diagram are positioned above the diagonal $y = x$. Long segments in a barcode typically correspond to significant topological features, while short segments are often considered as noise. Similarly, in a persistence diagram, features far from the diagonal are usually considered significant, and those closer to the diagonal are often treated as noise. However, this assumption might not hold when considering physical, chemical and biological data, where short segments or features near the diagonal might carry important physical information such as bond lengths, benzene ring size, and other properties [313]. There is also a Grassmannian persistence diagram construction [141, 142] for multiparameter persistence. Persistence barcodes/diagrams can effectively capture the topological features of the data, and have been used in the study of image patches [2, 49], hand-drawn letters [75], and unusual events in viral evolution [54].

Persistent Betti numbers/ Betti curves

Persistent Betti numbers [112] and Betti curves are multiscale analogs of the Betti numbers that count the number of topological features that persist in the filtration. Persistent Betti numbers are defined as the ranks of persistent homology groups and are often visualized as Betti curves, which offer a step function-based representation of persistent homology. The representation transforms persistence barcodes/diagrams into functional representations. The vector space structure of functions then enables basic operations like addition and scalar multiplication, making Betti curves suitable for statistical tasks such as computing the means and variances. These operators are not naturally defined for persistence barcodes/diagrams.

Betti curves are simple and easy to compute. In addition, as small perturbations often introduce short lifetimes that have minimal impact on the overall Betti number distribution, they are also robust to noise. However, one drawback of Betti curves is their limited expression power, as they only contain counts of topological features. In addition, the mapping from a diagram to a curve is not injective [68], which means that different diagrams can lead to the same curve.

Persistence landscape

A stable representation of persistent homology, named persistence landscapes [34, 35], offers a more structured and expressive representation of persistence

4.2.2 持久拉普拉斯算子的表示

拉普拉斯算子的表示主要涉及其谱性质，即其特征值和特征向量，它们编码了底层数据集的拓扑结构以及组合或几何结构的基本信息。另一个重要方面是，霍奇拉普拉斯算子能够通过霍奇分解将向量场分解为三个或五个正交分量，为分析数据中的流、梯度和循环提供了强大工具。接下来，我们将详细描述这些表示。

特征值表示

拓扑拉普拉斯算子的一个基本性质是其核与相应上同调群之间的同构。在实践中，计算通常是在域系数下进行的，在此系数下上同调与同调同构。这意味着拓扑拉普拉斯算子的零特征值的重数等于相应维度的贝蒂数。这一结果可推广到持久情形：持久拉普拉斯算子表现出类似的同构，其零特征值的重数对应于持久贝蒂数 [217]。因此，持久拉普拉斯算子的零特征值完整编码了持久同调信息。相比之下，非零特征值提供了底层数据结构的额外几何与组合洞察。拉普拉斯矩阵是对称且半正定的，这保证了所有非零特征值均为正。对于流形上的霍奇拉普拉斯算子，较大的特征值与相应特征函数的高频分量相关联。这些高频模式反映了精细尺度的结构变化或局部不规则性，可能代表噪声，尤其是当流形表现出非光滑特征（如尖锐拐角）时。将拉普拉斯算子的特征值与机器学习技术相结合，已成功应用于多个领域，包括药物设计与发现、计算生物学与化学以及材料科学。

特征向量表示

众所周知，与拓扑拉普拉斯算子相关联的特征向量捕获了关于底层对象的丰富拓扑与几何信息。例如，在形状分析中，全局点签名（GPS）[262]——一种利用拉普拉斯-贝尔特拉米算子的缩放特征函数定义的曲面表示——已被证明携带了足够的信息，可用于各种形状处理任务，包括形状分类、分割和对应。霍奇拉普拉斯算子的特征向量，包括在某些边界条件下的调和特征场、梯度特征场和旋度特征场，已被应用于生物大分子的分析 [325]。这些特征向量既可以单独作为机器学习模型的输入，也可以作为特征值特征的补充。

对于持久拓扑拉普拉斯算子，虽然已知其特征值能够捕获过滤过程中的拓扑与几何变化，然而，其特征向量的解释仍不明确，需要进一步研究。

diagrams while maintaining compatibility with statistical and machine learning frameworks. They were introduced to convert persistence diagrams into sequences of piecewise linear functions, which can be viewed as a horizontal version of persistence diagrams. Compared to Betti curves, which track only the count of topological features, persistence landscapes encode both the birth-and-death pairs and their persistence and maintain a bijective correspondence with persistence diagrams, making the latter preferable when detailed topological analysis is needed. Recently, it has been extended to the multiparameter persistence landscape (MPPL) [296] for the multiparameter case.

A key feature of persistence landscapes is their hierarchical structure, where different levels of functions capture topological features with varying significance. The function with the lowest lifespan subscript captures the most persistent topological features, while those with higher lifespan-subscript capture less persistent features. Persistence landscapes have been used to study conformational changes in protein binding[178], Microstructure Analysis [109], time-dependent functional networks constructed from time series [279], and image classification tasks and orbit classifications [173].

Persistence image

Persistence images [4] were introduced as an alternative stable representation that transforms persistence diagrams into fixed-size, grid-based functional representations, making them particularly well suited for convolutional neural networks (CNN). Persistence images are constructed by computing the integral of a persistence surface on a grid obtained by discretizing a subdomain of $\mathbb{R}^2$. Here, a persistence surface is defined as a weighted sum of Gaussian functions centered at each point in the persistence diagram, with the weighting function controlling the significance of each point. Persistence images have been proven to be stable with respect to the 1-Wasserstein distance between persistence diagrams in the same paper.

A key advantage of persistence images is that they are easy to compute and offer flexibility in three choices: the resolution, the distribution (and its associated parameters), and the weighting function. However, these choices are noncanonical, meaning that there are no guidelines for setting these parameters to achieve the best performance in machine learning tasks. PI-Net [276], a simple one-step differentiable architecture, is the first to propose the use of deep learning for computing persistence images directly from data. The persistence image has also been extended to a multiparameter setting, leading to the Multiparameter Persistence Image (MPPI) [52].

Persistent Betti number image

The notion of multidimensional persistence was proposed in [314] for transforming dynamical biomolecular data into persistent Betti number images by encoding the topological evolution of molecular systems over time. Specifically, given a sequence of data frames representing a dynamical process, persistent homology is applied to each frame, and the resulting persistent Betti numbers

Hodge 分解表示

Hodge 分解将向量场正交地分解为梯度、旋度和调和分量。这种分解通过提取不同的动态特征，使得对向量场的分析更加细致，特别有助于理解场的底层机制。它在电磁学 [222]、流体与动力系统 [318, 320]、几何建模 [304] 以及谱数据分析 [171] 中有众多应用。

最广为人知的 Hodge 分解是三分量 Helmholtz-Hodge 分解 [181]。对于欧几里得空间中的紧致区域，存在一种更精细的五分量向量场 Hodge 分解 [272]，它保持了底层区域的拓扑结构。离散的五分量 Hodge 分解已被提出用于曲面三角形网格 [249]、四面体网格 [248, 324]，以及由笛卡尔网格中水平集函数定义的区域 [281, 283]。Hodge 分解已成功应用于单细胞 RNA 速度的研究，通过利用细胞的基因表达来提取其不同的动态信息 [280]。此外，基于 Hodge 分解技术的流形拓扑深度学习 (MTDL) [207] 最近被提出用于医学图像分析。

4.2.3 持久狄拉克算子的表示

在组合情形下，持久狄拉克算子可被视为持久组合拉普拉斯算子 [9, 308] 的推广。它捕获了与持久组合拉普拉斯算子相同的拓扑信息，其特征值对应于所有维度上持久组合拉普拉斯算子特征值的平方根。然而，作为一阶算子，与二阶拉普拉斯算子相比，它对局部特征更为敏感，并且其矩阵耦合了不同的拓扑维度。

持久狄拉克算子的一个显著优势在于它能够实现高效的量子计算，相比经典的拓扑数据分析算法提供指数级加速。该算子不仅能够从持久同调中恢复拓扑输出，还能导出点云数据的谱特征。持久狄拉克算子也可以类似地表述于微分流形，从而推广微分拓扑设置中的霍奇-狄拉克算子，便于分析流形上的数据。

4.2.4 多尺度高斯环绕积分的表示

多尺度高斯环绕积分方法 [270] 是为分析一维曲线数据（包括纽结、链环和缠结）而开发的。该方法首先将曲线分解为一系列短线段，然后计算所有线段对之间的成对高斯环绕积分。这些积分构成一个分割矩阵，其条目量化了曲线段之间的拓扑环绕信息。接着，将线段之间的成对距离作为权重纳入条目中，得到一个精化的分割矩阵，该矩阵编码了跨越多个空间尺度的环绕结构。通过系统地改变尺度参数，可以构建一系列这样的矩阵，

are encoded as one-dimensional vectors. By stacking these vectors across all time steps, a two-dimensional image is constructed that reflects the temporal topological changes of the system. The multiscale multidimensional persistence proposed in the same paper further encodes molecular structures at various spatial scales within a single image. In particular, the flexibility-rigidity function with tunable scale parameters is used to generate a volumetric density representation of the molecule. For specific chosen scale parameters, the persistent Betti numbers are computed and recorded as one-dimensional vectors, then a 2D, 3D, or high-dimensional image is formed by assembling these vectors across all scales.

Colored persistence barcode

Several methods have been proposed to enrich persistence barcodes with additional information, resulting in the colored persistence barcode. For example, in the persistent enriched barcode approach [39], persistent cohomology is first computed, and supplementary weight information associated with simplices is incorporated into the cohomology representatives through specific functions to assign colors to the corresponding bars. In the persistent Jones polynomial method [277], each simplex is assigned a Jones polynomial, and the polynomial associated with the positive simplex corresponding to a bar is evaluated at a selected value to determine its color. These enhanced barcodes effectively encode background-related information, thereby providing a more informative topological characterization.

Graded persistence barcode

The most commonly used persistence barcode is derived from single-graded persistent homology. However, there are also methods that provide multi-graded persistent homology representations. For example, in the persistent Mayer homology method [269], the additional grading in the N -chain complex determines a specific type of boundary operator. By varying the grading index, a series of chain complexes can be constructed, resulting in a sequence of persistent homologies, i.e., a bigraded persistent homology. Similarly, in the persistent Khovanov homology method [197], an additional quantum grading that is determined by the smoothing type of the crossings, is introduced into the cochain complex, resulting in a bigraded persistent homology that captures richer structural information. These graded persistent homologies naturally give rise to graded persistence barcode representations.

5.2.2 Representations of persistent Laplacians

The representation of Laplacians primarily involves their spectral properties, that is, their eigenvalues and eigenvectors, which encode essential information about the topological, and the combinatorial or geometric structures of the underlying datasets. Another important aspect is that the Hodge Laplacian enables the decomposition of vector fields into three or five orthogonal components through the Hodge decomposition, providing a powerful tool for

每个矩阵捕获不同分辨率下的局部拓扑环绕信息。这些矩阵共同为底层曲线结构提供了全面的多尺度拓扑表示。

4.2.5 多尺度/持久琼斯多项式的表示

多尺度与持久琼斯多项式 [277] 将经典琼斯多项式 [20] 推广到多尺度设置, 从而能够分析不相交开曲线和闭曲线集合的局部与全局纠缠。给定特定的曲线分割, 多尺度琼斯多项式框架提供一个琼斯多项式矩阵, 其中每一行对应一个曲线段的特征向量, 该向量由在不同距离范围内计算的琼斯多项式确定。特征向量的每个条目是以下曲线集合的琼斯多项式: 该集合由该曲线段以及所有落在与该段特定距离范围内的其他曲线段组成。持久琼斯多项式建立在由曲线段的成对距离矩阵构造的 Vietoris-Rips 复形之上, 这引出了复形中面的出生、消亡和持久性信息。通过纳入各个面的琼斯多项式, 该框架产生一个加权条形码, 其权重由相关面的琼斯多项式给出。

4.2.6 软件与代码资源

表 1 列出了几种广泛用于计算持续同调、持续拉普拉斯算子及霍奇分解的软件包。除此之外, 另一个流行的工具是 Mapper [274], 它能够以单纯复形的形式为高维数据集提供简化表示。以点云数据为例, 首先使用聚类算法将数据划分为若干簇。随后基于这些簇构建一个单纯复形: 每个簇对应一个顶点, 而一组 n 个簇若其交集非空, 则构成一个 $(n-1)$ -单形。所得单纯复形的一维骨架通常被称为 Mapper 图, 它提供了对数据的简洁描述, 并捕捉了其拓扑结构的重要信息。已有多个软件工具实现了 Mapper 算法, 包括 KeplerMapper [294]、giotto-tda [289]、GUDHI [211] 以及 Mapper Interactive [328]。

4.3 特征化方法

持久同调和持久琼斯多项式可以编码为持久性条形码, 其由一系列(出生, 死亡)对组成, 表示拓扑特征在过滤过程中的生命周期。相比之下, 多尺度高斯链环积分和多尺度琼斯多项式通常由分割矩阵表示, 其中条目量化了底层数据段之间的局部链环信息。对于持久拉普拉斯算子和持久狄拉克算子, 谱信息(特征值和特征向量)通常用于特征提取。这些表示产生了三种类型的拓扑描述符: 持久性条形码、持久性谱和分割矩阵。除此之

analyzing flows, gradients, and cycles in data. In the following, we describe these representations in detail.

Eigenvalue representation

A fundamental property of topological Laplacians is the isomorphism between their kernel and the corresponding cohomology groups. In practice, computations are typically performed with a field coefficient, under which cohomology is isomorphic to homology. This implies that the multiplicity of zero eigenvalues of the topological Laplacian is equal to the Betti number in the corresponding dimension. This result extends to the persistent setting: the persistent Laplacian exhibits a similar isomorphism, where the multiplicity of its zero eigenvalues corresponds to the persistent Betti numbers [217]. Consequently, the zero eigenvalues of the persistent Laplacian fully encode the persistent homology information. In contrast, the nonzero eigenvalues provide additional geometric and combinatorial insights of the underlying data structure. Laplacian matrices are symmetric and semi-positive definite, which guarantees that all nonzero eigenvalues are positive. For Hodge Laplacians on manifolds, the larger eigenvalues are associated with higher-frequency components of the corresponding eigenfunctions. These high-frequency modes reflect fine-scale structural variations or local irregularities and may represent noise, especially when the manifold exhibits non-smooth features such as sharp corners. The integration of the eigenvalues of Laplacians with machine learning techniques has been successfully applied in a variety of domains, including drug design and discovery, computational biology and chemistry, and material science.

Eigenvector representation

It is known that the eigenvectors associated with topological Laplacians capture rich topological and geometric information about the underlying objects. For example, in shape analysis, the global point signature (GPS) [262], a representation of surfaces defined using scaled eigenfunctions of the Laplace-Beltrami operator, has been shown to carry sufficient information of surfaces for various shape processing tasks, including shape classification, segmentation, and correspondence. The eigenvectors of Hodge Laplacians, including the harmonic eigenfields, the gradient eigenfields, and the curl eigenfields under certain boundary conditions, have been applied to the analysis of biological macromolecules [325]. These eigenvectors can be used either individually as inputs in machine learning models, or as a supplement to eigenvalue features.

For persistent topological Laplacians, while the eigenvalues are known to capture the topological and geometric changes during the filtration process, the interpretation of the eigenvectors, however, remains unclear and requires further investigation.

Hodge decomposition representation

Hodge decomposition orthogonally splits vector fields into gradient, curl, and harmonic components. This decomposition enables a more detailed analysis of

表 1 用于计算持续同调和持续拉普拉斯算子的软件包。

软件	单纯复形	立方体复形	持续同调	持续上同调	持续拉普拉斯算子	霍奇分解
https://appliedtopology.github.io/javaplex/ [3]	是		是			
https://people.maths.ox.ac.uk/nanda/perseus/index.html [224]	是	是	是			
https://mrzv.org/software/dionysus2/ [229]	是		是	是		
https://godhi.inria.fr/ [211]	是	是	是	是		
https://riper.scikit-tda.org/en/latest/ [22]	是		是	是		
https://github.com/xoltar/phant [24]	是	是	是			
https://github.com/DIPHA/dipha/ [23]	是	是	是			
https://cran.r-project.org/web/packages/TDA/index.html [121]	是	是	是	是		
https://giotto-ai.github.io/gtda-docs/0.5.1/library.html [289]	是	是	是			
https://github.com/rodrigo/OpenPH [218]	是		是			
https://github.com/appliedtopology/ctf	是	是	是			
https://github.com/henselman-petrusek/Eirene.jl [152]	是	是	是			
https://bitbucket.org/hubwag/cubicle/src/master/ [297]		是	是			
https://github.com/shizuo-kaji/CubicalRipser_3dim [166]		是	是			
https://topology-tool-kit.github.io/ [290]	是	是	是			
https://www.sthu.org/code/libstick/ [114]	是		是			
https://weilab.math.msu.edu/HERMES/ [303]	是				是	
https://github.com/ndag/Persistent-Laplacian [217]	是				是	
https://github.com/zhesu1/HHD [280]		是				是
https://github.com/zhesu1/5ComponentHD [281]		是				是
https://github.com/rdzhao/3DHodgeDecomposition [324]	是					是

外，霍奇拉普拉斯算子实现了向量场的三分量和五分量正交分解，这自然地在欧拉设定中为原始向量场提供了多通道图像表示。接下来，我们将描述将这些拓扑描述符转化为适用于机器学习算法的特征向量的方法。

4.3.1 函数化表述

持久性条形码/图不能直接应用于机器学习任务，因为它们本身不具备固定大小特征向量的结构。持久性条形码包含数量可变、长度不同的区间，而持久性图则包含数量可变的无序点。向量化持久性条形码/图的一种直接方法是使用其函数化表述，这些表述可以通过在选定点对所得函数进行采样，直接离散化为机器学习输入。常用的函数化表述包括贝蒂曲线、持久性景观 [34]、持久性曲面和持久性图像 [4] 等。

4.3.2 分箱计数

另一种向量化持久性条形码的方法是分箱法 [40]。在该方法中，将等长的过滤范围划分为一组箱子，并统计每个箱子内的出生数、死亡数和持久数。这一过程产生三个相应的特征向量：出生计数、死亡计数和持久计数。该方法有效地捕捉了结构特性，例如键长，包括生物分子中的非共价相互作用，并已在各种拓扑数据分析研究中被广泛采用 [37, 38, 233]。

a vector field by extracting different dynamic features, making it particularly useful in understanding the underlying mechanisms of a field. It has numerous applications in electromagnetism [222], fluids and dynamic systems [318, 320], geometric modeling [304], and spectral data analysis [171].

The most well-known Hodge decomposition is the 3-component Helmholtz-Hodge decomposition [181]. For compact domains in Euclidean spaces, there is a refined 5-component Hodge decomposition of vector fields [272], which preserves the topology of the underlying domains. The discrete 5-component Hodge decomposition has been proposed for surface triangle meshes [249], for tetrahedral meshes [248, 324], and for domains defined by level set functions in Cartesian grids [281, 283]. Hodge decomposition has been successfully applied to the study of single-cell RNA velocities for extracting different dynamic information of cells by utilizing their gene expressions [280]. In addition, manifold topological deep learning (MTDL) [207] has been recently proposed based on the Hodge decomposition technique for medical image analysis.

5.2.3 Representations of persistent Dirac operator

In the combinatorial case, the persistent Dirac operator can be seen as a generalization of the persistent combinatorial Laplacians [9, 308]. It captures the same topological information as the persistent combinatorial Laplacians, with its eigenvalues corresponding to the square root of the eigenvalues of the persistent combinatorial Laplacians in all dimensions. However, as a first-order operator, it is more sensitive to local features compared to the second-order Laplacians, and its matrix couples different topological dimensions.

One significant advantage of the persistent Dirac operator is that it enables efficient quantum computation, providing exponential speed-up over classical TDA algorithms. This operator not only recovers the topological outputs from persistent homology but also leads to spectral features of the point cloud data. The persistent Dirac operator can also be formulated similarly for differential manifolds, extending the Hodge-Dirac operator in the differential topology setting, thus facilitating the analysis of data on manifolds.

5.2.4 Representations of multiscale Gauss linking integral

The multiscale Gauss linking integral (mGLI) method [270] is developed for the analysis of one-dimensional curve data, including knots, links, and tangles. The method begins by decomposing the curve into a collection of short segments, after which the pairwise Gauss linking integrals are computed between all segment pairs. These integrals form a segmentation matrix whose entries quantify the topological linking information between curve segments. Then pairwise distances between segments are incorporated as weights into the entries, yielding a refined segmentation matrix that encodes linking structures across multiple spatial scales. By systematically varying the scale parameter, a sequence of such matrices can be constructed, each capturing localized topological linking information at a distinct resolution. Collectively, these matrices provide

4.3.3 统计特性

对于持久性条形码, 一种基础且广泛使用的向量化方法是提取其统计特性 [8, 14, 40, 42]。例如, 可以计算条形长度、出生时间、死亡时间和中位数的描述性统计量, 包括最大值、最小值、总和、中位数、均值、标准差、方差和熵。这将产生一个维度为 $4k$ 的特征向量, 其中 k 表示四种量中每种量的统计度量数量。通过将过滤范围划分为 s 个固定区间并在每个区间内计算统计特性, 可以获得更细粒度的表示。例如, 统计出生-死亡区间覆盖每个区间的条形数量会产生一个 s 维向量, 而记录每个区间内的最大和最小出生值则会产生一个 $2s$ 维向量。

对于持久谱, 常见的做法是将过滤范围划分为均匀区间, 并在每个区间计算相应的拉普拉斯矩阵, 然后提取这些矩阵的特征值和特征向量以形成特征表示 [202, 219]。基本的统计度量, 如最大值、最小值、总和、中位数、均值、标准差、方差和熵, 可以应用于每个拉普拉斯矩阵的特征值, 从而产生捕捉跨尺度拓扑、几何和组合特征的特征向量。然而, 当拉普拉斯矩阵较大时, 计算完整谱的计算成本很高。在这种情况下, 通常只保留最小的 k 个非零特征值以及贝蒂数, 它们捕捉了最全局和最稳定的谱分量。这种截断的谱表示不仅计算高效, 而且对噪声具有鲁棒性。

分割矩阵编码了给定曲线对之间各分段的成对链接信息。一种直接的向量化策略是计算矩阵所有条目的全局统计特性, 例如最大值、最小值、总和、中位数、均值、标准差、方差和熵, 以构建特征向量。或者, 可以通过对每行或每列的条目求和来捕捉局部链接模式, 从而量化单个分段的总体链接贡献。然后, 可以对这些跨所有分段的局部值应用统计度量, 以生成保留局部结构信息的特征向量。

4.3.4 代数形式化

对于持久性条形码, 代数向量化主要通过从条形码构造的多项式映射实现。例如, 在 Adcock-Carlsson 坐标方法 [6] 中, 给定一个包含 n 个区间的条形码, 会构造一个代数函数环, 即满足特定性质的 $2n$ 个变量的多项式。然后通过从该环中选择 k 个代数函数, 并在 n 个 (出生, 死亡) 对上求值, 得到一个 k 维特征向量。在热带坐标方法 [167] 中, 对于一个包含 n 个区间的条形码, 仅使用最大值、最小值、加法和减法运算构造一个涉及 $2n$ 个变量的函数。该函数在变量置换下保持不变。特征向量通过对

a comprehensive multiscale topological representation of the underlying curve structure.

5.2.5 Representations of multiscale/persistent Jones polynomial

The multiscale and persistent Jones polynomials [277] extend the classical Jones polynomial [20] to a multiscale setting, enabling the analysis of local and global entanglements for collections of disjoint open and closed curves. Given a specific curve segmentation, the multiscale Jones polynomial framework offers a matrix of Jones polynomials, where each row corresponds to a feature vector of a curve segment determined by the Jones polynomials computed across various distance ranges. Each entry of the feature vector is the Jones polynomial of the set of curves consisting of that curve segment and all other segments that fall within a specific distance range from that segment. The persistent Jones polynomial builds on a Vietoris-Rips complex constructed from the pairwise distance matrix of curve segments, which leads to the birth, death, and persistence information of facets in the complex. By incorporating the Jones polynomials of facets, this framework leads to a weighted barcode, with weights given by the Jones polynomials of the associated facets.

5.2.6 Software and code resources

Table 1 lists several widely used software packages for computing persistent homology, persistent Laplacians, and Hodge decompositions. In addition to these, another popular tool is Mapper [274], which provides a simplified representation of high-dimensional datasets in the form of simplicial complexes. Taking point cloud data as an example, the data is first divided into clusters using a clustering algorithm. A simplicial complex is then constructed from these clusters: each cluster corresponds to a vertex, and a set of n clusters forms an $(n-1)$ -simplex if their intersection is nonempty. The one-skeleton of the resulting simplicial complex is commonly referred to as the Mapper graph, which provides a simple description of the data and captures important information about its topological structures. Several software tools have implemented the Mapper algorithm, including KeplerMapper [294], giotto-tda [289], GUDHI [211], and Mapper Interactive [328].

5.3 Featurization methods

Persistent homology and persistent Jones polynomial can be encoded as persistence barcodes, which consist of a collection of (birth, death) pairs representing the lifespan of topological features across the filtration. In contrast, the multiscale Gauss linking integral and multiscale Jones polynomial are typically represented by segmentation matrices, where the entries qualify the local linking information between segments of the underlying data. For persistent Laplacians and persistent Dirac operator, the spectral information (eigenvalue and eigenvector) is usually used for feature extraction. These representations

所有 n 个（出生，死亡）对或（出生，持久性）对求值该函数获得。还有一些方法利用定义在条形码区间上的复多项式来生成特征向量 [105, 125]。

对于持久性谱，向量化主要涉及对相关拉普拉斯矩阵的特征值应用函数。例如，在持久属性方法 [219] 中，将过滤范围划分为均匀的区间，并在每个区间内计算特征值的谱指数。这些谱指数 [252] 包括拉普拉斯图能量、广义图能量、广义平均图能量、谱半径、谱直径、Ivanciuc 矩阵谱算子、谱矩、Lovász-Pelikán 指数和拟维纳指数。除了这些谱描述符外，其他谱性质，如代数连通性、模块性、Cheeger 常数、顶点/边扩展，以及从流、随机游走和热核导出的量，也可以纳入向量化过程。除了这些经典谱性质外，更高级的函数，如黎曼 zeta 函数 [203] 和对称函数 [25]，也已被应用于特征值以进行特征提取。

4.3.5 多维公式化方法

多种公式化方法将多个参数或索引纳入特征生成过程。例如，Mayer 同调在过滤过程中包含一个额外的索引 [269]，从而产生三维持久性条形码。此外，多维持久同调为将动态分子数据转换为图像表示提供了一个有效框架，而多尺度多维持久性进一步在单张图像内编码了不同空间尺度下的分子结构 [314]。

对于持久谱，生成多维特征向量的一种方法是使用过滤值作为 x 轴，并在 y 轴上绘制前 k 个特征值或整个特征值谱的统计特性。另一种策略是将持久图像的思想扩展到谱特征。具体而言，对于每个过滤值，计算前 k 个特征值或完整谱的 k 个统计量，从而得到 k 个（过滤值，特征属性）对。这些对可以类比于持久性条形码中的（出生，死亡）对进行处理，从而允许构建持久图像。如果在整个过滤过程中收集所有这样的对，则可以得到二维图像表示。或者，如果在每个过滤值处构建单独的图像，并沿过滤轴将它们连接起来，则可以导出三维图像。

此外，当将霍奇拉普拉斯算子应用于笛卡尔网格时，可以通过霍奇分解直接获得图像表示。例如，在网格上应用三分量分解会产生无旋分量、无散分量和谐波分量，每个分量都具有与原始一形式相同的空间维度。然后可以通过沿通道轴连接这三个分量来构建多通道图像。

Table 1 Software packages for computing persistent homology and persistent Laplacians.

Software	Simplicial Complex	Cubical Complex	Persistent Homology	Persistent Cohomology	Persistent Laplacian	Hodge Decom.
JavaPlex [3]	Yes		Yes			
Perseus [224]	Yes	Yes	Yes			
Dinoyus [229]	Yes		Yes	Yes		
GUdhi [211]	Yes	Yes	Yes	Yes		
Rips [22]	Yes		Yes	Yes		
PHAT [24]	Yes	Yes	Yes			
DIPHA [23]	Yes	Yes	Yes			
R-TDA [121]	Yes	Yes	Yes	Yes		
Giotto-tda [289]	Yes	Yes	Yes			
OpenPH [218]	Yes		Yes			
CTL	Yes	Yes	Yes			
Eirene [152]	Yes	Yes	Yes			
Cubicle [297]		Yes	Yes			
CubicalRips [166]		Yes	Yes			
TTK [290]	Yes	Yes	Yes			
libstick [114]	Yes		Yes			
HERMES [303]	Yes				Yes	
PersistLap [217]	Yes				Yes	
HHD [280]		Yes				Yes
5ComponentHD [281]		Yes				Yes
3DHodgeDecom [324]	Yes					Yes

give rise to three types of topological descriptors: persistence barcode, persistence spectral, and segmentation matrix. In addition to these, the Hodge Laplacians enable three- and five-component orthogonal decompositions of vector fields, which naturally provide a multi-channel image representation of the original vector field in the Eulerian setting. In the following, we describe methods for transforming these topological descriptors into feature vectors suitable for machine learning algorithms.

5.3.1 Functional formulations

Persistence barcodes/diagrams can not be directly applied to machine learning tasks, as they are not naturally structured as fixed-size feature vectors. Persistence barcodes consist of varying numbers of intervals with different lengths, while persistent diagrams contain varying numbers of unordered points. One straightforward approach for vectorizing persistent barcodes/diagrams is to use their functional formulations, which can be directly discretized as machine learning inputs by sampling the resulting functions at selected points. Commonly used functional formulations include Betti curves, persistence landscape [34], persistence surfaces and persistence images [4], etc.

5.3.2 Counting in bins

Another method for vectorizing persistence barcode is the binning-based approach [40]. In this method, the uniform-length filtration range is divided into a collection of bins, and the numbers of births, deaths, and persistences within each bin are counted. This process yields three corresponding feature vectors: birth count, death count, and persistence count. This approach effectively captures structural properties such as bond lengths, including those

4.3.6 基于核的模型

基于核的机器学习模型依赖核函数将数据隐式映射到高维特征空间，从而能够建模复杂的非线性关系，而无需显式计算这些空间中的坐标。这些方法的核心组件是核函数，它充当所有数据点对之间的相似性度量。

为了将持久性图等拓扑表示整合到基于核的学习框架中，必须定义此类图之间的可比性概念。一种自然的方式是在持久性图的空间上定义度量结构。这些度量结构导出的距离可以量化持久性条形码/图之间的差异。已经研究了几种度量，例如豪斯多夫距离、瓶颈距离 [73] 和瓦瑟斯坦距离 [40, 221]，并证明它们对小扰动具有鲁棒性。然而，这些距离的计算，特别是对于大型数据集，可能在计算上非常昂贵 [105]。定义一个合适的平均过程也很具挑战性，因为局部均值可能不唯一 [221, 232]。特别是，这些度量主要用于基于距离的学习方法，例如 k-近邻 (KNN) 和一些聚类算法。因此，它们不适用于需要希尔伯特空间结构或固定大小特征输入的机器学习技术，例如支持向量机 (SVM)、主成分分析 (PCA)、决策树分类和深度神经网络。

基于核的模型通过定义一个将持久性图映射到希尔伯特空间的核来解决这个问题，从而允许它们直接与基于核的机器学习技术（如 SVM 和 PCA）集成。该核通过在希尔伯特空间中计算内积来编码持久性图之间的相似性。基于这种构造，已经提出了各种基于持久同调的核，例如持久尺度空间核 [255]、持久加权高斯核 [180]、切片瓦瑟斯坦核 [53] 和持久费希尔核 [183]，并且如相关论文所述，它们在形状分类、形状分割、纹理识别等应用中取得了成功。

4.3.7 持久同调的统计框架

过去十年中，通过统计方法研究持久性图取得了显著进展，解决了由其无序性和可变基数带来的挑战，尤其是在存在小数据扰动的情况下。图空间的度量结构，特别是在 Wasserstein 距离或瓶颈距离下，自然地引出了集中趋势的统计概念，例如 Fréchet 均值和中位数 [73, 221, 293]。

为了促进经典统计原理在拓扑数据分析中的应用，持久性图越来越多地被建模为随机变量。将持久性图嵌入到更易处理的数学空间中的函数表示法的发展，在推动这一进展中发挥了关键作用。例如，持久景观 [34] 将持久性图嵌入到巴拿赫空间中，允许直接计算均值、方差和置信带，从而将拓扑摘要更直接地整合到既定的统计方法中。类似地，持久图像 [4] 提

of noncovalent interactions in biomolecules, and has been widely adopted in various topological data analysis studies [37, 38, 233].

5.3.3 Statistical property

For persistence barcode, a basic and widely used approach for vectorization is to extract their statistical properties [8, 14, 40, 42]. For example, one can compute descriptive statistics of the bar lengths, birth times, death times, and midpoints, including the maximum, minimum, sum, median, mean, standard deviation, variance, and entropy. This yields a feature vector of dimension $4k$, where k denotes the number of statistical measures for each of the four quantities. A more fine-grained representation can be obtained by partitioning the filtration range into s fixed bins and computing statistical properties within each bin. For instance, counting the number of bars whose birth-death intervals cover each bin produces an s -dimensional vector, while recording the maximum and minimum birth values in each bin results in a $2s$ -dimensional vector.

For persistence spectra, a common practice is to divide the filtration range into uniform intervals and compute the corresponding Laplacian matrices at each interval, the eigenvalues and eigenvectors of these matrices are then extracted to form feature representations [202, 219]. Basic statistical measures, such as maximum, minimum, sum, median, mean, standard deviation, variance, and entropy, can be applied to the eigenvalues of each Laplacian matrix, yielding feature vectors that capture topological, geometric, and combinatorial characteristics across scales. However, when the Laplacian matrices are large, computing the full spectrum becomes computationally expensive. In such cases, it is common to retain only the smallest k nonzero eigenvalues, as well as the Betti numbers, which capture the most global and stable spectral components. This truncated spectral representation is not only computationally efficient but also robust to noise.

The segmentation matrix encodes the pairwise linking information between segments of a given pair of curves. A straightforward vectorization strategy is to compute global statistical properties, such as the maximum, minimum, sum, median, mean, standard deviation, variance, and entropy, over all entries in the matrix to construct feature vectors. Alternatively, localized linking patterns can be captured by summing the entries of each row or column, thereby quantifying the overall linking contribution of individual segments. Statistical measures can then be applied to these localized values across all segments to generate feature vectors that preserve local structural information.

5.3.4 Algebraic formulations

For persistence barcode, algebraic vectorization is primarily achieved through polynomial maps constructed from the barcode. For example, in the Adcock-Carlsson coordinate method [6], given a barcode with n intervals, a ring of algebraic functions, polynomials in $2n$ variables that satisfy certain properties, is constructed. A k -dimensional feature vector is then obtained by selecting k algebraic functions from this ring and evaluating them on the n (birth, death)

供了一种互补的方法, 将图转换为稳定的、有限维的向量表示, 从而易于支持标准的统计分析。这些嵌入促进了在 Wasserstein 距离下持久性图的中心极限定理 (CLT) 以及各种函数表示法的大数定律 (LLN) 类似物的建立 [34, 108], 加强了 TDA 作为一个更稳健、更具统计基础的领域, 并提升了其在更广泛的科学和工程领域的适用性。此外, 还发展了自助法来评估拓扑摘要的总体水平差异 [55, 57], 通过构建置信集和分析拓扑特征的渐近行为, 实现组间的正式比较 [120]。

充分统计量概括了样本中与基础概率分布相关的所有信息。一个基于热带几何的框架为持久同调构建了充分统计量 [227]。所开发的模型将条形码映射到单射且稳定的热带坐标中。这种通过热带几何嵌入到欧几里得空间的方法, 保留了参数建模所需的完整拓扑信息, 同时实现了与经典统计技术的无缝集成。

基于核密度估计 (KDE) 的框架直接在持久性图空间上工作。一些方法利用点过程理论, 将期望图定义为拉东测度空间中的平均对象 [107], 而另一些方法则利用 Wasserstein 距离和瓶颈距离来捕捉拓扑特征的内在几何结构 [213]。这些方法无需将图转换为向量表示即可估计拓扑特征的分布, 从而提供了对其变异性的更内在且几何忠实度更高的表征。

Maroulas 等人 [213] 引入了利用随机有限集框架为持久性图量身定制的非参数密度估计技术。他们使用概率假设密度 (PHD) 来估计拓扑特征的分布, 无需限制性假设, 从而能够稳健地将有意义的拓扑特征与噪声分离, 这对于分析复杂或嘈杂的数据至关重要。他们提供了理论收敛保证, 包括平均绝对瓶颈偏差的界限, 确保密度估计收敛到真实分布。在另一个互补的方向上, 已发展出一种用于持续同调的贝叶斯框架, 其中持续图被建模为来自具有先验强度的泊松点过程的实现, 从而支持对拓扑特征分布进行后验推断, 并为拓扑摘要提供概率解释 [214]。在一项专注于生成建模的相关进展中, Papamarkou 等人 [242] 引入了一种基于成对交互点过程和 RJ-MCMC 采样来生成持续图的方法, 能够在捕获特征间依赖关系的同时, 实现对拓扑特征分布的真实模拟。这有助于在拓扑数据分析中进行统计分析, 特别是在数据有限的情况下, 例如在材料科学中。一个相关模型将此概率框架扩展到生物网络的分类, 通过将持续图建模为独立同分布的聚类点过程, 使用高斯混合模型来捕捉空间分布, 并使用二项分布来建模拓扑特征数量的变异性 [212]。此外, 贝叶斯拓扑信号处理进一步推进了

pairs. In the tropical coordinate method [167], for a barcode with n intervals, a function involving $2n$ variables is constructed using only the operations of maximum, minimum, addition, and subtraction. This function is invariant under permutation of the variables. The feature vector is obtained by evaluating the function on all n (birth, death) pairs or (birth, persistence) pairs. There are also methods that utilize complex polynomials defined on barcode intervals to generate feature vectors [105, 125].

For persistence spectra, vectorization primarily involves applying functions to the eigenvalues of the associated Laplacian matrices. For example, in the persistent attribute method [219], the filtration range is divided into uniform bins, and spectral indices of the eigenvalues are computed within each bin. These spectral indices [252] include the Laplacian graph energy, generalized graph energy, generalized average graph energy, spectral radius, spectral diameter, Ivanciuc matrix spectrum operators, spectral moments, Lovász–Pelikán index, and quasi-Wiener index. In addition to these spectral descriptors, other spectral properties, such as algebraic connectivity, modularity, Cheeger constant, vertex/edge expansion, and quantities derived from flow, random walks, and heat kernels, can also be incorporated into the vectorization process. Beyond these classical spectral properties, more advanced functions such as the Riemann zeta function [203] and symmetry functions [25] have also been applied to eigenvalues for feature extraction.

5.3.5 Multidimensional formulations

Several formulations incorporate multiple parameters or indices into the feature generation process. For example, Mayer homology includes an additional index in the filtration [269], resulting in 3D persistence barcodes. In addition, multidimensional persistent homology provides an effective framework for transforming dynamic molecular data into image representations, and the multiscale multidimensional persistence further encodes molecular structures at various spatial scales within a single image [314].

For persistence spectra, one approach to generate a multidimensional feature vector is to use the filtration values as the x -axis and plot either the first k eigenvalues or the statistical properties of the entire eigenvalue spectrum on the y -axis. An alternative strategy is to extend the idea of the persistence image to spectral features. Specifically, for each filtration value, one computes the first k eigenvalues or k statistical measures of the full spectrum, resulting in k (filtration, eigen-property) pairs. These pairs can be treated analogously to (birth, death) pairs in persistence barcodes, allowing for the construction of a persistence image. If all such pairs are collected across the entire filtration process, a 2D image representation is obtained. Alternatively, if a separate image is constructed at each filtration value and these are concatenated along the filtration axis, a 3D image can be derived.

Additionally, when applying Hodge Laplacians to Cartesian grids, image representations can be obtained directly via Hodge decomposition. For example, applying the three-component decomposition on a grid yields curl-free,

这项工作以处理时间序列数据，支持对噪声或演化信号中动态拓扑特征进行不确定性感知的分类 [236]。

5 结论性评述

在人工智能（AI）的助力下，拓扑数据分析（TDA）已成为应用数学和数据科学中一个快速发展的领域。TDA 的一项关键技术是持续同调，这是一种多尺度代数拓扑工具，已在科学、工程、医学、工业和国防等不同学科中展现出令人瞩目的成功。然而，基于单纯复形的持续同调面临若干局限性，包括：无法处理非几何信息、依赖对高阶环路的定性描述、无法追踪非拓扑变化、无法处理有向网络，以及对超图等结构化数据不敏感。此外，虽然代数拓扑技术通常分析点云数据，但它们不能直接应用于微分流形上的数据或嵌入三维空间的一维曲线。这些 TDA 中的挑战在过去十年中推动了重大的进展。

本文全面综述了为应对持续同调的局限性而发展的 TDA 技术，着重阐述了它们的数学基础和优势。一些技术将持续同调从单纯复形扩展到其他拓扑领域，例如胞腔复形、路径复形、有向旗复形、层复形和超（有向）图。另一些技术则引入了替代的拓扑表述，包括拉普拉斯算子 [302]、狄拉克算子 [9]、层理论、迈耶拓扑 [269] 和交互拓扑 [193]。对于微分流形上的数据，已发展出基于微分拓扑的方法，例如持续德拉姆上同调、持续霍奇拉普拉斯算子和霍奇分解 [64, 282]。对于三维空间中一维曲线上的数据，则提出了来自几何拓扑的工具，例如高斯环绕积分 [270]、琼斯多项式 [277] 和霍瓦诺夫同调 [197, 268]。

TDA 的快速增长是由机器学习驱动的，例如拓扑深度学习（TDL）[37] 和数学人工智能。然而，来自 TDA 的拓扑表示无法在没有适当向量化的情况下直接在机器学习中实现。本文还回顾了针对不同输入数据格式的拓扑算法选择、它们相关的拓扑表示，以及在机器学习背景下的拓扑向量化方法。

TDA 和 TDL 是动态且快速发展的领域。TDA 和 TDL 的未来发展将由理论探索 and 实际需求共同驱动，包括对可解释深度神经网络的需求。我们预见未来将在以下几个方向取得进展。

目前，单个数据点周围的局部几何和/或拓扑可以通过持续上同调 [39]、演化同调 [41]、交互拓扑 [193] 和持续层拉普拉斯算子 [309] 来描述。需要进一步努力完善局部拓扑方法以用于实际应用。

divergence-free, and harmonic components, each of which has the same spatial dimensions as the original one-form. A multi-channel image can then be constructed by concatenating these three components along the channel axis.

5.3.6 Kernel-based models

Kernel-based machine learning models rely on kernel functions to implicitly map data into high-dimensional feature spaces, enabling the modeling of complex nonlinear relationships without explicitly computing the coordinates in those spaces. The core component of these methods is the kernel function, which acts as a similarity measure between all pairs of data points.

To integrate topological representations such as persistence diagrams into kernel-based learning frameworks, it is essential to define a notion of comparability between such diagrams. A natural way is to define metric structures on the space of persistence diagrams. These metric structures induce distances that can quantify the differences between persistence barcodes/diagrams. Several metrics, such as Hausdorff distance, bottleneck distance [73], and Wasserstein distance [40, 221], have been studied and shown to be robust to small perturbations. However, the computation of these distances, especially for large datasets, can be computationally expensive [105]. It is also challenging to define an appropriate averaging process, as the local mean may not be unique [221, 232]. In particular, these metrics are primarily used for distance-based learning methods such as k-nearest neighbors (KNN) and some clustering algorithms. Therefore, they are unsuitable for machine learning techniques that require a Hilbert space structure, or fixed-size feature inputs, such as support vector machines (SVM), Principal Component Analysis (PCA), decision tree classification, and deep neural networks.

A kernel-based model addresses the issue by defining a kernel that maps persistence diagrams into a Hilbert space, allowing them to be directly integrated with kernel-based machine learning techniques, such as SVM and PCA. The kernel encodes the similarity between persistence diagrams by computing an inner product in the Hilbert space. Based on this construction, various persistent homology-based kernels, such as persistence scale space kernel [255], persistence weighted Gaussian kernel [180], sliced Wasserstein kernel [53], and persistence Fisher kernel [183], have been proposed and demonstrated success in applications as reported in their papers, including shape classification, shape segmentation, texture recognition, etc.

5.3.7 Statistical Frameworks for Persistent Homology

The study of persistence diagrams through statistical methods has advanced significantly in the past decade, addressing the challenges posed by their unordered nature and variable cardinality, especially in the presence of small data perturbations. The metric structure of diagram space, particularly under Wasserstein or bottleneck distances, naturally leads to statistical notions of central tendency, such as Fréchet means and medians [73, 221, 293].

目前大多数 TDA 方法是为点云数据开发的。然而，序列数据在 DNA 测序、自然语言处理和脑电图（EEG）分析等应用中被广泛使用。这些领域需要新颖的 TDA 方法论。

在微分几何中，阿蒂亚-辛格指标理论建立了一个拓扑指标。针对紧致流形上的数据，可以发展用于椭圆微分算子族的持续指标理论。低维拓扑方法的发展对于分析嵌入三维空间的一维曲线（例如纠缠聚合物和神经网络）至关重要。持久弗洛尔同调可能为模拟蛋白质-蛋白质结合相互作用提供一个有效的框架。此外，稳健的计算算法用于持久科瓦诺夫同调和持久科瓦诺夫拉普拉斯算子仍然是紧迫的优先事项。

最后，大语言模型（LLMs）预示着人工智能的新时代，为各学科创造了机遇。像 DeepSeek 和 ChatGPT 这样的聊天机器人可以将抽象数学理论——包括计算拓扑学——转化为实用工具 [196]。预计未来几年，人工智能辅助的拓扑学以及拓扑学增强的人工智能将快速增长。

TDA 是一个广阔的跨学科领域，涵盖代数拓扑、微分拓扑和几何拓扑；计算算法；人工智能驱动的方法论；以及现实世界的应用。尽管本文旨在全面综述超越持久同调的 TDA 和 TDL，但由于该领域的广度，我们不可避免地遗漏了许多重要贡献。我们注意到，自 2017 年 TDL 引入以来 [37]，人们在开发拓扑神经网络方面付出了巨大努力。为了专注于 TDA，我们限制了自己详细阐述这一发展。感兴趣的读者可以参考最近一篇关于 TDL 各个方面的综述 [243]。我们希望这项工作能为探索拓扑技术的研究人员提供指导，并帮助他们为分析不同领域的特定问题选择最佳方法。

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To facilitate the application of classical statistical principles within topological data analysis, persistence diagrams have been increasingly modeled as random variables. The development of functional representations that embed persistence diagrams into more tractable mathematical spaces has played a key role in driving this progress. For example, persistence landscapes [34] embed persistence diagrams into Banach spaces, allowing for the direct computation of means, variances, and confidence bands, thus integrating topological summaries more directly into established statistical methodologies. Similarly, persistence images [4] offer a complementary approach, transforming diagrams into stable, finite-dimensional vector representations that readily support standard statistical analyses. These embeddings have facilitated the establishment of Central Limit Theorems (CLT) for persistence diagrams under Wasserstein distances and analogues of the Law of Large Numbers (LLN) for various functional representations [34, 108], strengthening TDA as a more robust and statistically grounded field, and enhancing its applicability across a wider range of scientific and engineering domains. Furthermore, bootstrap methods have been developed to assess population-level differences in topological summaries [55, 57], enabling formal comparisons between groups by constructing confidence sets and analyzing the asymptotic behavior of topological features [120].

Sufficient statistics encapsulate all the information in a sample that is relevant to the underlying probability distribution. A tropical geometry-based framework constructs sufficient statistics for persistent homology [227]. The developed model maps barcodes into tropical coordinates that are injective and stable. This embedding into a Euclidean space via tropical geometry retains the complete topological information required for parametric modeling while enabling a seamless integration with classical statistical techniques.

Kernel Density Estimation (KDE)-based frameworks work directly on the space of persistence diagrams. Some approaches define expected diagrams as mean objects in a space of Radon measures, drawing on point process theory [107], while others utilize Wasserstein and bottleneck distances to capture the intrinsic geometry of topological features [213]. These methods estimate the distribution of topological features without converting diagrams into vector representations, offering a more intrinsic and geometrically faithful characterization of their variability.

Maroulas et al. [213] introduced nonparametric density estimation techniques tailored for persistence diagrams using the framework of random finite sets. Using Probability Hypothesis Density (PHD), they estimate the distribution of topological features without restrictive assumptions, enabling a robust separation of meaningful topological features from noise, which is essential to analyze complex or noisy data. They provide theoretical convergence guarantees, including bounds on the mean absolute bottleneck deviation, ensuring that density estimates converge to the true distribution.

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In a complementary direction, a Bayesian framework for persistent homology has been developed in which persistence diagrams are modeled as realizations from Poisson point processes with prior intensities, enabling posterior inference over topological feature distributions and supporting a probabilistic interpretation of topological summaries [214]. In a related development focused on generative modeling, Papamarkou et al. [242] introduced a method to generate synthetic persistence diagrams based on pairwise interacting point processes and RJ-MCMC sampling, enabling realistic simulation of topological feature distributions while capturing dependencies among features. This helps perform statistical analysis in topological data analysis, especially when data is limited, such as in materials science. A related model extends this probabilistic framework to the classification of biological networks by modeling persistence diagrams as independent and identically distributed cluster point processes, using Gaussian mixtures to capture spatial distribution and binomial distributions to model variability in the number of topological features [212]. Additionally, Bayesian topological signal processing further advances this line of work to handle time-series data, supporting uncertainty-aware classification of dynamic topological features in noisy or evolving signals [236].

6 Concluding remarks

Aided by artificial intelligence (AI), topological data analysis (TDA) has become a rapidly growing field in applied mathematics and data science. A key technique of TDA is persistent homology, a multiscale algebraic topology tool that has demonstrated impressive success across diverse disciplines in science, engineering, medicine, industry, and defense. However, persistent homology with simplicial complex faces several limitations, including: inability to handle non-geometric information, reliance on qualitative descriptions of high-order cycles, inability to track non-topological changes, incapability in handling directed networks, and insensitivity to structured data such as hypergraphs. Additionally, while algebraic topology techniques typically analyze point-cloud data, they are not directly applicable to data on differentiable manifolds or one-dimensional curves embedded in 3-space. These challenges in TDA have motivated significant advancements in the past decade.

This paper provides a comprehensive review of TDA techniques developed to address persistent homology's limitations, emphasizing their mathematical foundations and strengths. Some techniques extend persistent homology from simplicial complexes to other topological domains, such as cell complexes, path complexes, directed flag complexes, cellular sheaves, and hyper(di)graphs. Others introduce alternative topological formulations, including Laplacians [302], Dirac operators [9], sheaf theory, Mayer topology [269], and interaction topology [193]. For data on differentiable manifolds, methods rooted in differential topology, such as persistent de Rham cohomology, persistent Hodge Laplacians, and Hodge decomposition, have been developed [64, 282]. For data on one-dimensional curves in 3-space, tools from geometric topology, such as the

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Gauss linking integral [270], Jones polynomial [277], and Khovanov homology [197, 268], have been proposed.

The rapid growth of TDA is driven by machine learning, such as topological deep learning (TDL) [37] and mathematical AI. However, topological representations from TDA cannot be directly implemented in machine learning without appropriate vectorization. This paper also reviews the selection of topological algorithms for different input data formats, their associated topological representations, and methods for topological vectorization in machine learning contexts.

TDA and TDL are dynamic and rapidly evolving fields. The future development of TDA and TDL will be driven by both theoretical curiosity and practical needs, including the demand for interpretable deep neural networks. We envision future advancements in the following directions.

Currently, the local geometry and/or topology surrounding individual data points can be described by persistent cohomology [39], evolutionary homology [41], interaction topology [193], and persistent sheaf Laplacian [309]. Further efforts are needed to refine local topology methods for practical applications.

Most current TDA approaches were developed for point-cloud data. Sequential data, however, are widely used in applications such as DNA sequencing, natural language processing, and electroencephalogram (EEG) analysis. These areas require novel TDA methodologies.

In differential geometry, the Atiyah–Singer index theory establishes a topological index. A persistent index theory for families of elliptic differential operators could be developed for data on compact manifolds.

The advancement of low-dimensional topology methods is critical for analyzing 1D curves embedded in 3-space (e.g., entangled polymers and neural networks). Persistent Floer homology may offer an effective framework for modeling protein-protein binding interactions. Additionally, robust computational algorithms for persistent Khovanov homology and persistent Khovanov Laplacians remain an urgent priority.

Finally, large language models (LLMs) herald a new era in AI, creating opportunities across disciplines. Chatbots like DeepSeek and ChatGPT can transform abstract mathematical theories—including computational topology—into practical tools [196]. Rapid growth is anticipated in AI-assisted topology and topology-enhanced AI in the coming years.

TDA is a vast interdisciplinary field, spanning algebraic, differential, and geometric topology; computational algorithms; AI-driven methodologies; and real-world applications. While this paper aims to comprehensively review TDA and TDL beyond persistent homology, we inevitably omit many significant contributions due to the field's breadth. We note that tremendous efforts have been made to develop topological neural networks since the introduction of TDL in 2017 [37]. To focus on TDA, we have restricted ourselves from elaborating on this development. The interested reader is referred to a recent review on various aspects of TDL [243]. We hope this work serves as a guide

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for researchers exploring topological techniques and helps them select optimal approaches for analyzing specific problems across diverse fields.

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